Heat & hot water

Sprinter hydronic heat and hot water — design, living with it, and install

September 2026

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Source Register

Part one: safety → design → living with it → install. Part two is the technical record.

How to Use This Book

Read in this order: Safety → The sofa → System Design → The Numbers → User's Guide → Current Status → Builder's Guide.

STAGE	CHAPTER
Bay the tanks bolt into	The sofa
Architecture	System Design
Day-to-day (after Acceptance)	User's Guide
Install	Builder's Guide

Skim The Numbers as a short card, then keep going. While you build, return to Safety, Controls, and The Parts whenever a HOLD points there. Do not light the heater from the User's Guide until Acceptance.

Two finish lines. Diesel cabin heat and diesel hot water can pass before HOLD 9. The complete system, including Isotemp AC, needs HOLD 9. The install map lives at the top of the Builder's Guide.

Lookups. For numbers and citations, use The Numbers, the Isotemp value study, Sources, Decisions, Questions for You, Open Questions, Hazards, and the Locked facts checklist. Part names live in The Parts. What is on the van versus still in the garage lives in Current Status. Part two is reference, not the work order. DANGER, CAUTION, and MUST boxes override surrounding prose.

When sources disagree

1. SRC-009 – Hydronic D5S fuel, exhaust, wiring, and first fire
2. SRC-019 – hose practice, header height, and bleed
3. This book – locked van layout (DEC-009 · DEC-010 · DEC-004 / DEC-013 / DEC-020)
4. OPEN items – do not invent an answer; prefer Questions for You

Diagrams are not to scale. Land every terminal from SRC-009, SRC-003, and SRC-002 on the bench.

Safety

Read this chapter before any procedure. Anyone helping needs to read it too.

Jump: System Design · User's Guide · Builder's Guide · Hazards · The Numbers · Isotemp value study

DANGER

Carbon monoxide. The exhaust must end in open air, away from the combustion-air intake and from cabin openings. Stop work if exhaust can reach breathing air. Never run the Hydronic D5S diesel heater in a closed garage. → HAZ-002

DANGER

Fire and fuel. Keep the fuel line off the exhaust. Keep the engine and heater off during fuel work. Inspect the fuel system before first fire. Do not light the heater to "test" a wet line. Use diesel EN 590; up to 20% FAME is acceptable; pure biodiesel is not (SRC-009). → HAZ-001 · HAZ-008

DANGER

Scalding. Tap water can track glycol temperatures of about 150–185°F. Use only the AM100-1LF thermostatic mixing valve on the Duda B3-12DW-20 plate heat exchanger, set near 120°F and labeled. Draw taps from the mixed outlet only. The Isotemp factory mixer stays capped forever – never for taps (DEC-008). → HAZ-005

DANGER

Shock and fire – Isotemp element. The 750 W / 115 VAC element lands on Paneltronics **WATER HEATER** only—never on 12 V (DEC-004 / DEC-020). Never energize an empty chamber. Confirm upstream GFCI/ELCI and cable route before first use (HOLD 9). → HAZ-006 · HAZ-009

CAUTION

Unattended Isotemp AC. Paneltronics **WATER HEATER** is not cut by the master switch. Do not leave the inverter and **WATER HEATER** on for long unattended periods unless you intend a deliberate chamber top-off on shore or abundant solar, the chamber is full, and the GFCI path is verified (HOLD 9). → HAZ-006 · HAZ-012 · User's Guide – AC chamber

CAUTION

Hot glycol and steam. Keep the loop full and bled. The heater limit is 2.5 bar; the expansion header cap relieves near 1.2 bar. Never run the pump dry. Check the expansion header only when cool. Treat running hoses as burn-hot. A lukewarm cabin heater while the burner runs usually means air or a kink (minimum flow about 250 L/h). → HAZ-003

CAUTION

Air lock, overheat, and lockout. Use reinforced hose, avoid forced kinks, and put bleeders at high points. Do not thrash the EasyStart Timer off and on more than twice into a fault; clear locks with diagnostics (SRC-009). After an overheat, wait until coolant is below about 70°C. Work on the exhaust only when the exhaust is cold. → HAZ-010

MUST

Keep glycol and drinking water apart; heat crosses only at the Duda double wall (DEC-003). Cap unused ports. Mix at most 50% propylene glycol, one brand only. Keep heater supply voltage about 10.5–16 V. Do not run overnight heat above about 5,000 ft without Altitude kit 22 1000 33 22 00 (DEC-006). → HAZ-007 · HAZ-011

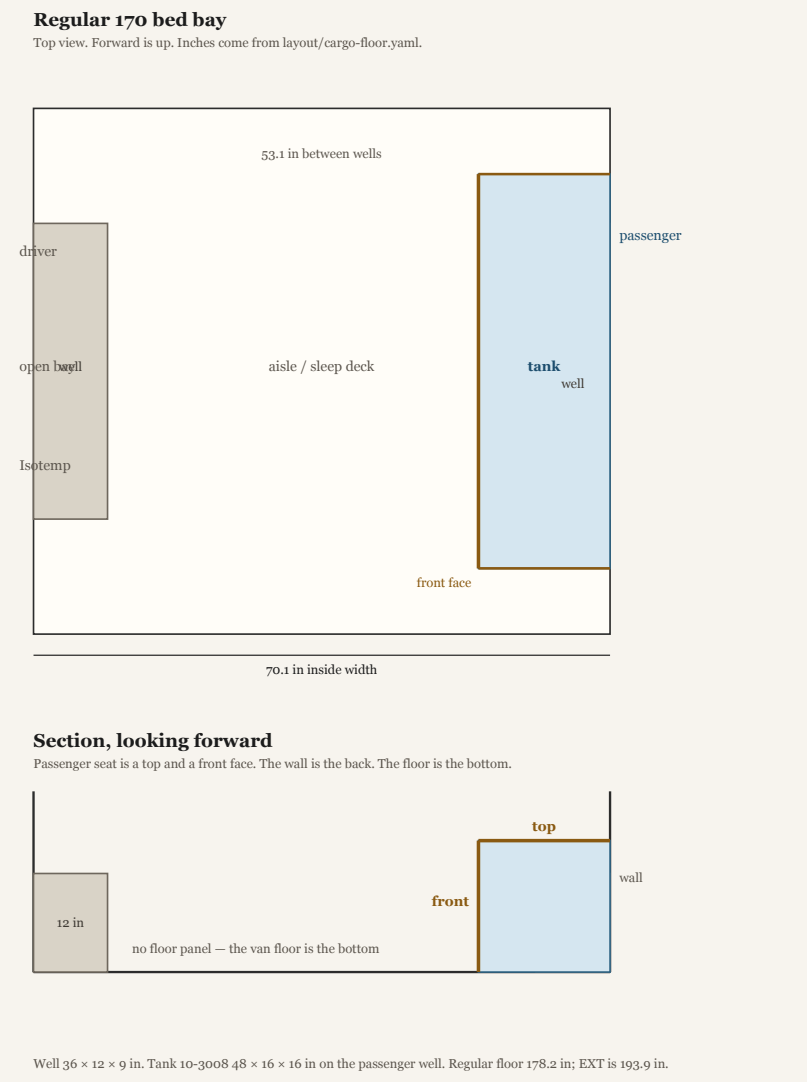
The sofa

The sofa is the cargo floor over the rear wheel wells. The current bed comes out first. That bay blocks the cabin mounts: do not bolt the freshwater tank, the freshwater strainer, the Isotemp Slim Square 4.2 gal heat battery, or the 5 L expansion header until the seat-base frame gives them a place. Fuel and exhaust under the van can proceed at the same time.

Cargo floor

The body is the regular 170, not the 170 EXT. Both use the 170-inch wheelbase and the same wheel-well envelope. The EXT cargo floor is longer by about 15.7 inches, and that extra length is behind the rear axle (SRC-043). A bed drawn on an EXT plan will run long on this van.

Change inches in `book/layout/cargo-floor.yaml`, then run `./book/build.sh`. The script rewrites the drawing. Do not edit the SVG by hand.



Seat base on the cargo floor

Inside width at the floor	70.1 in
Width between wheel wells	53.1 in
Wheel well, regular 170, including 2017	36 in along the wall × 12 in high × 9 in into the cabin (SRC-042)
Cargo floor, regular 170	178.2 in
Cargo floor, 170 EXT	193.9 in
Freshwater tank, ordered	S&B **10-3008**, passenger well, 48 × 16 × 16 in, ±0.5 in, about 35 gal (SRC-044)

The 36 × 12 × 9 envelope is the published upfit box for a single-rear-wheel van. Rattle Trap, the cork floor, and the curved shoulders of the hump change the cavity you can actually use. Measure the hump on this van before you cut tube. The distance from the rear doors to the hump is not in these sheets. Measure that too. The drawing does not place the bay along the van.

What the bay holds

The passenger well holds the S&B tank. The freshwater strainer sits on that tank's outlet. The driver bay holds the Isotemp, on its own aluminum cradle bolted to the floor. The expansion header goes in the backrest, high enough that its top stays the highest circulating glycol point. Confirm that height after the frame exists.

Day geometry, once the deck exists: outboard panels become seat backs, and inboard panels become seat butts, with an aisle between them. Panel count and hinge or slide hardware stay open until the frame is measured.

Frame first

Build the seat-base frame before the deck, the backrest, the cushions, or the sleeper hardware. The frame is EZTube 1 in square tube (0.050 in wall, 6063-T5) with Rostek 3-way corners (drawing) at the outer corners. Crossmembers need their own T fittings. The frame bridges each wheel well and lands on the floor. Feet go at the ends of the well and along the aisle side, once those pads are confirmed. Do not put seat load on the wheel-well sheet metal or on the freshwater tank.

Buy one Rostek fitting and try it on the actual tube before you buy a set. Do not overtighten a set screw into the 0.050 in wall. The connector drawing does not give a load rating for this seat. The corner stands above the 1 in tube. Pocket the underside of the sapele so the sheet bears on the tube. Measure that stand-off on the trial fitting before you rout the sheet. EZTube, the Rostek corners, and the sapele are the accepted options (DEC-021).

The deck and the day panels are Chesapeake Light Craft premium sapele marine plywood. Cut that sheet only after the frame footprint is measured.

	PROVISIONAL	STATUS
Length along the wall	about 75 in	Recalled. Confirm.
Height	about 21 in	Recalled. Not yet tied to the bare frame, the hump, or the finished seat.
Depth, wall to aisle	–	Not set.
Crossmember count and spacing	–	After the footprint is measured.

Bill of materials intent: EZTube and Rostek corners for the frame; T fittings for crossmembers; short T-slot or bolted aluminum for the Isotemp cradle; Chesapeake Light Craft premium sapele for the deck and for the ⅜–½ in day/night panels (**2–3** panels); hinges or drawer slides; foam tape where the frame meets metal.

Before you cut

1. Measure the sofa opening, the wheel-well length, projection, and height, and the driver-side bay.

2. Mark the Isotemp envelope on the driver-side floor with tape. Mark the passenger tank envelope on that well. The frame clears both and lands on the floor.
3. Confirm the shortest practical coil-hose route to the Duda under the sink and the Sure Marine cabin heater under the fridge. Keep the coil below the expansion header. Leave about 30 cm in front of the Isotemp element and thermostat. Protect the Paneltronics cable.
4. Support the Isotemp empty and full. Empty mass is about 35 lb. A full chamber adds about 16 L of static glycol. The cradle is the restraint. Hose fittings are not. Leave the fill port, the vent, the safety valve, the coil fittings, and the element connection reachable. Day panels, when they exist, must move without draining the loop.
5. Plan wall access hatches where bolts must reach metal through the Rattle Trap and the insulation. Do not gut a whole cavity to reach one bolt.
6. Fit-check one Rostek corner, then record the floor-foot locations. Do not cut the deck until that footprint is settled.

HOLD S — sofa bay

- ☐ Current bed removed; seat-base frame bridges the wheel well and lands on the floor (deck later)
- ☐ Passenger tank envelope marked; strainer has a place on the tank outlet
- ☐ Driver-side bay selected for the Isotemp; tape-fit done
- ☐ Isotemp full-load weight and restraint method documented (aluminum cradle and straps)
- ☐ Expansion header location remains the highest circulating point (exact mount named)
- ☐ Hybrid construction per DEC-021; floor-foot locations recorded
- ☐ Plywood access hatches planned where wall bolts need metal

Pass this HOLD before HOLD 3 and before HOLD 6. Fuel and exhaust under the van do not wait on it.

System Design

Stage 1 — Architecture. Read the glycol loop first, then the two jobs off that loop, then the part names, then how you call for heat. Next, skim The Numbers and continue to the User's Guide.

Jump: Safety · The glycol loop · The Parts · Controls · The Numbers · User's Guide · Current Status · Builder's Guide

Purpose

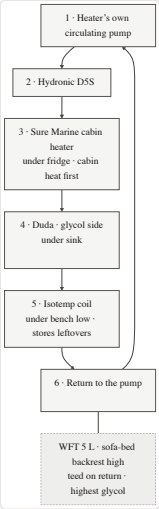
One closed propylene glycol loop heats cabin air and tap water. The Hydronic D5S sits under the van. The Sure Marine cabin heater, Duda, Isotemp, and expansion header sit in the cabin. Glycol never mixes with drinking water, never connects to the engine (DEC-009), and has no second pump (DEC-013).

Leftover heat can sit in the Isotemp static chamber. A planned 750 W / 115 VAC element on Paneltronics **WATER HEATER** can warm that chamber. The element does not pump the loop, and it does not put hot water at the taps by itself.

Stop and reject any proposal that would splice into the engine, tee glycol into drinking water, bypass cabin heat, add a second glycol pump, or land the 750 W element on the 12 V panel.

Day-to-day use is in the User's Guide. Keep-or-sell numbers are in the Isotemp value study.

The glycol loop



Glycol loop

Glycol loop layout

Glycol circulates in this order only (DEC-010):

1. Heater's own circulating pump
2. Hydronic D5S
3. Sure Marine cabin heater — cabin heat first
4. Duda (glycol side) — hottest fluid for sinks (~**80%** of hot-water use is sinks)
5. Isotemp coil — stores leftovers
6. Return to the pump

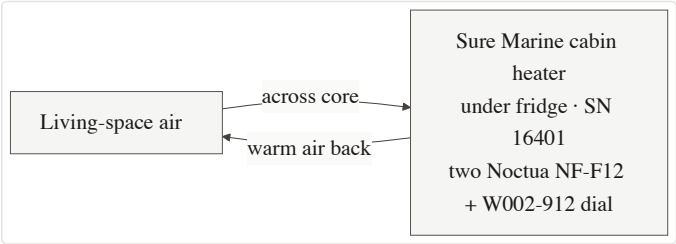
The expansion header tees into the **return** at the highest circulating point. It is not in series, and it is not teed off the Isotemp coil. Keep the expansion header highest and everything else below it. Leave an unrestricted path from that tee back to the pump. Put fill and drain at the low point. Mount the cabin heater level, bottom-in / top-out, with a bleeder on the outlet.

Where those parts sit. The Sure Marine cabin heater is under the fridge (SN 16401). The Duda is under the sink. The Isotemp sits driver-side under the bench (low). The expansion header sits in the sofa-bed backrest (high). Day panels map outboard to seat back and inboard to seat butt. The sofa/bed is hybrid (DEC-021): an EZTube seat-base frame with Rostek corners bridges each wheel well and lands on the floor; the deck is Chesapeake Light Craft premium sapele, cut after the frame; the Isotemp sits in an aluminum cradle. The body is the regular 170, not the 170 EXT. The freshwater tank occupies the passenger wheel well. That bay is The sofa. Build the frame before you bolt the tank, the strainer, the Isotemp, or the expansion header.

The two jobs

Cabin air

Living-space air crosses the Sure Marine cabin heater under the fridge. Two Noctua fans and one dial are on that unit. Combustion air and exhaust stay outside.



Cabin air

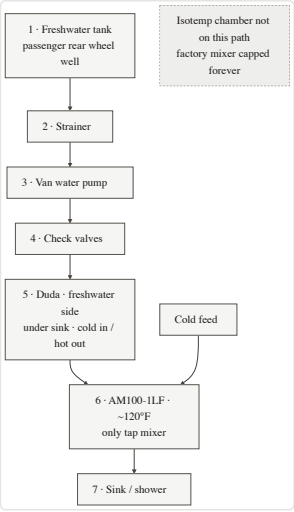
Sure Marine cabin heater

Tap water

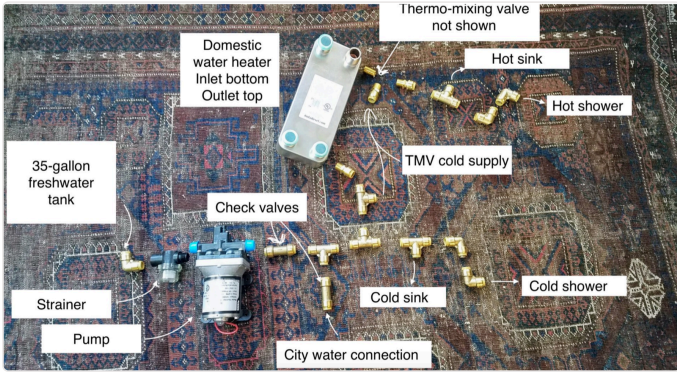
Freshwater runs in this order only (DEC-003 / DEC-008):

1. Freshwater tank (passenger rear wheel well)
2. Strainer
3. Van water pump
4. Check valves
5. Duda (freshwater side — cold in bottom, hot out top)
6. AM100-1LF (~120°F) — cold feed joins here
7. Sink / shower

Heat crosses the Duda double wall only. The Isotemp chamber is not on this path. The factory Isotemp mixer stays capped forever.



Fresh water system



Freshwater layout

The Parts

Meet each part once here. After this table, use only these official names or the short forms listed under each name. Do not invent aliases (buffer, Bosch, Espar, “the tank,” “the core,” “the plate”).

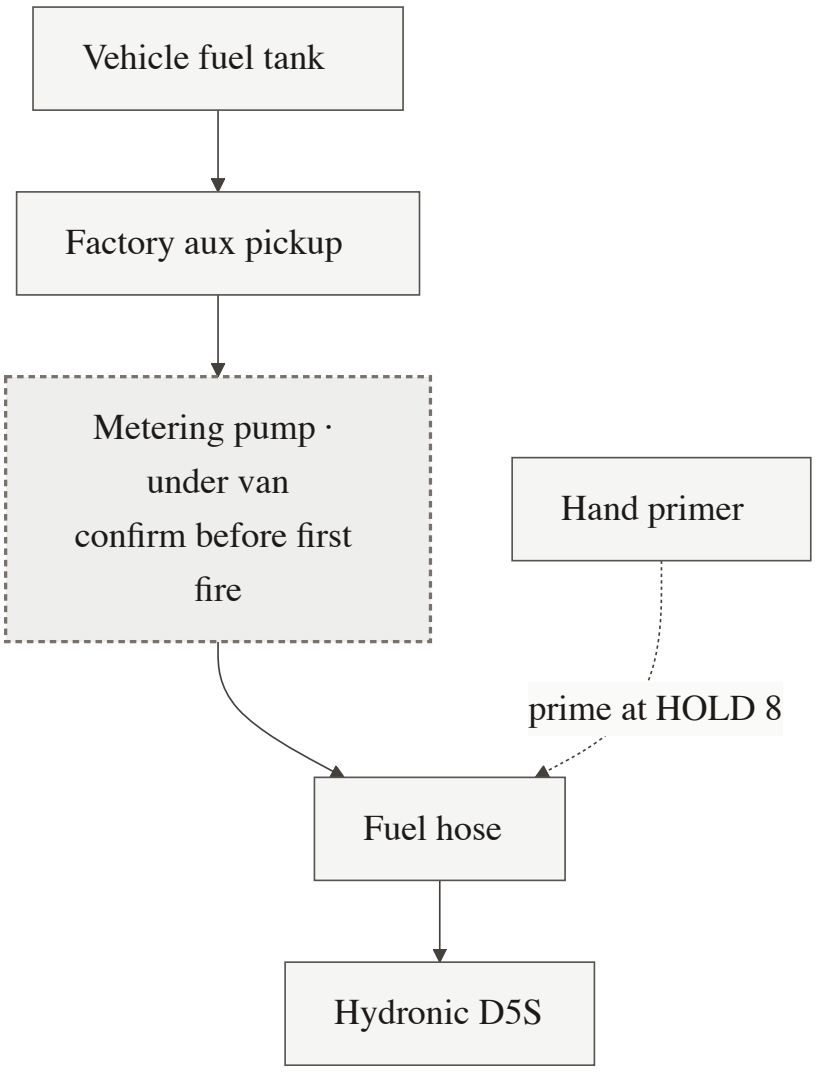
PART (OFFICIAL NAME)	PHOTO	DOES	DOES NOT
Hydronic D5S diesel heater *Short: Hydronic D5S; "the heater" only when unambiguous*		Burns diesel; own pump moves the loop; stages on coolant temperature (The Numbers; SRC-009)	Connect to the engine
Sure Marine cabin heater *Short: cabin heater*		Glycol → cabin air; two fans + one Low/Med/High dial (DEC-015); target both Noctua NF-F12	Pull combustion air into the cabin
Duda B3-12DW-20 plate heat exchanger *Short: Duda; Duda plate*		Glycol → freshwater across a double wall	Mix the two fluids
AM100-1LF thermostatic mixing valve *Short: AM100-1LF; the mixer*		Limits shower/sink temperature (~120°F)	Connect to a glycol fitting
Isotemp Slim Square 4.2 gal heat battery *Short: Isotemp; Isotemp coil; Isotemp chamber*		Softens cycling via coil ↔ static chamber; planned AC on **WATER HEATER**; factory mixer capped forever	Hold drinking water; replace the Duda; circulate glycol with the element; mix taps
5 L expansion header *Short: expansion header. The jug on hand is Water Flow Tank, MPN 65288A*		Takes expansion and collects air at the high point. Cap relieves near 1.2 bar	Sit in series in the loop; use the Isotemp safety valve as this cap
EasyStart Timer *Short: EasyStart*		Schedule / target / start-stop (SRC-003); bottom unit on the wood panel (under LinkPRO / PROwatt)	Replace the master lockout
Master switch *Representative Sure Marine W005-378K*		Off cuts 12 V to EasyStart, SC1600B, and cabin fans (DEC-012); garage unit not photographed yet	Control Isotemp AC

PART (OFFICIAL NAME)	PHOTO	DOES	DOES NOT
SC1600B thermostat *Short: SC1600B*		Heat **signal** only (R–W; no fan output — SRC-033); needs relay (Q-022)	Fan power; hot water; trusted auto until landing verified
Noctua NF-F12 PWM fans *Short: Noctua fans; cabin fans*		Move cabin air; power leads only (DEC-015)	Heat call
Fan speed controller *Short: fan dial; book pick Sure Marine W002-912*		Low/Med/High **without** hard Off (DEC-018)	Off / start the heater
Altitude kit 22 1000 33 22 00 *Short: altitude kit*		Cuts fuel delivery at altitude (SRC-002; DEC-006)	Replace EasyStart; needed only for high camps
Paneltronics AC panel *Branch label always **WATER HEATER***		**WATER HEATER** branch feeds Isotemp 750 W (DEC-020)	12 V heater control

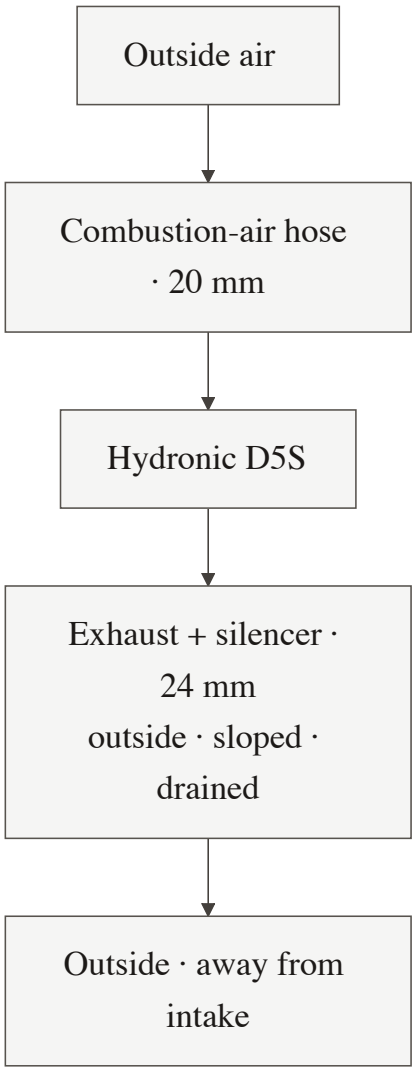
Fuel, air, exhaust, and fluids

Fuel. Fuel runs from the factory aux pickup through the metering pump to the Hydronic D5S. Prefer a metering-pump angle of 15°–35°. Keep the fuel line rising continuously, and never rest it on the exhaust (SRC-009). Prime with the hand bulb at HOLD 8, then remove it. Confirm the fuel pump under the van before first fire (Current Status).

Exhaust and combustion air. These are two separate paths to outside air. Neither path enters the cabin. Kit limits are in The Numbers. Confirm the install at HOLD 2.


Vehicle fuel tank
Factory aux pickup
Metering pump · under van confirm before first fire
Hand primer
prime at HOLD 8
Fuel hose
Hydronic D5S

Fuel system



Combustion air and exhaust

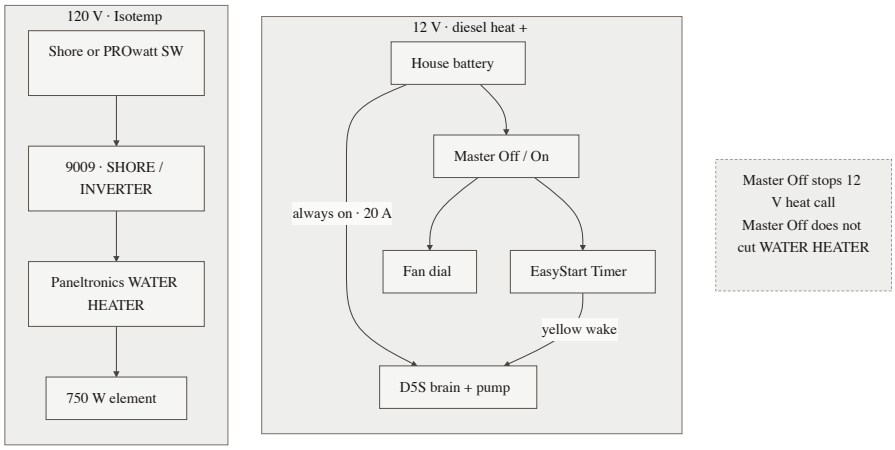
Glycol. Fill the circulating loop and the Isotemp chamber with water plus propylene glycol at **most 50%** glycol, **one brand only** (SRC-009; SRC-029 / SRC-030). Prefer a water-first leak check before you commit glycol. Never run the pump dry. The freshwater path is under Tap water.

Controls

Hydronic wiring is two separate worlds. They never share a fuse panel.

WORLD	POWER	WHAT IT DOES	PARKED LOCKOUT
12 V	House battery	Diesel heat call, heater pump, cabin fans	**Master Off**
120 V	Shore or PROwatt SW → Paneltronics	Isotemp chamber element only	**WATER HEATER** breaker Off

Master Off stops diesel heat and fans. Master Off does **not** cut Isotemp AC. Isotemp AC never lands on the 12 V panel.

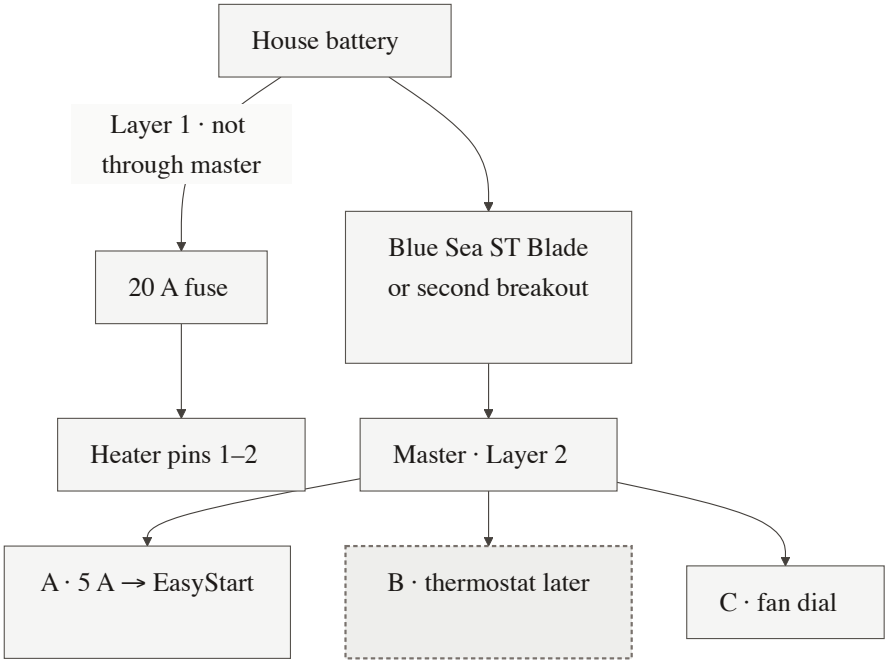


Day-one electrical

Day-one diesel heat (enough for first fire): turn the master **On**, start heat on the EasyStart Timer, and set the fan dial. Pin tables and fuse sizes live in Electrical and Controls. Still open: thermostat auto (Q-022), Isotemp GFCI (Q-009), and the altitude kit at HOLD 4 (Q-015).

12 V — three layers

Only the middle layer goes through the master switch.



Power

Layer 1 — heater always powered. Run a heavy pair from the house battery through a **20 A** fuse on the positive. Use **12 AWG** if the round trip is **20 ft / 6 m** or less. Land heater pins **1** (red +) and **2** (brown −). This feed does **not** go through the master. The heater brain stays powered so it can finish an after-run and remember faults.

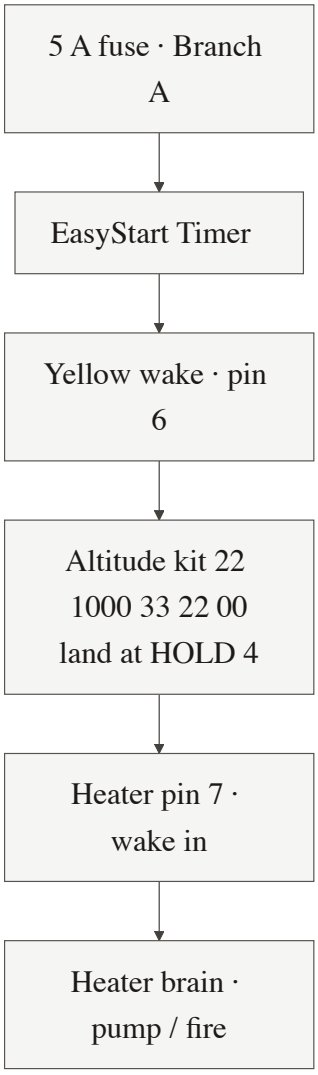
Layer 2 — master unlocks the cabin controls. Mount one Off/On switch next to the EasyStart. The master switches **positive only** to three small branches. Land them on the Blue Sea ST Blade, or on a **second breakout** if that block is full:

BRANCH	FUSE	FEEDS
A	**5 A** (insert **last**)	EasyStart Timer
B	size still OPEN	Later: thermostat → relay
C	size still OPEN	Fan dial → both Noctua fans

Master On does **not** start the heater. Master On only unlocks those branches so EasyStart (and later the thermostat) can call for heat. When you park or leave, turn the master **Off**.

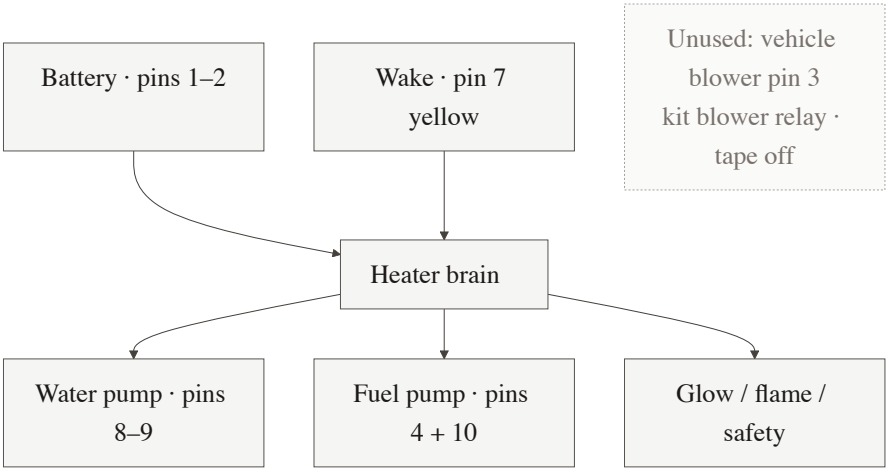
For an emergency stop, turn EasyStart off, then master Off, then pull the 20 A if you must, and disconnect the battery last. Do not cycle off and on more than twice into a fault (SRC-009).

Layer 3 — EasyStart wakes the heater.



Wake

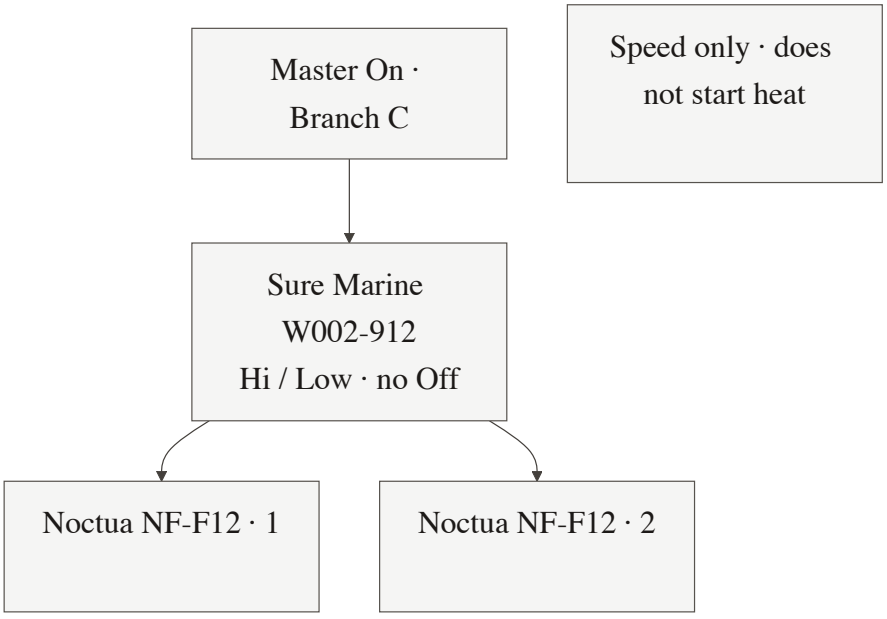
Branch A powers EasyStart pin **1** (+) and pin **3** (−). On start (or a schedule), EasyStart puts positive on pin **6** (yellow). That yellow wire becomes heater pin **7** (S+ — "switch on"). Once the heater sees S+, it decides whether to pump, glow, and fire. With wake present, the heater runs its water pump (pins **8–9**), the fuel metering pump (pins **4** and **10**), and glow / flame sensing. Tape off kit vehicle-blower wiring (pin **3**, blower relay, and the 25 A blower fuse). That path is for a car heater matrix, not the Sure Marine cabin heater.



Heater

Altitude kit. The kit plugs into the control harness as a whole adapter. Do not snip only the yellow wire. Land it at HOLD 4 while the EasyStart harness is open (DEC-006 / Q-015). It is required for overnight camps above roughly 5,000 ft.

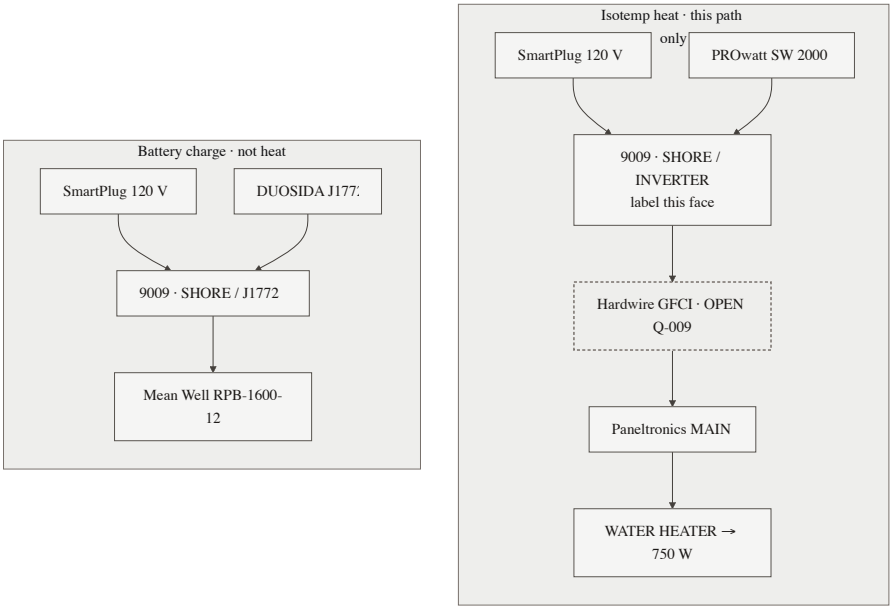
Cabin fans ride Branch C only. Book pick: Sure Marine **W002-912**. Do **not** use a Noctua NA-FC1. Both Noctua fans share red and black after the dial; leave PWM and tach unused. The dial sets **speed only**. It never starts the diesel. Until Q-022 closes, the fans run whenever the master is On and the dial is up.



Fans

Thermostat auto is not ready. The SC1600B **R** / **W** terminals are a dry contact. They need a **relay** for fan current (SRC-019). Do **not** land them on EasyStart pins **9–10** (those pins are for Eberspächer's optional temperature sensor). Until Q-022 names an approved wake path, start heat only from the EasyStart Timer. Relay notes live in Electrical and Controls.

120 V — Isotemp chamber only



AC

The 750 W element is a Paneltronics load. Power runs from **SmartPlug** shore or the **PROwatt SW 2000** through one **Blue Sea 9009** rotary (**SHORE** / **INVERTER**) into Paneltronics MAIN, then the **WATER HEATER** breaker, then the element. Label that 9009 face—it is poorly labelled today (`photos/ac-electrical.jpg`). Upstream hardwire GFCI/ELCI is still OPEN (Q-009). Inverter face GFCI outlets protect only loads plugged into them.

The element warms the Isotemp **static** chamber only. Circulating glycol still needs the diesel pump through the **coil**. The element alone does not make a hot shower. Route the element cable at HOLD 4. Leave **WATER HEATER** off until HOLD 9.

Second 9009 (not heat). A separate rotary selects SmartPlug shore versus **DUOSIDA** J1772 into the **Mean Well RPB-1600-12** charger. That path charges the house battery. It does not feed **WATER HEATER**.

This book does not rebuild the whole van electrical system. House inventory and photo extracts live in Working design understanding and Electrical and Controls. Wire sizes in this chapter are **AWG**; the manuals print mm² for the same conductors (SRC-009 / SRC-003).

→ The Numbers, then the User's Guide.

The Numbers

Skim this card after System Design, then continue to the User's Guide. Figures come from the Hydronic D5S diesel heater manual (SRC-009) unless noted. If you swap the heater, rewrite this page from that heater's manual first. Chamber sizing and keep-or-sell numbers live in the Isotemp value study.

Jump: System Design · User's Guide · Isotemp value study · Builder's Guide

Heat output (Power / High / Low)	5.2 / 5.0 / 2.1 kW (~17,740 / 17,060 / 7,165 Btu/h)
Fuel use at those settings	0.64 / 0.61 / 0.26 L/h
Coolant staging (approx.)	Power → High ~65°C · High → Low ~80°C · pause ~85°C · restart ~75°C
During control pause	Burner off; **pump keeps running**
Overheat restart	Coolant below 70°C (158°F)
Fluid in the heater itself	~0.18 L
Minimum flow through the heater	250 L/h
Heater pump (reference)	~680 L/h @ 0.1 bar; under ~12 W
Max propylene glycol overpressure	2.5 bar
expansion header relief	~1.2 bar
Isotemp potable PRV (not glycol)	~5.5 bar / 80 psi
Isotemp AC working thermostat	Cuts near **75°C / 167°F**; overheat ~95°C, manual reset (SRC-022)
Mix	≤50% Prestone LowTox AF555 only (DEC-017)
Voltage window	10.5–16 V
Electrical start / run (without pump)	~120 W start; ~40 / 37 / 12 W Power / High / Low
Fuel	Diesel EN 590; ≤20% FAME OK; pure biodiesel not
Metering pump angle	Prefer 15°–35°; flat 0°–15° forbidden
Metering pump heads	Pressure tank→pump max 3000 mm; suction pressureless max 1000 mm; suction with 0.03 bar cap max 400 mm
Exhaust	24 mm ID; ~0.2–2 m; downslope; ~5 mm drain
Combustion air	20 mm ID; kit hose max ~0.76 m
Altitude	Unlimited to ~5,000 ft without kit; kit 22 1000 33 22 00 for 5,000–10,000 ft
AM100-1LF	Set ~120°F, lock, label (DEC-008)
Isotemp factory mixer	**Unused forever**
Isotemp element	750 W ≈ **6.5 A** @ 115 V on **WATER HEATER** only
Paneltronics AC feed	One **9009**: shore ↔ PROwatt → **WATER HEATER**
Mean Well charge	Other **9009**: shore ↔ J1772 (not heat)
Diesel start lockout	≤2 off/on into a fault

Planning scale

Chamber charge cool → ~75°C	~**40–60 min** at 750 W; taps cold until pump runs
Holdover at Low (2.1 kW)	**Tens of minutes** – Isotemp value study
Dish rinse, warm loop	~**30–60 s** to hot; ~**2–5 min** rinse
Dish rinse, cold loop	Flame usually on; useful hot in ~**2–5 min**

User's Guide

Stage 2 — Use. Read this after System Design and The Numbers. After Acceptance, use this chapter as the day-to-day manual. Until Acceptance, treat the tables below as the operating target—not as permission to light the heater.

Diesel cabin heat and diesel hot water can pass before HOLD 9. The complete system, including Isotemp AC, needs HOLD 9. Keep-or-sell numbers stay in the Isotemp value study.

Jump: System Design · The Numbers · Current Status · Builder's Guide · Acceptance

After Acceptance

YOU WANT...	DO THIS
Parked / unattended	Master **Off**. Leave **WATER HEATER** off unless you deliberately want a shore or solar chamber top-off (AC chamber)
Cabin heat	Master **On** → EasyStart start → fan dial → wait a few minutes
Mild night, Low oversized	Burner may pause and restart; the chamber softens some short cycles
Shower / lots of hot water	EasyStart running; draw only through the AM100-1LF; expect the diesel to work
Charge chamber (after HOLD 9)	Set the Paneltronics **9009** to shore or inverter → turn **WATER HEATER** on (chamber must be full). Taps stay cold until EasyStart runs
Dish rinse after charge	Short EasyStart start so the **pump** runs – AC chamber and dishes
Pre-heat later	Master **On** → set an EasyStart schedule
Mountains overnight	Altitude kit already landed at HOLD 4; do not overnight above ~5,000 ft without it
Emergency stop	EasyStart off → master Off → Isotemp AC off → fuse or battery if needed. No more than two off/on cycles into a fault

Until Q-022 closes, start heat from the EasyStart Timer only—not from the SC1600B.

A heat call starts the Hydronic D5S **pump**. The **burner** fires only if coolant is cool enough. Cool cabin air for the first minute or two is normal. When the loop is warm enough, the flame may pause near **85°C** and restart near **75°C** while the pump keeps circulating (The Numbers). That is when leftover heat can move into or out of the Isotemp chamber.

Hard rules

The boxes in Safety still govern. For day-to-day use:

1. Turn the master Off when the van is parked or unattended.
2. Draw taps only through the AM100-1LF (about 120°F, labeled). To set the valve: loosen the handwheel screw, lift, turn, lock, and check at the faucet (SRC-036).
3. Do not run overnight heat above about 5,000 ft without the altitude kit (DEC-006).
4. Do not cycle the EasyStart off and on more than twice into a fault. Clear locks with diagnostics (SRC-009).
5. Check the expansion header level only when the system is cool.
6. Hot water at the taps needs circulating glycol from the diesel **pump** through the Duda. Isotemp AC heats the static chamber only.
7. The Isotemp element lands on Paneltronics **WATER HEATER** only—never on 12 V, and never into an empty chamber (DEC-020).
8. The factory Isotemp mixer stays capped forever (DEC-008).
9. Label the Paneltronics **9009** face **SHORE** / **INVERTER** before you rely on it.

What the Isotemp does day to day

The Isotemp is not a faucet tank. Planning numbers live in the Isotemp value study.

Cabin heat. The D5S Low stage is about **2.1 kW**. On mild nights Low is often oversized. The static chamber can soak up some of that surplus through the coil and give it back while the burner pauses. Expect the clearest cycling help around **40°F**, some help around **20°F**, and little help at **0°F** and colder. Holdover is **tens of minutes**, not hours.

Hot water. Heat at the sink or shower always means EasyStart → pump → Duda → AM100-1LF. The chamber can shorten a diesel run or feed a short dish rinse after an AC charge. It cannot replace the pump.

AC chamber and dishes

Use this only after HOLD 9 (chamber full; GFCI path verified).

The 750 W element is **120 V** only. The hydronic master does not cut it. Energy arrives from shore or the PROwatt SW through the Paneltronics-path **Blue Sea 9009** (`photos/ac-electrical.jpg`). Mark that face **SHORE** / **INVERTER**. Mark the other 9009 **SHORE** / **J1772** for Mean Well charge only.

Charge. Connect shore power, or run the inverter with enough headroom for about 750 W plus inverter idle. Turn Paneltronics **WATER HEATER** on. The element heats the static chamber only. From a cool chamber, plan about **40–60 minutes** to reach the working band near **75°C**. Taps stay cold during the charge because the pump is not running.

Leave WATER HEATER on? Leave it on when you are on shore power, or when solar clearly covers about 750 W plus inverter idle, and you want the chamber ready. Leave it off when you are saving the house battery, parking unattended for a long stretch, or HOLD 9 is still unfinished. Kill **WATER HEATER** (and the inverter if you are done) separately from the hydronic master.

Dish rinse after a charge. Turn the master On and make a short EasyStart start so the **pump** runs. If the glycol loop is still warm, the flame often stays off. Hot water at the sink usually arrives in about **30–60 seconds**, and you get about **2–5 minutes** of rinse. If the loop is cold, the flame usually lights, and useful hot water takes about **2–5 minutes**. This is a dish rinse, not a shower.

If something is wrong

1. Turn EasyStart off, then master Off, then Isotemp AC off.
2. Pull the fuse or disconnect the battery if needed (that counts toward the two off/on limit).
3. Before you restart, check for fuel wetness or smell, exhaust into the cabin, a glycol leak, a low expansion header while the system is cool, battery outside 10.5–16 V, or a lukewarm Sure Marine cabin heater while the burner runs. → Hazards
4. After an overheat, wait until coolant is below about 70°C.

Normal Power / High / Low staging is not a fault. See The Numbers.

Occasional checks

WHEN	CHECK
After loop work	Cool expansion header level; Prestone LowTox AF555 ≤50%
Before high camps	Altitude kit installed (landed at HOLD 4)
Early fires	Fuel and exhaust for wetness, rub, or smell
Before a heat call	Battery 10.5–16 V
Before Isotemp AC	Chamber full; **WATER HEATER** plus upstream GFCI/ELCI
Long storage	Master Off; Isotemp AC off
After first fire / after HOLD 9	Log starts/hour, pause length, dish rinse, AC charge minutes — Isotemp commissioning log

→ Status, then Make it.

Current Status

Read this after the User's Guide and before the Builder's Guide. Honest inventory as of September 2026. Part names match The Parts. The HOLD checklists still govern the work.

Jump: Project Phases · Builder's Guide · Questions for You · Isotemp value study

Next action

> **NEXT**
>
> **Next — two tracks.** Under the van, finish HOLD 1 and HOLD 2 against SRC-009. In the cabin, start The sofa. Do not bolt the freshwater tank, the strainer, the Isotemp, or the expansion header until HOLD S. **Future work before first fire:** get under the van and confirm the **fuel pump** (owner not yet confident); then prime at HOLD 8.

PHASE	STATUS
S. Sofa bay	**Do next, with A** — The sofa; blocks the tank, strainer, Isotemp, and expansion header
A. Fuel and exhaust	**Do next, with S** — HOLD 1–2 open
B. Mount and wire	After S and A — includes altitude kit at HOLD 4
C. Dry plumbing	Later
D. Fill and first fire	Last — HOLD 9 for complete system only

Ask once per part: if it is already on the van, inspect and finish the remaining work; if it belongs in the sofa bay, it waits on HOLD S. After diesel Acceptance, live from the User's Guide. Keep-or-sell numbers for the Isotemp are in the Isotemp value study.

Effort estimate (remaining)

ASSUMPTION — one person, DIY pace, parts already on hand. Not a quote. Heater underbody mount, fuel tap, exhaust rough-in, Duda/Sure Marine placement, EasyStart mount, and house AC/DC backbone are treated as **already spent** and are not in these totals.

BLOCK	LOW	TYPICAL	HIGH	NOTES
Phase A — fuel + exhaust walk (HOLD 1–2)	2 h	3 h	6 h	Mostly inspect/fix vs SRC-009
Sofa/bed rip + hybrid rebuild (HOLD S)	40 h	**55 h**	80 h	Largest block: frame, cradle, backrest expansion header, seat-back / seat-butt panels, hatches
Phase B — mounts + 12 V / altitude / cable (HOLD 3–4)	10 h	16 h	24 h	Isotemp and expansion header wait on HOLD S
Phase C — glycol + freshwater hose (HOLD 5–6)	10 h	16 h	28 h	Clamps, bleeders, labels; rework if a run fights furniture
Phase D — fill, bleed, first fire (HOLD 7–8)	6 h	10 h	16 h	Water-first then glycol; watchful first fire
HOLD 9 — Isotemp AC first energize	2 h	3 h	6 h	After chamber full; ground-fault story still open
Q-022 later — thermostat relay + wake	2 h	4 h	8 h	After diesel heat works
Contingency / fetch / rework (~20%)	14 h	20 h	30 h	

FINISH LINE	TYPICAL REMAINING	CALENDAR (WEEKENDS / EVENINGS)
Diesel cabin heat + diesel hot water (through HOLD 8)	**~110–120 h**	~4–8 weeks part-time
Complete system (+ HOLD 9 + light Q-022)	**~120–130 h**	add a few sessions

If two people share furniture and hose work, wall-clock time drops faster than man-hours (often ~0.7× calendar for the sofa and hose phases).

Already on the van

ITEM	NOW	FINISH AT
Hydronic D5S diesel heater	Mounted; **25 2526** / **H-Kit**	Confirm tilt (SRC-009)
Exhaust / combustion air	Owner thinks OK	Quick HOLD 2 look before fire
Fuel pickup + pump	Tap OK; hose likely run; **pump check = future work** under van	HOLD 1; prime at HOLD 8
Duda B3-12DW-20 + AM100-1LF mixer	Mixer on; one hose on; **under the sink** (as-built)	Glycol port; mixed and cold to taps; set ~120°F
Sure Marine cabin heater	REAL **16401**; **IN bottom / OUT top**; fans taped; **under the fridge**	Level, bleeder, fan power – HOLD 3 / 4
EasyStart Timer	On wood panel (with LinkPRO / PROwatt SW)	Land per SRC-003 / SRC-009 – HOLD 4
Xantrex LinkPRO + PROwatt SW	On wood panel with EasyStart; PROwatt face has dual GFCI outlets	House monitor + inverter; Paneltronics **WATER HEATER** path still HOLD 9
Blue Sea 12 V distribution block	On van (ST Blade + ML-RBS class)	Free slots **or** second breakout for master / EasyStart / fans – HOLD 4
Blue Sea 9009 rotaries (×2)	On van; Paneltronics path poorly labelled	Label SHORE/INVERTER and SHORE/J1772 at HOLD 4/9
Paneltronics AC panel	**72313** / **9972313B**: MAIN 30 A · BATTERY CHARGER · **WATER HEATER** (empty, reserved) · OUTLETS	Element cable + document protection at HOLD 9

Diesel-only Acceptance can finish without HOLD 9. Complete system Acceptance needs HOLD 9. After diesel first fire, only Q-022 and Q-009 may still wait; land the altitude kit at HOLD 4.

Still in the garage

ITEM	DO THIS
5 L expansion header	**Sofa-bed backrest (high)**; Isotemp under bench (low); tee on return; free path from tee to pump
Isotemp Slim Square 4.2 gal heat battery	Mount empty **driver side under new sofa/bed (low)**; **factory mixer capped forever**; element → **WATER HEATER** (dead)
Altitude kit 22 1000 33 22 00	Install and wire at HOLD 4 while the EasyStart harness is open (SRC-002)
Master switch	Mount next to EasyStart; Off cuts three 12 V branches (upstream of EasyStart); does **not** cut Isotemp AC
Hose, clamps, Prestone LowTox AF555 glycol	Reinforced hose; stainless clamps; ≤50% PG (DEC-017)
SC1600B thermostat, Noctua fans, speed dial (W002-912 pick)	Mount and wire; auto call waits on Q-022

Sofa/bed (discovery): current bed **coming out**. Rebuild hybrid per DEC-021: EZTube seat-base frame first (it bridges the wheel well and lands on the floor), aluminum Isotemp cradle, day panels with **outboard → seat back** / **inboard → seat butt**. Plan: The sofa. expansion header in **backrest**; Isotemp under bench. Sure Marine under fridge; Duda under sink. House DC: Blue Sea blade block may need a **second breakout** for hydronic fuses. Envelope: Rattle Trap + Thinsulate-class behind panels. Details: Working design understanding · photo extracts electrical · heater · insulation. Effort: Effort estimate.

Do not change the design into these

Second glycol pump	DEC-013
Engine coolant splice	DEC-009
Summer cabin-heater bypass	DEC-010
12 V feeding the 750 W element	DEC-004
Hot taps without the diesel pump	Heat in the Isotemp chamber is not heat at the Duda

Still open (check the manuals first)

May wait after a low-elevation diesel first fire

- Q-022 — SC1600 relay and approved wake landing; use EasyStart only until then
- Q-009 — upstream GFCI/ELCI and element route before HOLD 9

Do at HOLD 4 (not on the wait list)

- Q-015 — altitude kit install and wire (identity confirmed); required before high camps

Owner placement / electrical answers

- Working design understanding — live discovery (sofa, layout, 12 V story)
- Questions for You — remaining placement / GFCI / house snapshot (Q-025)

→ Builder's Guide.

Builder's Guide

Stage 3 — Make it. The sofa bay comes first in the cabin. Heat phases start after System Design, a skim of The Numbers, the User's Guide, and Current Status. Fuel and exhaust under the van can run while the sofa frame is going in.

Jump: The sofa · Project Phases · HOLD 0 · 1 Fuel · 2 Exhaust · 3 Mounts · 4 Wiring · 5 Glycol · 6 Fresh · 7 Fill · 8 First fire · 9 AC · Acceptance

Work in a simple loop: assemble, inspect, pass the HOLD, then continue. Do not skip a HOLD. If a part, port, or wire is unclear, stop. After Acceptance, use the User's Guide as the day-to-day manual.

Authority. SRC-009 governs the heater. SRC-002 and SRC-003 govern the altitude kit and EasyStart. SRC-019 governs hose, header, and bleed craft. System Design governs this van's locked layout.

MUST

Use reinforced heater hose, stainless clamps, and brass, copper, or stainless barbs (section 5). Garden hose and ordinary radiator hose fail under hydronic heat.

What "done" means

Cabin heat and diesel hot water can pass the diesel-only checks before HOLD 9. Diesel-only still means the Isotemp **coil is in the loop** and the **chamber is full of static glycol**—you just skip AC energize. Complete system acceptance requires the Isotemp AC circuit and first energization at HOLD 9.

Four Project Phases

Treat the install as the sofa bay plus four heat phases. The sofa bay runs with Phase A. It blocks Phase B and the freshwater tank. Inside a phase, finish each HOLD before you move on.

PHASE	WHAT YOU FINISH	STEPS	NOW
S. Sofa bay	Seat-base frame in place so the freshwater tank, strainer, Isotemp, and expansion header have a home	The sofa	← **with A**
A. Fuel and exhaust	Underbody fuel and exhaust checked safe	1–2	← **with S** (on van; HOLD 1–2 still open)
B. Mount and wire	expansion header and Isotemp mounted; EasyStart / heater 12 V and altitude kit landed; Isotemp AC cable routed but dead	3–4	After S and A
C. Dry plumbing	Glycol and freshwater hoses connected, still empty	5–6	Later; tank strap waits on S
D. Fill and first fire	Water-first loop check → glycol loop + Isotemp chamber → first fire	7–8 (+9)	Last

NEXT

Next — two tracks. Under the van, finish HOLD 1 and HOLD 2 against SRC-009. In the cabin, start The sofa. Do not bolt the freshwater tank, the strainer, the Isotemp, or the expansion header until HOLD S. Do not fill or fire yet.

What can wait until after first fire

After a low-elevation diesel first fire that delivers cabin heat and hot water at the Duda, only two items may still be open:

- Cabin thermostat auto-call (Q-022) — start heat from the EasyStart Timer
- Isotemp AC and upstream GFCI (Q-009) — HOLD 9 only

Everything else through step 8 is required for that first fire: a filled glycol loop, a filled static Isotemp chamber, EasyStart and heater 12 V, the master switch, and the cabin fans.

The altitude kit does not block a low-camp first fire. It is still not on the wait list. Land it at HOLD 4 while the EasyStart harness is open (Q-015 / DEC-006). You will need it before high camps.

STEP	WORK ON	DO NOT DO YET	DONE WHEN
0. Prepare	Design, status, manuals, parts	Cut hose / power "to see"	HOLD 0
1. Fuel	Inspect pickup, pump angle, clamps, rise	Dry start	HOLD 1
2. Exhaust / intake	Outlet, drain, supports, separation	Run heater	HOLD 2
3. Mount	expansion header highest; Isotemp empty; cabin heater level; label ports	Fill	HOLD 3
4. Electrical	EasyStart / heater 12 V; master; fans; cable to **WATER HEATER** dead; altitude kit	Heat call; energize element	HOLD 4
5. Glycol hose	Locked loop order; bleeders; lowest drain	Add glycol	HOLD 5
6. Freshwater	Tank → pump → Duda → mixer → taps	Trust hot temp yet	HOLD 6
7. Fill / bleed	Water-first → drain → ≤50% PG → bleed → static Isotemp fill	Fire; AC	HOLD 7
8. First fire	Prime; watch; hot inspect; shutdown; cool check	Call done with open required HOLD	Acceptance
9. AC commissioning	Upstream GFCI/ELCI; **WATER HEATER** on	Energize before complete system acceptance	HOLD 9

Before you start

1. Keep the engine off, the heater off, and no smoking or open flame.
2. Keep SRC-009, SRC-003, and SRC-002 within reach.
3. Read every DANGER, CAUTION, and MUST box before the work it governs.
4. Walk under the van with a flashlight.
5. Do not fill with glycol, energize Isotemp AC, or fire the heater except at their HOLDs.
6. **Label before you connect.** Tape every open port.

HOLD 0 — ready

- ☐ System Design check done
- ☐ User's Guide read
- ☐ The sofa read — cabin track is HOLD S, in parallel with Phase A
- ☐ Current Status skimmed — next actions are Phase S and Phase A
- ☐ Engine off, heater off, no flame/smoking
- ☐ Manuals on hand
- ☐ Underside walk done
- ☐ Masking tape + marker ready

Bring to the bay

MUST HAVE	WHY
Reinforced heater hose, stainless (prefer double-band) clamps, brass/copper/stainless barbs	SRC-019
Fill/drain with garden-hose thread	Lowest-point fill
expansion header + Isotemp from garage	Mount (cabin heater already on van)
Master switch from garage	Next to EasyStart
Prestone LowTox AF555 + water + marked bucket	≤50% (DEC-017); water-first
Priming bulb + adapters	HOLD 8 only
EasyStart / Hydronic diagrams; Eberspächer-style crimper	No generic-crimp kit pins
SC1600B; two Noctua NF-F12; 12 V fan dial	Cabin air
Multimeter	Prove master Off
Altitude kit 22 1000 33 22 00	Land at HOLD 4 (required before high camps)

Position the parts

Before you cut hose, lock relative heights. Wrong heights trap air and starve flow.

PART	POSITIONING RULE
5 L expansion header	**Highest** circulating-loop point. Top of tank above every hose high point, the Sure Marine cabin heater, the Duda B3-12DW-20 plate heat exchanger, and the coil. Expansion volume and air "burp" tank.
Hydronic D5S diesel heater + its pump	Below expansion header fluid level so exchanger and pump self-vent. Pump not at the absolute lowest point (dirt settles there).
Sure Marine cabin heater	Below expansion header. Mounted **level**. Propylene glycol **in bottom, out top**. Outlet-side bleeder.
Duda B3-12DW-20 plate heat exchanger	Below expansion header. Glycol ports on the circulating loop; freshwater ports never shared with glycol.
Coil in the Isotemp Slim Square 4.2 gal heat battery	Below expansion header, or with a high-point bleeder on those hoses. Chamber is static – not teed into loop flow.
Return path expansion header tee → pump	Unrestricted. No Sure Marine cabin heater, plate, or coil between the expansion header tee and the Hydronic D5S diesel heater's own pump.
Fill / drain	Lowest point of the circulating loop.

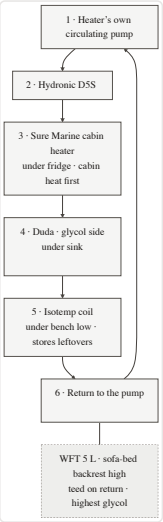
Locked circulating flow path:

Return → Hydronic D5S diesel heater's own pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger (glycol side) → coil in the Isotemp Slim Square 4.2 gal heat battery → return. 5 L expansion header tees into the return at the high point; it is not in series.

Forbidden while installing:

- Engine coolant splice
- Summer bypass around the Sure Marine cabin heater
- Second glycol pump / solar-shore glycol-assist circuit

- Landing the Isotemp 750 W element on the 12 V panel
- Teeing the Isotemp chamber into circulating flow
- Using the Isotemp factory mixer as the tap mixer



Glycol loop

Glycol loop layout

HOLD A – positions locked

- ☐ Planned mounts meet the height table
- ☐ Return path from expansion header tee to pump is unrestricted on paper and in the bay
- ☐ No forbidden path is planned
- ☐ Tape labels planned for every open port

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1. Fuel system

The tank's factory extra fuel connection is already in use. That is better than drilling the tank. It is not finished work.

Fuel this heater accepts (SRC-009): ordinary diesel EN 590. Up to 20% FAME blend allowed. Pure biodiesel as the only fuel is not. Switch the engine and the Hydronic D5S diesel heater off before any fuel work. No naked flame, no smoking.

Walk the existing install against every rule below. Fix failures before you leave this section.

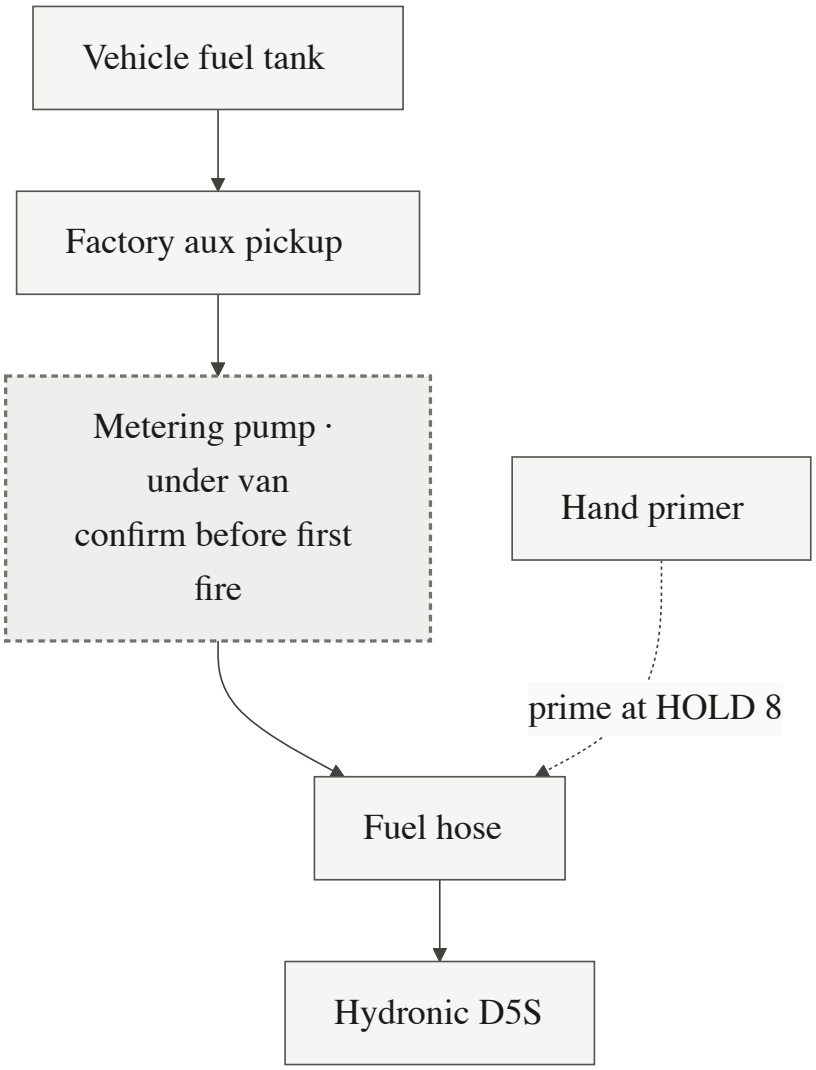
1. Find the pickup at the tank, the metering pump, and the line to the heater. Every fuel part is mounted on its own – the hose does not carry the weight of a filter, valve, or pump.
2. Metering pump sits about **15°–35°** from flat (preferred). Allowed **15°–90°**. Flat **0°–15°** is forbidden.
3. Heads (SRC-009): pressure head tank → pump max **3000 mm**; suction head pressureless tank max **1000 mm** diesel; suction head with 0.03 bar tank-cap valve max **400 mm** diesel. Measure if anything looks marginal.
4. From pump to heater, the line rises the whole way. No dips that trap air.
5. Fuel line never rests on the exhaust. Crossings have air space and a heat shield. Drips and vapour cannot collect on hot parts or electrical gear.
6. Clamp lines about every **50 cm**. Cut hose with a sharp knife – no crushed or burred ends. Where two pipes join, butt the metal inside the hose so the joint does not make bubbles.

7. Stainless clamps on hose barbs. Do not overtighten and crush the line. Do not use ordinary Teflon tape on fuel fittings — shreds can foul the burner. Use fuel-rated thread sealant only where a threaded joint needs it.
8. No wetness, smell, rub marks, or drip paths onto hot parts or wires.

> **DANGER**
> **If anything in the fuel path looks wrong, fix fuel before you plumb propylene glycol. Do not light the Hydronic D5S diesel heater to "test" a bad fuel line.**

Before first fire (section 8), the whole fuel line must be filled and vented. Do **not** prime yet in this section — only inspect and fix. Priming is a numbered step immediately before first fire so you do not burn start attempts on a dry line. On diesel, the control may restart after about a **70-second** miss and then lock after **two further** failed attempts. Do not click off and on more than twice into a fault.

Priming tool (use in section 8 only): temporary hand-bulb siphon with hose-barb adapters. Confirm joint and adapter sizes now so you are ready later. Do not leave the bulb in the permanent circuit.



Fuel system

HOLD 1 — fuel

- [ ] Pickup, pump angle, rise, clamps, and exhaust clearance all pass
- [ ] No wetness or smell
- [ ] Priming adapters confirmed for section 8; tool not left in the line
- [ ] Lockout rule understood (≤2 off/on into a fault)

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2. Exhaust and combustion air

Already installed. You are verifying, not redesigning. Numbers are from SRC-009 p. 24–25.

Exhaust — every item must be true:

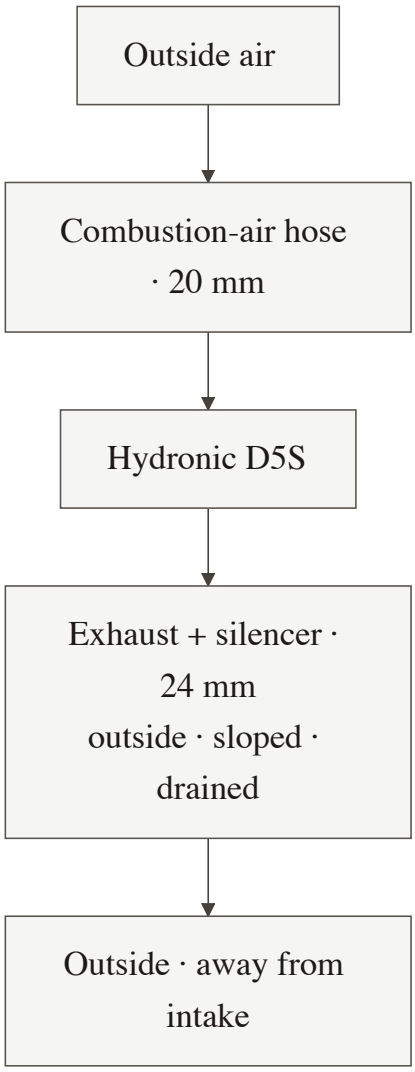
1. Ends in open air outside the van: not under a door, not into a wheel well, not near a vent that can feed cabin air, and not at the combustion-air intake. Does not stick out past the vehicle side.
2. Whole run about **0.2 m minimum and about 2 m maximum**. Flexible exhaust inner diameter about **24 mm**. Silencer fixed to the vehicle on its bracket (bracket torque about 6+0.5 Nm; pipe clips about 7+0.5 Nm per SRC-009), not hanging on the pipe alone.
3. Pipe slopes slightly down; a small drain (~**5 mm**) at the lowest point for condensate.
4. Supports about every **50 cm** against vibration. Clearance from fuel lines, electrical cables, brake hoses, and anything else heat can damage.
5. Mouth not aimed into the airstream, not clogged with dirt or snow, and not recirculating into the intake (silencer clearances on the order of **150–200 mm** from the intake opening — confirm on the SRC-009 figure).
6. Tail after the silencer is shorter than the heater-to-silencer run.

Combustion air — every item must be true:

1. Drawn from outside the cabin, from a place that stays cool (kit intent ≤ about **25°C**), not spray- or dust-laden, and not downstream of the exhaust.
2. Kit hose inner diameter about **20 mm**, **maximum about 0.76 m**. May be shortened; cut clean so chips cannot jam the combustion fan. Hose clip about 3+0.5 Nm per SRC-009.
3. Opening always free; not into a wind blast; slope down with a small drain if condensate can collect.

> **DANGER**
> If exhaust can reach cabin air, stop. Carbon monoxide is the failure mode. Do not run the Hydronic D5S diesel heater in a closed garage. Do not work on exhaust while the heater is running — wait until it is cold.

> **CAUTION**
> A stainless 90° elbow at the heater (field note) can reduce hanging height and give condensate a low point. If one is fitted, it still must obey slope, drain, clearance, and max-length rules in SRC-009. Confirm what is actually on this van before redesigning.



Combustion air and exhaust

HOLD 2 — exhaust / intake

- ☐ Exhaust open-air, sloped, drained, supported, clear of fuel/electrical/brake lines
- ☐ Intake outside, length and diameter within kit limits, opening free
- ☐ No recirculation path into cabin or intake

---

3. Mount the tanks and cabin heater

Pass HOLD S before you drill for the Isotemp or the expansion header. Work inside the van. Re-check the positioning table before you drill.

5 L expansion header

Not mounted yet. Must be the **highest** circulating glycol point. **Owner:** mount in the **sofa-bed backrest (high)** with the Isotemp under the bench (low). Confirm the expansion header top still wins after framing; raise the backrest cradle or fall back to a higher wall mount only if it loses the height check.

1. Mount so the **top** of the expansion header is the highest propylene glycol point in the whole loop. Plan was about 20–30 inches above the floor. Verify nothing else will sit higher once hose is run.

2. Later you bleed air here. You fill from the lowest drain until this tank takes fluid.
3. Cap is the ~1.2 bar relief. Do not substitute a different cap without confirming relief pressure stays below the heater's 2.5 bar limit.
4. Arrange relief discharge into a **catch bottle**, drip pan, or hose to a safe path — never onto bedding or wiring (DEC-019).

Isotemp Slim Square 4.2 gal heat battery

Still in the garage. **Driver side under the new sofa/bed** (owner). Envelope about 22-1/4 × 7-3/16 × 15-3/4 in; empty weight about 35 lb; full chamber adds about 16 L of fluid mass — mount to structure rated for that. Favor easy service access over cosmetic hiding.

1. Mount so its drain / safety valve is the lowest point on the Isotemp.
2. Leave service access in front (~**30 cm**).
3. Coil fittings below the 5 L expansion header. If not, add a bleeder at the high point of those hoses.
4. Keep coil hoses short (heater minimum flow is 250 l/h — every restriction counts).
5. Bench-fit and label the chamber connections outside the van if that makes the port identities clear. Mount the Isotemp empty on the **driver side under the new sofa/bed**. Do not fill the main chamber until the Isotemp is secured, the coil is connected, and the water-only circulating-loop leak check in section 7 is complete.
6. During chamber fill (section 7B), use the Isotemp main-chamber fill and vent ports from the Isotemp installation diagram, **not** the hydronic coil fittings. Add the **same** propylene glycol mix as the loop until the chamber is full and air is expelled, then cap. Leave this volume sitting still. It is never drinking water.
7. Do not pipe the Isotemp "fresh water" ports to the shower. Cap the factory mixer on the Isotemp **forever**. AM100-1LF is the only tap mixer (DEC-008).
8. Route the Isotemp electric-element cable to the Paneltronics **WATER HEATER** breaker. Do not connect it to the 12 V panel or energize it before the chamber and AC safety checks pass (HOLD 9).

Duda B3-12DW-20 plate heat exchanger (already in the van)

AM100-1LF thermostatic mixing valve already on the hot-water outlet (top). Cold water in through the cabinet (bottom). One lower port already has a hose. Mixed-hot to the taps still open. One upper fitting still open for propylene glycol.

Before any new hose: put tape labels on every open fitting:

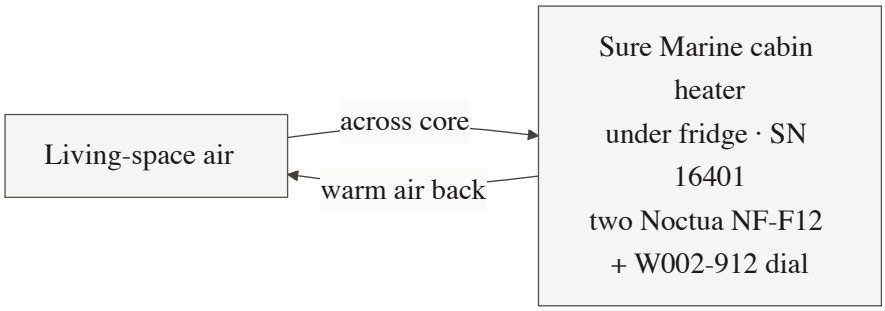
LABEL	MEANING
FRESH IN	Cold freshwater into the Duda (already through cabinet)
HOT TO MIXER	Duda hot out into the AM100-1LF thermostatic mixing valve (already)
MIXED HOT	From mixer to sink/shower (still to run)
COLD TO MIXER	Cold into the mixer's cold inlet (still to run)
GLYCOL A / GLYCOL B	The two propylene glycol ports only — never the mixer, never the freshwater inlet

1. Do not move the AM100-1LF thermostatic mixing valve onto a propylene glycol fitting.
2. Keep remaining hoses short.
3. Finish propylene glycol only on the ports labeled GLYCOL A and GLYCOL B.
4. Later (section 6): MIXED HOT to sink and shower; COLD TO MIXER to the mixer cold inlet.

Sure Marine cabin heater (already on the van)

REAL unit, SN **16401**. Label: **IN — bottom fitting · OUT — top fitting**. Fan power leads are taped, not finished. Location: **under the fridge** (owner/Tyrah: fits).

1. Confirm it sits **level**. An angled mount traps air away from the bleeder.
2. Glycol **bottom in, top out** (matches the Duda orientation). Keep that orientation if you reverse the cabin heater for fit (SRC-032).
3. Use the outlet bleeder (or add one on the outlet / high side). Bleed later until a steady stream (SRC-019).
4. Keep the Sure Marine cabin heater below the expansion header once the expansion header is mounted.
5. Cabin air only across the Sure Marine cabin heater — not from a fume-laden bay. Not ignition-protected for gasoline engine bays (ABYC note on the nameplate) — fine in this living-space install.
6. Finish **two** fans on the outlets (DEC-015). Target: both Noctua NF-F12 PWM (a mixed pair is on the unit today — replace or confirm intent). Wire fans **in parallel** to **one** inline **12 V** Low/Med/High dial on the shared positive (DEC-018). Power leads only; PWM/tach unused. No NA-FC1.
7. Dial is Low/Med/High **without** hard Off. Heat on/off is master + EasyStart (SC1600 only after Q-022). Book pick for the dial: Sure Marine **W002-912**.



Cabin air

HOLD 3 — mounts

- ☐ expansion header is the highest circulating point; relief has a catch path (DEC-019)
- ☐ Isotemp mounted empty; coil below expansion header or bled; factory mixer capped forever; element cable toward **WATER HEATER** (dead)
- ☐ Duda plate ports identified; mixer still on freshwater hot only
- ☐ Sure Marine cabin heater level, bottom-in/top-out, outlet bleeder fitted or planned
- ☐ Two fans + one speed dial installed (confirm both Noctua NF-F12); PWM/tach unused

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4. Wiring and controls

Two worlds. HOLD 4 lands **12 V** diesel heat (heater feed, master, EasyStart, fans, altitude kit) and routes the Isotemp **element cable** dead to Paneltronics. **120 V** energize is HOLD 9 only. Story and diagrams: Controls. Pin colours and AWG: Electrical and Controls. Land every connector from SRC-009 / SRC-003 on the bench — kit pins need the Eberspächer-style crimper.

Heater battery feed and master

1. Disconnect house-battery positive before cutting heater power leads.
2. Photograph the existing **Blue Sea ST Blade** fuse block (used vs free slots). Owner: positions may be **full** — if so, add a **second blade-fuse breakout** fed from a spare bus / high-amp fuse with its own feeder, then land hydronic branches there (photo-extract-electrical).
3. Heater battery **+** (red) and **−** (brown) at **12 AWG** if the round trip is **20 ft / 6 m** or less (SRC-009 p. 33 — manual says 4 mm²). Longer → go up one AWG size so drop stays ≤ 0.5 V.

4. Land the **20 A** main fuse on that positive → heater pins **1–2**. This feed does **not** go through the master.
5. From the battery (switched + only – typically after **Blue Sea ML-RBS** if that switch is still in the positive path), mount the **master Off/On** next to EasyStart. After the master, three fused branches:

BRANCH	FUSE	GOES TO
A – EasyStart	**5 A** – insert **last**	EasyStart pin 1 red
B – thermostat path	Small (size OPEN)	Later relay / SC1600 signal
C – cabin fans	Sized for two Noctuas (OPEN)	Fan dial → both fans

Meter-prove master **Off** kills A, B, and C. Heater pins 1–2 may still be hot through the 20 A – that is expected. Emergency: EasyStart off → master Off → pull fuse → battery. ≤ two off/on cycles into a fault.

EasyStart Timer and heater harness

EasyStart is **already mounted** on the wood panel. Landing is unfinished.

1. Mate the 10-pin connector per the pin table in Electrical and Controls (SRC-003).
2. Land the heater connector: water pump pins **8–9**, metering pump **4** / **10**, battery **1–2**, wake **7**.
3. Yellow wake: EasyStart pin **6** → harness → heater pin **7**, with the altitude kit in line on that harness (SRC-002).
4. Tape off heater pin **3** and the kit blower-relay leads. Seal empty chambers.
5. Put the **5 A** fuse in **last**. Battery must sit in the heater window (~10.5–16 V).

Fans and SC1600 (auto waits)

1. Mount SC1600B out of sun, vents, footwells. No mercury thermostat.
2. Wire fans: dial → both Noctuas on power leads only. Low/Med/High **without** hard Off. Meter that Low still spins.
3. Until Q-022: start heat from EasyStart only. Do not splice SC1600 into EasyStart pins 9–10.

Altitude kit 22 1000 33 22 00

When: land at HOLD 4 with the EasyStart harness. Required for overnight camps above ~5,000 ft; do not leave the kit in the garage just because a low-elevation first fire can run without it.

What: indoor air-pressure sensor that trims fuel. From ~4,600 ft the heater starts cutting capacity (~9% per 3,300 ft). Kit manual max ~20,000 ft (SRC-002).

1. Confirm box P/N **22 1000 33 22 00** and heater **H-Kit** sticker.
2. Mount indoors: plug facing **down**, not airtight, away from vents.
3. Plug the adapter into the control harness per the Hydronic diagram in SRC-002 (including the S+ / yellow path). Insulate unused leads.
4. Check: start → metering pump runs → shut off → wait for after-run. Full factory proof needs EDiTH – without that tool, mark **installed, not service-verified**.

HOLD 4A – altitude kit

- ☐ Correct P/N and H-Kit recorded
- ☐ Sensor mounted (down plug, not airtight, away from vents)
- ☐ Adapter landed per SRC-002
- ☐ Unused leads insulated
- ☐ Functional check done (or marked installed, not service-verified)

Isotemp AC cable (route now; live at HOLD 9)

1. Label the Isotemp 750 W element cable.
2. Route it to Paneltronics **WATER HEATER** only — never the 12 V panel.
3. Cap / lock out until HOLD 9.
4. Element heats the static chamber only. Circulating glycol still runs through the coil when the diesel pump is on.

Dry check before plumbing

1. Master **Off** kills EasyStart, thermostat path, and fans (meter each).
2. Master **On** restores those three without a blown fuse or hot wire.
3. Fan dial: Low/Med/High change speed; Low still spins.
4. Do not call for heat until fuel and coolant are ready (section 8).
5. Do not energize the Isotemp element yet.

> **MUST**
> Isotemp 750 W is a Paneltronics AC load. It heats the static chamber only. It does not pump the loop. Circulating glycol still runs through the coil when the diesel pump runs. Do not land it on 12 V. Do not energize until HOLD 9.

HOLD 4 — 12 V electrical (AC element stays dead)

- ☐ Blue Sea blade block photographed; free slots named **or** second breakout planned/installed for hydronic branches
- ☐ Heater feed 12 AWG (≤20 ft round trip) and 20 A fuse landed
- ☐ Master Off kills EasyStart / thermostat path / fans
- ☐ EasyStart landed; 5 A fuse inserted last
- ☐ Heater harness landed; blower leads taped off
- ☐ Yellow wake EasyStart pin 6 → heater pin 7 via altitude kit
- ☐ Fan dial Low/Med/High confirmed (Low still spins)
- ☐ Isotemp cable to **WATER HEATER**, labeled, locked out
- ☐ Altitude kit HOLD 4A complete
- ☐ Q-022 still OPEN unless closed against the manuals

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5. Propylene glycol plumbing

Hose and fittings

Use **reinforced heater hose** sized for the ports (this build already has 3/4 in on a landed Duda B3-12DW-20 plate heat exchanger barb). Maximum recommended single-loop hose run in marine practice is about 100 ft — keep this van's runs far shorter.

> **CAUTION**
> Do not use garden hose or ordinary car radiator hose for long runs. They kink and fail under sustained hydronic heat.

- Brass, copper, or stainless barbs and unions.
- **Stainless** hose clamps; double-banded clamps preferred so band holes do not cut the hose.

- Hose fully over the barb; clamp on the barb, not past it.
- On beaded (smooth) fittings: thin hose adhesive on the inside of the hose and the outside of the fitting; let it tack a few minutes; assemble; still use a clamp. Do not put hose adhesive on fuel hose.
- Support long runs with stainless or nylon clamps. About **1/8 in** clearance through bulkhead holes so the hose cannot chafe (hole about 1/4 in larger than the hose).
- Metal elbow for a tight 90°; do not force a kink into the hose.
- In-line bleeder at any hose high point that can trap air.
- Fill/drain at the **lowest** circulating-loop point (garden-hose thread is fine).
- Brass or stainless on the coil in the Isotemp Slim Square 4.2 gal heat battery — not ordinary plastic. Coil connections are 1/2 in BSPP class — use heat-rated hose and metal fittings.

Connect in locked order

Dry-fit first. Cut hose long, trial-fit every run without clamping, check for kinks and height mistakes, then shorten and clamp.

0. Install a fill/drain valve with garden-hose thread at the **lowest** circulating-loop point. Label it FILL/DRAIN.
1. Return line to the Hydronic D5S diesel heater's own circulating pump.
2. Pump discharge to the Hydronic D5S diesel heater's water inlet. Keep this short and free of the Sure Marine cabin heater, plate, and coil. Avoid tight elbows immediately at the pump.
3. Hydronic D5S diesel heater water outlet to the Sure Marine cabin heater (bottom in, top out).
4. Through the propylene glycol ports on the Duda B3-12DW-20 plate heat exchanger (GLYCOL A / GLYCOL B only).
5. Through the coil in the Isotemp Slim Square 4.2 gal heat battery.
6. Back to the return line. Tee the 5 L expansion header into that return at its highest point; not a branch off the Isotemp coil.

The coil stays in the loop always. No bypass around it. Do not open the four-gallon chamber into the flowing loop.

Bleed order (later, in section 7): (1) 5 L expansion header, (2) Sure Marine cabin heater outlet, (3) any other high-point bleeder you installed.

> **MUST**
> **Never run the Hydronic D5S diesel heater's pump with the loop dry. Never connect any propylene glycol hose to a drinking-water fitting.**

HOLD 5 — glycol plumbing dry

- ☐ Hose, clamps, supports, and elbows match the rules above
- ☐ Flow path matches the locked order
- ☐ expansion header tee on return high point; path from tee to pump unrestricted
- ☐ No glycol hose on a freshwater fitting
- ☐ Bleeders present at expansion header and Sure Marine cabin heater outlet

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6. Fresh water plumbing

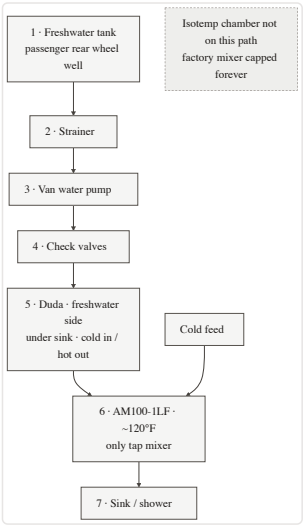
Pass HOLD S before you strap the freshwater tank. The strainer mounts on that tank's outlet.

Intended path: water tank → strainer → pump → check valves → Duda B3-12DW-20 plate heat exchanger (in bottom, out top). City water can tee in. Hot sink and hot shower from the AM100-1LF thermostatic mixing valve. Cold sink, cold shower, and the AM100-1LF thermostatic mixing valve's cold inlet share the cold side.

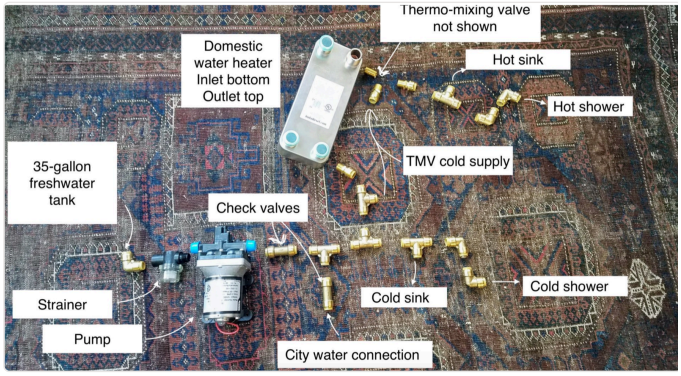
DANGER

A diesel boiler can produce very hot domestic water. An anti-scald valve is mandatory. In this build that valve is the AM100-1LF thermostatic mixing valve. Tap hoses must come from the mixed outlet only. Unmixed plate outlet water can track about 150–185°F.

1. Fresh water in is already through the cabinet. Do not add a second inlet on a propylene glycol port.
2. Hot out is already into the AM100-1LF thermostatic mixing valve. Run hose from the **mixed** outlet to the taps.
3. Cold also feeds the AM100-1LF thermostatic mixing valve's cold inlet.
4. Set the AM100-1LF before anyone showers (~**120°F**; standard range 70–145°F per SRC-036):
   - Apply thermostrip (or thermometer) on the **Mix** outlet pipe.
   - Loosen the handwheel screw → lift the handwheel → turn to the desired temperature → reseat and retighten to **lock**.
   - Confirm with a thermometer at a faucet. Water above 120°F can scald — write the setting on the CAUTION label and stick it on the valve (SRC-036).
5. Tap temperature is the **AM100-1LF** on the Duda only. Cap anything unused on the Isotemp (including the factory mixer) so nobody mistakes it for a tap control.
6. Cap anything you are not holding. Keep propylene glycol fittings and water fittings obviously different while you work.



Fresh water system



Freshwater layout

HOLD 6 — freshwater

- ☐ Tap hoses from mixed outlet only
- ☐ Cold feed to mixing valve present
- ☐ Mixer set ~120°F per SRC-036 handwheel procedure and labeled
- ☐ Isotemp unused ports capped (factory mixer included — never for taps)
- ☐ No glycol/fresh cross at any open fitting

7. Fill and bleed

Two separate fills. Do not confuse them.

A. Circulating loop — estimate, water-check, then propylene glycol

Order-of-magnitude capacities (measure your actual hose lengths; do not buy concentrate from a guess alone):

PIECE	APPROX. VOLUME
Hydronic D5S diesel heater	~0.18 L
Sure Marine cabin heater (REAL-class)	~0.2 L
3/4 in heater hose	Measure actual inside diameter and installed length
5 L expansion header	up to ~5 L when full (not all of that is circulating)
Coil in the Isotemp Slim Square 4.2 gal heat battery	Measure / add on fill
Isotemp chamber	~4.2 gal **static** — separate fill in 7B

Water-first leak check (mandatory before committing glycol):

1. Cap off the 5 L expansion header so air can leave. Fill from the lowest drain with **plain water** until the expansion header shows fluid (about a quarter up is enough for a leak hunt).
2. Do **not** energize the pump yet if hose cuttings or sawdust may still be in the loop.
3. Walk every joint for drips and every hose for kinks. Fix now, while you are only wasting water. If you find a leak or kink, drain, repair, and repeat this water fill.
4. Drain that water completely.

Propylene glycol fill:

5. Mix **Prestone LowTox AF555** in a clean marked bucket (DEC-017). **Simple recipe:** equal parts concentrate and water by volume (50% max). Example: 1 gal concentrate + 1 gal water. Prefer slightly under 50%. Do not pour undiluted. Cap the Star brite jug — do not use it in this loop.
6. Fill with mixed propylene glycol from the FILL/DRAIN valve until the expansion header takes fluid.
7. With the system **cool** and the expansion header cap off, open bleeders in this order: expansion header, then Sure Marine cabin heater outlet, then any other high-point bleeder. Leave each open until a **steady stream** comes out. Sputtering means air is still present.
8. Run the Hydronic D5S diesel heater's pump briefly (loop must already be wet — never dry). Bleed again in the same order.
9. Confirm heater, pump, and coil sit below expansion header fluid level. Recheck expansion header level when cool. Aim for a stable cool level with head on the pump (about half full is a common target after bleed — top up as needed).

> **CAUTION**
> Never check the expansion header level when the system is hot. Escaping fluid or steam can scald.

> **MUST**
> Never run the pump dry. Water alone is not the permanent coolant — after the leak check, operate on ≤50% propylene glycol mix.

The expansion header cap relieves at about 1.2 bar (below the heater's 2.5 bar limit). Do not treat the Isotemp potable safety valve as a propylene glycol setting.

B. Isotemp Slim Square 4.2 gal heat battery chamber (static)

Do this after the Isotemp is mounted, coil connections are complete, and the water-only circulating-loop leak check above has passed, and before first fire.

1. Use the Isotemp main-chamber fill and vent ports from the Isotemp installation diagram, not the hydronic coil fittings.
2. Add the **same** propylene glycol mix as the circulating loop until the chamber is full and air is expelled.
3. Cap the ports. Leave the volume sitting still.
4. Keep the safety-valve drain aimed somewhere a spill is acceptable.
5. Route the element cable to the Paneltronics AC panel; do not energize it yet.
6. Optional: exercise the PRV lever once with the drain aimed safely (glycol, not potable, will come out).

HOLD 7 — filled and bled

- ☐ Water leak check passed and drained
- ☐ Circulating loop filled with ≤50% Prestone LowTox (DEC-017)
- ☐ Bleeders yielded steady stream; pump brief run; re-bleed done
- ☐ Cool expansion header level confirmed; heater/pump/coil below expansion header fluid
- ☐ Isotemp chamber filled static with same mix; element routed to Paneltronics and still de-energized

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8. Commissioning (first fire)

Do this only after HOLDs 1–7 are true and you can see no wet joints.

Prime fuel now (not earlier)

1. Confirm adapters fit the chosen joint.
2. Open the joint, pull diesel with the hand bulb past the metering pump to the heater inlet until fuel arrives wet at the heater end.
3. Remove bulb, union, and adapters. Restore the normal line.
4. Wipe and inspect for wetness. Do not leave any priming part in the permanent circuit.

First heat

First heat is inspection under temperature, not a casual warm-up.

1. Work outdoors or with the garage fully open. Never in a closed garage.
2. Master switch **On**. Start from the EasyStart Timer, not by poking wires.
3. Stay with the van for the whole first fire. Watch under the van and every new hose joint.
4. Check fuel, exhaust, and clamps for wetness and smell while it runs.
5. Listen for the heater's own pump. The loop must move; air-locked heaters overheat. Minimum flow through the heater is **250 l/h**.
6. Walk every propylene glycol hose for kinks and every fitting for seepage once the loop is hot.
7. Expect normal staging: Power → High ~65°C → Low ~80°C → pause ~85°C; restart from pause ~75°C.
8. Optional flow hint: supply-to-return ΔT near ~10°F suggests healthy circulation; much larger ΔT (above ~17°F) suggests low flow. Confirm against lukewarm Sure Marine cabin heater while the burner runs.

9. If it faults, do not click off and on more than twice. Use the EasyStart Timer's diagnostic readout.
0. After shutdown, wait until the exhaust is cold before you touch it.
1. Open a tap. Hot, not scalding. If you cannot read a setpoint on the AM100-1LF thermostatic mixing valve, do not hand someone a shower.
2. Recheck the 5 L expansion header level after the first **cool-down**. Top up and bleed again if it dropped.

If it will not start, work this list in order: fuel prime → air in the propylene glycol loop → battery voltage (about 10.5–16 V) → Altitude kit wiring if already installed.

> **DANGER**
> **Do not run the Hydronic D5S diesel heater in a closed garage. Exhaust is carbon monoxide.**

HOLD 8 – first fire

- ☐ First run watched end-to-end
- ☐ No unexplained wetness or smell
- ☐ Pump heard / flow evidenced; no overheat lockout from air
- ☐ Mixer setpoint confirmed before any shower
- ☐ Cool expansion header level rechecked after shutdown

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9. First energize of Isotemp AC element (HOLD 9)

Do this only after HOLD 7 (chamber full) and after you have closed Q-009 far enough to name the actual Paneltronics breaker, GFCI/grounding, cable size, and how shore / PROwatt SW feed Paneltronics MAIN (SRC-040 / SRC-041 / Q-025). A diesel-only first fire may precede HOLD 9, but complete system acceptance requires HOLD 9.

1. Confirm the Isotemp chamber is full of the same propylene glycol mix; element never runs dry (SRC-022).
2. Confirm the element cable lands only on the Paneltronics AC panel — never the 12 V panel.
3. Verify breaker rating, GFCI or equivalent ground-fault protection, grounding/bonding, polarity, strain relief, and any enable switch. Confirm which **Blue Sea 9009** feeds Paneltronics; label its face **SHORE** / **INVERTER** (owner: poorly labelled today).
4. Read the Isotemp operating instructions for element use and overheat reset.
5. Energize briefly while present. Confirm the chamber warms; confirm no smell, trip, or hot cable.
6. Shut AC off. To move stored heat toward the taps later: EasyStart Timer call so the Hydronic D5S diesel heater's pump runs, then open a tap through the AM100-1LF thermostatic mixing valve.

> **DANGER**
> **Do not energize the 750 W / 115 VAC element on an empty chamber. Do not land it on 12 V. Treat live AC as shock/fire hazard until HOLD 9 checks are written down and true.**

HOLD 9 – Isotemp AC

- ☐ Chamber full
- ☐ Paneltronics breaker / GFCI / grounding / cable documented (Q-009 / Q-016 closed or explicitly deferred with circuit locked out)

- ☐ Paneltronics **9009** face labeled **SHORE** / **INVERTER** (Mean Well **9009**: **SHORE** / **J1772**)
- ☐ First AC run watched; no fault
- ☐ Element left off when unattended unless you deliberately want an AC charge

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Acceptance

There are two finish lines. Do not confuse them.

Diesel cabin heat and diesel hot water at the Duda — safe to park and use when all of these are true:

- ☐ Fuel line inspected and dry; primed before first fire
- ☐ Exhaust ends outside, away from the intake and cabin openings; slope and drain present
- ☐ Circulating propylene glycol full and bled; heater and coil below expansion header fluid level
- ☐ Coil in the Isotemp Slim Square 4.2 gal heat battery in the loop; 4.2 gal chamber full of the same mix, sitting still
- ☐ Duda B3-12DW-20 plate heat exchanger: propylene glycol one side, water the other; AM100-1LF thermostatic mixing valve on hot-water out, set and labeled ~120°F; taps from mixed outlet only
- ☐ EasyStart Timer command verified
- ☐ Master switch Off removes power from all three intended downstream branches and is meter-verified as a hard lockout
- ☐ Sure Marine cabin heater fans and speed controller installed; dial is speed only
- ☐ SC1600B automatic heater/fan behavior is either verified against the EasyStart Timer manual or remains explicitly OPEN under Q-022 — do not call it complete on assumption
- ☐ First commissioning run completed without unexplained wetness or smell
- ☐ Whole-system inspection after install, before day-to-day diesel use

Complete system (diesel plus Isotemp AC) — also requires HOLD 9: chamber full, GFCI/ELCI and Paneltronics **WATER HEATER** path verified, first AC energize watched. After that, live from the User's Guide, including AC chamber and dishes.

May stay open after a successful low-elevation diesel first fire

- Q-022 — SC1600B auto call (start heat from EasyStart Timer until closed)
- HOLD 9 — required before using the element and before calling the build fully complete; not required for diesel cabin heat or diesel hot water at the Duda
- Sure Marine cabin heater final cosmetic mount — must still be plumbed, level, and bled before relying on cabin air

Altitude kit 22 1000 33 22 00 — does not block a low-elevation first fire, but land at HOLD 4 while the EasyStart harness is open; required before high camps (Q-015 / DEC-006).

Technical appendix

This is the controlled reference record for the build: specifications, sources, hazards, decisions, open questions, and verification. It is not a second installation sequence. The sofa bay is The sofa. The heat order is in the Builder's Guide. Day-to-day operation is in the User's Guide. Architecture for Tyrah is in System Design.

How this appendix maps to part one

IF YOU NEED...	START HERE	THEN
Locked loop, forbidden designs, part glossary	System Design / The glycol loop / The Parts	System Architecture · Decision Log
Staging, fuel, altitude, mixer numbers	The Numbers	Verified Requirements · Source Register
Living with cabin heat, dishes, AC	User's Guide	Isotemp value study
What is on the van vs garage	Current Status	Parts Inventory · BOM
Serial install HOLDs	Builder's Guide	Installation · Commissioning
Dual finish lines (diesel vs complete)	How to Use · Acceptance	HOLD 9 · Q-009
Keep-or-sell Isotemp numbers	Isotemp value study	Coolant Loop · DEC-007
Locked facts (contradiction hunt)	Locked facts checklist	Fix Design → Status → Build → appendix in one pass
Photo knowledge extracts	electrical · heater · insulation	Cross-check Q-025 · HOLD 4 fuse breakout · HOLD S hatches
The sofa bay, cargo-floor inches, and the drawing	The sofa	Cargo floor · HOLD S · DEC-021
House / sofa working draft	Working design understanding	Promote into Controls · HOLD S · Q-009
Owner answers still open	Questions for You	full Open Questions table
Stop rules	Safety	Hazard Register

Finish lines (same as part one)

- **Diesel cabin heat and diesel hot water at the Duda** can pass after first fire without HOLD 9.
- **Complete system** (including Isotemp AC) requires HOLD 9.
- After a low-elevation diesel first fire, only Q-022 and Q-009 / HOLD 9 may still wait. The altitude kit is **not** on that wait list—land it at HOLD 4 (Q-015 / DEC-006).

The Isotemp value study is the keep-or-sell and cycling analysis for the Slim Square. Read it when you need the numbers behind the User's Guide.

Working drafts that are not yet locked live under `book/research/` (for example Working design understanding). Promote VERIFIED rows into part one and the controlled appendix; do not invent a second report.

Spec and sources

This section is part of the technical appendix. Use it for the build snapshot, architecture, sources, and hold points. It is not the order of work.

Hand this file with the PDFs in `sources/hydronic-install/` and `sources/coolant/`. This book cites those documents. It does not replace them.

Authority order: manufacturer manuals (SRC-009, SRC-002, SRC-003, …) win over blogs and memory. On-van photos and owner answers close *location* questions. Unresolved items stay OPEN — do not invent pinouts or hose routes.

LABEL	MEANING
VERIFIED	Confirmed by an authoritative source or inspection
ASSUMPTION	Provisional, waiting on confirmation
MEASURE	Must be taken on the actual vehicle or part
OPEN	Unresolved, and it may block design or commissioning

Snapshot

One loop. Hydronic D5S → Sure Marine cabin heater → Duda → Isotemp coil → return. No engine. No second pump. Isotemp chamber = static glycol; factory mixer unused forever (AM100-1LF is the only tap mixer). Drinking water only through the Duda + AM100-1LF.

House electrical (context only). Two worlds: **12 V** diesel heat from the **200 Ah** house battery (master Off is the parked lockout); **120 V** Isotemp 750 W on Paneltronics **WATER HEATER** via one **Blue Sea 9009** (shore ↔ **PROwatt SW 2000**). A second **9009** charges only (SmartPlug / DUOSIDA J1772 → Mean Well)—not heat. Blade block may need a **second breakout** for hydronic fuses. GFCI on Isotemp still OPEN (Q-009). Story: Controls. Photos: electrical.

Two finish lines. Diesel cabin heat / diesel hot water can pass before HOLD 9. Complete system (Isotemp AC) needs HOLD 9. After a low-elevation diesel first fire, only Q-022 and Q-009 may still wait; land the altitude kit at HOLD 4.

Remaining effort (ASSUMPTION). About **110–120 person-hours** typical through diesel first fire (sofa rebuild is the largest block); about **120–130 h** with HOLD 9 and light thermostat auto. Breakdown: Effort estimate.

Where to read. Part one: The sofa → System Design → Numbers → User's Guide → Status → Builder. Part two: this appendix, starting at the Technical appendix map.

Vehicle and heater

Vehicle	2017 Mercedes-Benz Sprinter 4×4, regular 170 body (not 170 EXT), 170″ wheelbase
Heater	Eberspächer Hydronic D5S diesel heater, 12 V, order no. **25 2526 05 00 00**
Heat output	5.2 / 5.0 / 2.1 kW (Power / High / Low), SRC-009
Nameplate	Hydronic II · D5S · **25 2526** · **H-Kit** confirmed
Purchase	Installation kit for Hydronic II D4S/D5S · EasyStart Timer · £902.98 GBP on the heater line
Not this heater	Hydronic II C (25 2506 / 4.8 kW) · Hydronic D5WS

Use **SRC-009** for heater specs, wiring, fuel, exhaust, and coolant rules.

Architecture (locked)

Independent living-space glycol loop (DEC-009). No engine connection. Design intent from the owner's 2018 notes (SRC-004): diesel reliability for cabin heat and hot water; double-wall plate so freshwater stays drinkable; thermal mass to soften cycling; heater under the van so combustion stays outside the cabin. There is no second glycol pump (DEC-013). The Isotemp 750 W element is a Paneltronics **WATER HEATER** AC load for the static chamber only (DEC-004 / DEC-020).

Glycol flow path (DEC-010), not the 2018 blog order:

Return line → Hydronic D5S diesel heater's own circulating pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger (glycol) → coil in the Isotemp Slim Square 4.2 gal heat battery → return line

5 L expansion header tees at the return high point of the circulating loop (not off the coil). The coil always stays in the loop. No summer bypass valve.

PIECE	JOB	WHERE
Duda B3-12DW-20 plate heat exchanger	Freshwater heat exchanger.	On van — **under the sink**
AM100-1LF thermostatic mixing valve	On Duda hot outlet. Scald control. Tap hoses not run.	On van
Sure Marine cabin heater	Only cabin heat. REAL SN **16401**; IN bottom / OUT top.	On van — **under the fridge**
5 L expansion header	Expansion / bleed high point.	Garage → **sofa-bed backrest (high)**
Altitude kit 22 1000 33 22 00	Required for high camps (**ASAP**). P/N + H-Kit confirmed.	Garage → wire per SRC-002 at HOLD 4
EasyStart Timer	Schedule / start-stop.	Mounted on wood panel; not fully wired
Blue Sea 12 V distribution	House ST Blade fuse block (+ ML-RBS in build photos)	On van — hydronic may need a **second breakout** at HOLD 4
Blue Sea 9009 (×2)	Rotary transfers: Paneltronics (shore/inverter); Mean Well (shore/J1772)	On van — label faces; Paneltronics one poorly labelled
SmartPlug + DUOSIDA J1772	120 V shore inlet + 32 A / 240 V Level-2 inlet	On van (build photos)
Xantrex PROwatt SW 2000	Inverter for Paneltronics when 9009 selects it	Wood panel; P/N 806-1220
Paneltronics AC (72313 / 9972313B)	MAIN 30 A · BATTERY CHARGER · **WATER HEATER** (empty, for Isotemp) · OUTLETS	On van
Isotemp Slim Square 4.2 gal heat battery	Circulating glycol in the coil. Static glycol in the chamber. Not DHW. Factory mixer unused forever.	Garage → **driver side under bench (low)**

The heater itself holds only ~0.18 L. The Isotemp mass damps cycling.

Vendor pages and manuals: Hydronic D5S · D5S manual (PDF) · EasyStart Timer (PDF) · Altitude kit (PDF).

What is already on the van

ITEM	STATUS
Hydronic D5S diesel heater (underbody)	Done; 25 2526 / H-Kit
Fuel pickup (factory aux)	Inspect at HOLD 1
Exhaust / combustion air	On van; re-check at HOLD 2
Duda B3-12DW-20 + AM100-1LF	On van **under the sink**; taps + one glycol barb still open; factory Isotemp mixer unused forever
Sure Marine cabin heater	On van **under the fridge**; SN 16401; fans unfinished
EasyStart Timer	Mounted; landing unfinished
Paneltronics AC (72313 / 9972313B)	Face known; **WATER HEATER** empty for Isotemp

Still in the garage

ITEM	STATUS
5 L expansion header	→ **sofa-bed backrest (high)**; confirm top wins height after framing
Isotemp Slim Square 4.2 gal heat battery	→ **driver side under bench (low)**
Altitude kit 22 1000 33 22 00	Wire per SRC-002 at HOLD 4 – first high camp **ASAP**
Master switch	Mount next to EasyStart at HOLD 4
Hose / clamps / Prestone LowTox AF555	≤50% PG for this loop (DEC-017)
SC1600B / Noctua fans / speed dial / fan relay	Both Noctua; voltage dial; relay TE V23134-A3052-X540 or equiv.; auto waits on Q-022

Part-one inventory with finish-at HOLDs: Current Status.

Hold points before first fire

1. Fuel walk at HOLD 1 (HAZ-001).
2. EasyStart / heater 12 V landed enough to start and stop (HOLD 4 / Q-013). Land the altitude kit at HOLD 4 while the harness is open (Q-015 / DEC-006) – required before high camps; prefer not to defer it past HOLD 4.
3. Fill and bleed circulating glycol including the Isotemp coil; fill chamber separately (static). Heater and coil below expansion header.
4. Route Isotemp element to Paneltronics **WATER HEATER**; document protection on that path before energize (HOLD 9 / Q-009).
5. AM100-1LF set ~120°F and tap hoses landed before anyone showers.

Where leftover comparison material mentions Hydronic II C or D5WS, treat it as **ruled out** (DEC-000). Blog SRC-004/005 is design history, not current BOM or loop order.

Hazard Register

Detailed hazard register for the practical guidance in System Design, User's Guide, and Builder's Guide. Manufacturer rules outrank blogs.

ID	HAZARD	CAUSE	CONSEQUENCE	PREVENTION/MITIGATION	VERIFICATION	STATUS
HAZ-001	Fuel leak	Damaged/poorly routed fuel line or pump	Fire / injury	SRC-009 fuel rules: no exhaust contact, clamped lines (~50 cm), butt joints, continuous rise pump→heater	Leak inspection before first fire	OPEN – routing/clamps still need a look
HAZ-002	Exhaust-gas intrusion	Exhaust into interior, intake, or openings	CO exposure	Open-air termination; slope + drain; no recirculation (SRC-009 p. 24–25); never run in a closed garage	Routing check before first fire	OPEN – owner believes complete
HAZ-003	Hot-coolant release	Overpressure, failed hose, or air-locked overheat	Burn	≤ 2.5 bar; 5 L expansion header cap ~1.2 bar; min flow 250 l/h; self-vent height; heater ~0.18 L – loop must stay full and bled (SRC-009)	Pressure/leak test; bleed high points	OPEN until commissioned
HAZ-004	Electrical overcurrent (12 V heater / fan)	Undersized conductor or fuse	Fire / damage	Fuse and wire from EasyStart Timer + heater diagrams (SRC-003 / SRC-009); fuse timer and fan separately after master switch (DEC-012)	Correct conductor and fuse selection	OPEN – wiring unfinished
HAZ-005	Scalding DHW	Unmixed Duda B3-12DW-20 plate heat exchanger outlet tracks glycol temp; mixer missing, bypassed, or set too high	Burn	AM100-1LF thermostatic mixing valve on the Duda B3-12DW-20 plate heat exchanger hot outlet (DEC-008 / SRC-024); range 70–145°F; set ~120°F; tap hoses from mixed outlet only	Land taps; label setpoint before showers	OPEN – taps not run; setpoint unset
HAZ-006	Dry-fire / shock on Isotemp Slim Square 4.2 gal heat battery element	115 VAC on empty chamber; missing GFCI/ground; element in static glycol (DEC-007)	Fire, tank damage, shock; possible early wear	Fill chamber first; land on Paneltronics **WATER HEATER** (72313 / 9972313B – DEC-020); **upstream GFCI/ELCI** (not on panel); never a 12 V feed; isolate when servicing (SRC-022)	Fill-then-power; AC safety test (Q-009)	OPEN – GFCI/route pending (face known)

ID	HAZARD	CAUSE	CONSEQUENCE	PREVENTION/MITIGATION	VERIFICATION	STATUS
HAZ-007	Glycol / potable cross-contamination	Failed Duda B3-12DW-20 plate heat exchanger or wrong plumbing	Ingested coolant	Duda B3-12DW-20 plate heat exchanger double-wall is the only glycol↔fresh interface (DEC-003); Isotemp Slim Square 4.2 gal heat battery never holds potable (DEC-007)	Confirm fittings; never tee propylene glycol into water	OPEN — fittings still open on van
HAZ-008	Fuel takeoff quality	Pre-documented tap-in; owner once flagged concern	Fuel leak / fire	Factory **aux** connection used (lower risk than drilled tap). Still inspect routing/clamps vs SRC-009 before fire (Q-010)	Physical inspection with photos	OPEN — inspect before proceeding
HAZ-009	Wrong power for element	Treating 12 V panel slot as 750 W feed	Fire / damage	Element is 115 VAC / ~6.5 A on Paneltronics AC only (DEC-004 / DEC-020). 12 V is never the load	Verify AC branch before energizing (Q-016)	OPEN until HOLD 9
HAZ-010	Control-box lockout / dry starts	Repeated off/on into faults; dry fuel line	Locked heater; failed trips	Prime fuel before first fire (SRC-005 tool; SRC-009 vent rule). ≤2 off/on cycles; EasyStart Timer diagnostics (SRC-009 p. 32)	First-fire checklist	OPEN until first fire
HAZ-011	High-altitude operation without kit	Sustained use above ~1500 m without air-pressure sensor	Poor combustion / faults	Kit **22 1000 33 22 00** wired per SRC-002; H-Kit on nameplate (DEC-006 / SRC-009 p. 32). Prefer land at HOLD 4 while EasyStart harness is open	Install from garage at HOLD 4; required before high camps (Q-015)	OPEN — kit in garage (identity confirmed)
HAZ-012	Unattended Isotemp AC	Inverter + Paneltronics **WATER HEATER** left on; master Off does not cut AC	Battery drain; shock/fire if path unfinished	Kill **WATER HEATER** / inverter separately unless deliberate shore/solar top-off with full chamber and verified GFCI (HOLD 9)	User's Guide AC chamber; HOLD 9	OPEN until HOLD 9

System Requirements

Scope

Design constraints for this build. Heater is Hydronic D5S diesel heater, order **25 2526 05 00 00** (DEC-000). Specs from **SRC-009** only.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-009	p. 12–13, 15, 24–25, 31–32	D5S heat/fuel/flow/voltage; install; exhaust; altitude; staging
SRC-002	Whole	Altitude kit 22 1000 33 22 00
SRC-003	Whole	EasyStart Timer (DEC-002)
SRC-004 / SRC-005	Design intent / field notes	Independent loop goals; not current BOM
SRC-021 / SRC-022	Product + I&O	Isotemp Slim Square 4.2 gal heat battery (DEC-001 / DEC-007)
SRC-023	Order	Heater + kit + EasyStart Timer purchase
SRC-024 / SRC-026	Photos	Duda B3-12DW-20 plate heat exchanger + AM100; intended freshwater path
DEC-009 … DEC-013	Decision log	No engine; loop order; master switch; no electric-assist path

Content

Functional intent

Diesel hydronic space heat plus on-demand DHW on an **independent living-space propylene glycol loop** (DEC-009) — **no engine connection**.

ROLE	PART
Heat in (diesel)	Hydronic D5S diesel heater underbody
Space heat out	Sure Marine cabin heater
DHW out	Duda B3-12DW-20 plate heat exchanger + AM100-1LF thermostatic mixing valve (DEC-003 / DEC-008)
Thermal mass	coil in the Isotemp Slim Square 4.2 gal heat battery (circulating) + ~4.2 gal static chamber (DEC-007)
Expansion / bleed	5 L expansion header (DEC-005)
Control	EasyStart Timer + SC1600B thermostat + master Off/On (DEC-002 / DEC-012 / DEC-014)
Hot water	Diesel-heated glycol through the Duda B3-12DW-20 plate heat exchanger; Isotemp static chamber can also be heated by its 750 W / 115 VAC element from the Paneltronics AC panel
Electric element	750 W / 115 VAC Isotemp element; Paneltronics AC-panel load, not a 12 V load. Breaker, GFCI/grounding, cable route, and operating control remain to be verified

Loop flow path (DEC-010): Hydronic D5S diesel heater's own circulating pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger → coil in the Isotemp Slim Square 4.2 gal heat battery → return to pump. 5 L expansion header tees at the return high point. Coil always in series — no summer bypass valve.

Heater performance — Hydronic D5S diesel heater (SRC-009 p. 13)

ORDER NO.	HEAT FLOW (POWER / HIGH / LOW)	FUEL USE	MIN FLOW	VOLTAGE
25 2526 05 00 00	5200 / 5000 / 2100 W	0.64 / 0.61 / 0.26 l/h	250 l/h	10.5–16 V

Tolerances ±10% at rated voltage / Esslingen reference altitude. Staging ~65 / 80 / 85 / 75 °C; during pause the circulating pump continues; overheat restart <70 °C (SRC-009 p. 31–32). Plain-language card: The Numbers. Cycling estimates: Isotemp value study.

Hard constraints

- Coolant: water + propylene glycol, **≤50%** coolant; ≤ 2.5 bar (SRC-009). Header cap ~1.2 bar.
- Heater and pumps below the 5 L expansion header fluid level (SRC-009 p. 15).
- Exhaust and combustion air outside the cabin (SRC-009 p. 24–25).
- Sustained use above ~1500 m needs Altitude kit 22 1000 33 22 00 (DEC-006). Prefer land at HOLD 4 while the EasyStart harness is open (Q-015).
- Underbody mount allowed if install rules are met (SRC-009 p. 15) — this van: driver-side / fuel-fill area.

Out of scope / ruled out

- Hydronic II C and D5WS heaters (DEC-000)
- Engine-coolant splice / combination valve (DEC-009)
- Bosch buffer; plumbing summer valve (DEC-001 / DEC-010)
- OEM Sprinter factory hydronic (SRC-014); Vito tandem (SRC-015)
- Webasto/TSL boilers as the heater (SRC-018 / SRC-019 — loop practice only)
- Airtronic D2 (SRC-007); floor stack (SRC-020)

Verification

- ☐ Requirements traced to SRC-009 / DECs
- ☐ Layout checked on the vehicle
- ☐ Open items below still marked honestly

Open items (blockers)

- ☐ **OPEN:** Q-010 fuel-line routing inspection before first fire
- ☐ **OPEN:** Q-013 EasyStart Timer wiring finish
- ☐ **OPEN:** Q-015 Altitude kit 22 1000 33 22 00 land at HOLD 4 (required before high camps; preferred with harness open)
- ☐ **OPEN:** Q-009 Paneltronics AC path (HOLD 9). Q-016 architecture closed (AC-only); remaining work is Q-009.
- ☐ **CLOSED:** Q-019 / DEC-017 — Prestone LowTox AF555 ≤50% for this loop
- ☐ **CLOSED:** Q-021 / DEC-018 — fan dial Low/Med/High without hard Off (meter at HOLD 4)

System Architecture

Scope

Identify candidate hydronic architectures and which sources apply to each. Heater is Hydronic D5S diesel heater (DEC-000). Isotemp Slim Square 4.2 gal heat battery is the buffer (DEC-007). Duda B3-12DW-20 plate heat exchanger is the freshwater heat exchanger (DEC-003). See the Isotemp value study for the keep-or-sell analysis, energy limits, and commissioning measurements.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-004 / SRC-005	Whole posts	Independent van living-space loop; Bosch buffer withdrawn
SRC-021 / SRC-022	Product + I&O	Isotemp Slim Square 4.2 gal: coil in circulating glycol; chamber used as static mass (DEC-007)
SRC-009	p. 20–23	Engine-coolant integration options (inline, thermostat, 5- and 6-port combination valve)
SRC-010	p. 10–12	D5WS-era combination-valve layouts; 6-port Order No. 330 00 176
SRC-001	Installation location / coolant	Hydronic II C same engine-bay integration family
SRC-008	Product page	6-port valve candidate P/N 25 2014 80 62 00
SRC-014	Whole sheet	Factory Sprinter/Crafter layout (MB pump and controls)
SRC-018 / SRC-019	Water system	Marine parallel-manifold / summer-valve practice (reference only)

Content

Candidate A — Independent living-space loop `[DEC-009]`

Design intent (SRC-004, Mar 2018): fossil-fuel reliability plus optional electric (solar / alternator / shore) for both space heat and DHW; double-wall PHE so the 35 gal freshwater tank stays drinking water; thermal mass to slow heater cycling; heater outside under the driver so combustion noise stays away from sleeping. Those goals still hold.

Original blog circuit order (SRC-004 — superseded for plumbing): Espar pump → D5 → Duda B3-12DW-20 plate heat exchanger → Sure Marine cabin heater → Bosch buffer → expansion, with a planned summer valve to bypass the Sure Marine cabin heater, and engine connection deferred. **Do not install that order.**

Locked now:

Hydronic D5S diesel heater circulating pump → Hydronic D5S diesel heater (EasyStart Timer + Altitude kit 22 1000 33 22 00) → Sure Marine cabin heater → **Duda B3-12DW-20 plate heat exchanger (glycol side) → coil in the Isotemp Slim Square 4.2 gal heat battery** → return; 5 L expansion header on the return high point. No auxiliary glycol pump. The Isotemp 750 W / 115 VAC element is a separate Paneltronics AC-panel load; it does not create a second circulating loop. During control pause the burner is off and the circulating pump keeps running (SRC-009; The Numbers).

GOAL FROM SRC-004	HOW THIS BUILD MEETS IT
Diesel energy in	Hydronic D5S diesel heater underbody
Electric energy in	Paneltronics AC panel → Isotemp 750 W / 115 VAC element; exact protection/control verification remains open
Space heat out	Sure Marine cabin heater
DHW out	Duda B3-12DW-20 plate heat exchanger + AM100 mixer
Thermal mass / less cycling	Isotemp Slim Square 4.2 gal heat battery static glycol chamber (DEC-007), not Bosch ~8.5 gal claim
No engine splice	Locked DEC-009 (blog had already deferred engine)
Summer without blowing cabin heat	Short timer call with cabin thermostat above setpoint — **not** a plumbing summer valve (rejected DEC-010), **not** an electric-assist circuit (DEC-013 withdrawn)

Roles (do not mix)

- **Isotemp Slim Square 4.2 gal heat battery:** used almost as designed. Circulating glycol through the coil. The ~4 gal chamber holds **static glycol**, not freshwater. Slows Hydronic D5S diesel heater Power/High/Low/Off swings. Bosch is gone. Do not tee that chamber into loop flow, and do not use it for tap water. Factory mixer unused.
- **Duda B3-12DW-20 plate heat exchanger:** the freshwater heat exchanger. Cold in through the cabinet, hot out through the AM100-1LF thermostatic mixing valve (SRC-024). Unmixed outlet tracks glycol temperature. Double-wall = drinking-water separation (SRC-004 intent).
- **AM100-1LF thermostatic mixing valve:** already on the Duda B3-12DW-20 plate heat exchanger potable hot outlet (DEC-008 / SRC-024). Not the Isotemp Slim Square 4.2 gal heat battery factory mixer. Mixed-out not yet run to taps.
- **750 W element on the Isotemp Slim Square 4.2 gal heat battery:** separate Paneltronics AC load. It heats the static chamber only; circulating glycol still runs through the coil via the D5S pump. The element is not the DHW exchanger and does not replace that pump.

Loop flow path locked (DEC-010): Hydronic D5S diesel heater's own circulating pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger (glycol) → coil in the Isotemp Slim Square 4.2 gal heat battery → return to pump. 5 L expansion header tees at the return high point. Coil always stays in the loop (no summer bypass). Intended DHW path is the **35 gal freshwater tank** → pump → Duda B3-12DW-20 plate heat exchanger → AM100-1LF thermostatic mixing valve → taps (SRC-026). The Isotemp Slim Square 4.2 gal heat battery chamber is not on that path.

Planned locations (updated from SRC-004 "all under sink"): heater + metering pump underbody (driver side / fuel-fill area — SRC-005). Duda B3-12DW-20 plate heat exchanger **under the sink** (as-built). Sure Marine cabin heater **under the fridge** (owner/Tyrah: fits). Isotemp Slim Square 4.2 gal heat battery **driver side under new sofa/bed** (current bed coming out; hybrid per DEC-021; EZTube seat-base frame bridges the well and lands on the floor; day geometry **outboard → seat back / inboard → seat butt**). Regular 170 body, not the 170 EXT. Freshwater tank on the **passenger** rear wheel well. 5 L expansion header in the **sofa-bed backrest (high)**; confirm top is highest circulating glycol point after framing. Cabin envelope already has **FatMat Rattle Trap** + Thinsulate-class insulation behind grey panels / L-track (photo-extract-insulation). Owner priorities for sofa packaging: ease of access and ease of construction.

Candidate B — Engine-coolant integration `[NOT SELECTED — DEC-009]`

Not used. Owner locked **no engine connection**. The SRC-009 / SRC-010 combination-valve material below is retained only as rejected reference.

1. Inline + non-return valve (heat to engine first).
2. Thermostat + T-piece (small circuit to cabin HX until ~70–75 °C, then large circuit including engine).
3. Combination valve, 5-port + T-piece, or 6-port without T-piece when flow and return run in parallel (SRC-009 p. 23). SRC-010 6-port Order No. 330 00 176. SRC-008 lists a later 6-port P/N 25 2014 80 62 00.

SRC-014 is the factory Sprinter NCV3 / Crafter version of this idea. Not this van's plan.

Candidate C — Marine-style parallel heating loops `[REFERENCE ONLY]`

SRC-018 / SRC-019 (Webasto / TSL-17, not this heater): supply/return manifolds, summer valve to bypass space-heat loops for DHW-only, bleeders at high points. Useful plumbing practice; not a spec source for the Eberspächer.

Shared heater-side interfaces

Regardless of architecture family, the Eberspächer has ports: exhaust (A), fuel (B), combustion air (V), water inlet (WE), water outlet (WA) (SRC-009 p. 14).

The Isotemp Slim Square 4.2 gal heat battery has a 316 SS hydronic coil on the circulating glycol loop and a ~4 gal chamber of **static glycol** (DEC-007). Factory design was coil + potable water. This build keeps the coil/static split and puts glycol in both. Those two volumes must not mix except through the coil wall. Do not use the chamber as freshwater.

What is not this system

- SRC-007 Airtronic D2 (air heater).
- SRC-015 Vito tandem D5WS + D5WZ (wrong platform — this van is 2017 Sprinter 170, Q-005).
- SRC-018 / SRC-019 as the boiler.
- SRC-016 SureCal tank (not selected).
- SRC-020 floor insulation (out of hydronic scope).
- Bosch electric glycol buffer (withdrawn, DEC-001).
- Plumbing summer valve / blog loop order Duda B3-12DW-20 plate heat exchanger→Sure Marine cabin heater→Bosch (SRC-004 superseded by DEC-010 / DEC-001).

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- ☐ **OPEN:** (loop order locked DEC-010; remaining electrical Q-009 for HOLD 9)

Parts Inventory

Scope

Document-implied and owner-selected components. This is not a complete physical count on the vehicle.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-009 / SRC-010 / SRC-012	Scope of supply / parts list	Heater-family spare and kit parts
SRC-002 / SRC-003	Whole	Altitude kit 22 1000 33 22 00 and EasyStart Timer
SRC-008 / SRC-017	Whole	Valve, Heatmiser
SRC-004	Architecture list	Blog-claimed van parts, Bosch withdrawn
SRC-021 / SRC-022	Product + I&O	Selected Isotemp Slim Square (static glycol mass)

Content

Status

No full on-vehicle inventory has been performed. Floor insulation is out of scope.

Where things are (owner, 2026-09-19)

ITEM	LOCATION	NOTES
Hydronic D5S diesel heater	On van, underbody	25 2526 / H-Kit confirmed
Fuel pickup / metering pump	On van	Factory aux; inspect at HOLD 1
Exhaust / combustion air	On van	Verify at HOLD 2
Duda B3-12DW-20 + AM100-1LF	On van	**Under the sink**; mixer on; one 3/4 in hose; taps + other glycol barb open
Sure Marine cabin heater	On van	REAL SN **16401**; **under the fridge**; IN bottom / OUT top; fans unfinished
EasyStart Timer	On van	Mounted on wood panel; not fully wired
Paneltronics AC	On van	**72313** / 9972313B; **WATER HEATER** empty for Isotemp
Isotemp Slim Square 4.2 gal heat battery	Garage	Mount empty **driver side under new sofa/bed (low)**
5 L expansion header	Garage	→ **sofa-bed backrest (high)**; confirm top wins height after framing
Altitude kit 22 1000 33 22 00	Garage	P/N + H-Kit confirmed; install/wire before high camps
Master switch	Garage	Mount next to EasyStart at HOLD 4
Glycol, hose, clamps	On hand	Prestone LowTox AF555 ≤50% (DEC-017); 3/4 in hose on Duda
SC1600B	On hand	Face Off/Heat; auto landing Q-022

Selected

ITEM	SOURCE	NOTES
Isotemp Slim Square 4.2 gal heat battery, 115 V / 750 W	SRC-021 / SRC-022; DEC-001 / DEC-004 / DEC-007	Coil = circulating; chamber = static. Element → Paneltronics **WATER HEATER**. Garage.
Duda B3-12DW-20 plate heat exchanger	SRC-004; SRC-024; DEC-003	On van **under the sink**.
AM100-1LF thermostatic mixing valve	DEC-008; SRC-024	On Duda hot outlet.
Sure Marine cabin heater	SRC-004; photos 2026-09-19, S/N 16401	**On van under the fridge.**
5 L expansion header	DEC-005	Garage → **sofa-bed backrest (high)**.
Altitude kit 22 1000 33 22 00	SRC-002; DEC-006	Garage; P/N confirmed; wire per SRC-002 (Q-015).
Paneltronics 9972313B / face 72313	DEC-020; SRC-038	**WATER HEATER** reserved for Isotemp.
Glycol, hose, clamps	Owner 2026-09-19	Prestone LowTox AF555 ≤50%.
SC1600B Digital Thermostat	DEC-014	On hand; signal only – not fan power.
Fuel priming bulb + barbs	SRC-005	On hand; HOLD 8 only.

Document-implied parts (presence unconfirmed)

ITEM	SOURCE	NOTES
Hydronic D5S diesel heater	DEC-000	On van
Heater water pump	SRC-009 p. 13	Hydronic D5S diesel heater: <12 W, 680 l/h @ 0.1 bar
Metering (dosing) pump	SRC-009 fuel chapter	Orientation and suction-head limits
EasyStart Timer 22 1000 34 15 00	SRC-003; DEC-002	Mounted on wood panel; not fully wired (Q-013)
1/2 in female-to-PEX swivel fittings	SRC-021	VLO recommendation for USA BSP-to-NPT adapters

Withdrawn / not selected

ITEM	SOURCE	NOTES
Bosch electric glycol buffer	SRC-004	Replaced by SRC-021 (DEC-001)
SureCal calorifier	SRC-016	Not the selected tank
6-port combination valve / engine splice	SRC-008 / SRC-009	Not this build (DEC-009)
Plumbing summer valve around Sure Marine cabin heater	SRC-004	Rejected (DEC-010)

D5WS spare-parts catalogue

SRC-012 is an exploded-view catalogue for D3WZ/D5WZ/B5WS/D5WS. **Not this heater** (DEC-000). Do not order from it for the Hydronic D5S diesel heater.

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded

- [ ] Relevant inspection hold points completed

Open items

- [ ] **OPEN:** Q-015 Altitude kit — install and wire (P/N already confirmed)
- [ ] **MEASURE:** Sure Marine cabin heater output rating on the physical unit
- [ ] Photograph remaining hydronic parts in boxes

BOM and Missing Parts

Scope

Build a working BOM for the independent loop (DEC-009). Isotemp Slim Square selected as static glycol mass (DEC-001 / DEC-007); Duda is the DHW exchanger (DEC-003).

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-001 / SRC-009 / SRC-010	Scope of supply	What a heater kit is supposed to include
SRC-002 / SRC-003	Whole	Altitude kit 22 1000 33 22 00 and timer
SRC-004	Architecture list	Candidate living-loop parts; Bosch withdrawn
SRC-021 / SRC-022	Product + I&O	Isotemp Slim Square 4.2 gal heat battery
`03-parts-inventory.md`	Whole	Selected vs unconfirmed list

Content

No approved heater-side BOM. Do not purchase from mixed Eberspächer manuals.

Selected Isotemp / Duda line

ITEM	QTY	SPEC	SOURCE
Isotemp Slim Square 4.2 gal heat battery	1	4.2 gal static glycol mass + coil in circulating loop; **not** the DHW heat exchanger	SRC-021 / DEC-001 / DEC-007
Duda B3-12DW-20 plate heat exchanger	1	Double-wall glycol-to-fresh-water exchanger, mounted	DEC-003; SRC-024
AM100-1LF thermostatic mixing valve	1	On the Duda B3-12DW-20 plate heat exchanger hot outlet. Tap runs not landed	DEC-008; SRC-024
Hydronic coil hoses (Isotemp Slim Square 4.2 gal heat battery)	on hand	Owner has hose; remaining stock ID / material `[OPEN]`	Owner 2026-09-19
Duda B3-12DW-20 plate heat exchanger potable + glycol hoses	partial	Mixer on the Duda B3-12DW-20 plate heat exchanger; freshwater in through cabinet; one 3/4 in heater hose on a lower port; mixer-to-tap and remaining glycol barb open	SRC-024; SRC-026
Glycol	on hand	**Prestone LowTox AF555** concentrate only for this loop (DEC-017); ≤50%. Cap Star brite Bio-Safe	Owner; SRC-027–031; DEC-017
Clamps	on hand	Confirm type on the joints that matter	Owner 2026-09-19
SC1600B Digital Thermostat	1	Sure Marine cabin heater / cabin heat control — not hot water (DEC-014; earlier DEC-013 assist-circuit pump withdrawn)	DEC-014
Master Off/On 12 V switch	1	Gates EasyStart Timer + Sure Marine cabin heater circuit	DEC-012
Altitude kit 22 1000 33 22 00	1	Required; garage — land at HOLD 4	DEC-006 / Q-015

Known documentation gaps

GAP	WHY IT BLOCKS A BOM	SOURCE
Loop architecture	Combination valve vs independent loop	**Resolved DEC-009:** independent loop only; no engine connection
Duda B3-12DW-20 plate heat exchanger keep/drop	Extra exchanger vs coil in the Isotemp Slim Square 4.2 gal heat battery only	**Resolved DEC-003:** Duda B3-12DW-20 plate heat exchanger is the DHW heat exchanger
DHW mixing valve unidentified	Scald control on the Duda B3-12DW-20 plate heat exchanger outlet	**Resolved DEC-008 / SRC-024:** AM100-1LF thermostatic mixing valve is on the Duda B3-12DW-20 plate heat exchanger
Control stack unknown	EasyStart Timer vs Heatmiser vs both	**Resolved DEC-002:** EasyStart Timer only
Loop flow path / summer valve	Blog order vs locked series	**Resolved DEC-010:** Hydronic D5S diesel heater's own pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger → Isotemp Slim Square 4.2 gal heat battery → return; no summer valve
6-port valve P/N unconfirmed	SRC-008 vs SRC-010 historical 330 00 176	**N/A (DEC-009):** no engine splice
Expansion-tank volume	5 L expansion header + coil/hose circulating fill; 4.2 gal chamber separate static	DEC-005; DEC-007
Remaining hose stock ID	Landed Duda B3-12DW-20 plate heat exchanger hose is 3/4 in / 16 bar (SRC-024); leftover coil hose still open	Owner 2026-09-19
Glycol product / mix	Prestone LowTox AF555 concentrate ≤50% (DEC-017); do not mix brands	DEC-017; SRC-029

Missing identification work (do this first)

1. Photograph heater nameplate, pump, metering pump, timer, altitude sensor, and any valve stamp.
2. Photograph the Isotemp Slim Square 4.2 gal heat battery nameplate/P/N and the Sure Marine cabin heater if present.
3. Duda B3-12DW-20 plate heat exchanger is on the van (SRC-024). Do not re-buy it.
4. Match each photo to an SRC ID before adding a BOM line.
5. Install Altitude kit 22 1000 33 22 00 from the garage at HOLD 4; wire per SRC-002 (Q-015 — P/N and H-Kit already confirmed). Required before high camps.

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- ☐ **OPEN:** Q-009 (Isotemp Slim Square 4.2 gal heat battery AC circuit / GFCI sizing)
- ☐ **CLOSED:** Q-019 / DEC-017 — Prestone LowTox AF555 ≤50%
- ☐ **CLOSED:** Q-021 / DEC-018 — fan dial without hard Off
- ☐ **MEASURE:** Remaining hose stock ID / material

Mechanical Layout

Scope

Heater, pump, and Isotemp Slim Square 4.2 gal heat battery mounting rules. Vehicle floor insulation is out of hydronic scope.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-009	p. 15–16	Hydronic II location and positions (**this heater**)
SRC-005	Mount notes	Underbody behind fuel fill / driver wall — as-built location
SRC-022	§2–3 locating / mounting	coil in the Isotemp Slim Square 4.2 gal heat battery below header; safety valve lowest; ~30 cm front service
SRC-018	Component positioning	Marine boiler location practice — reference only
SRC-019	Fan heaters / expansion p. 5–7, 24–27	Cabin heater level, bottom-in/top-out; expansion header as highest "burp" point — workmanship only

Content

Heater and water pump `[manufacturer]`

- Install below the minimum coolant level (5 L expansion header / radiator / heat exchanger) so the heater exchanger and pump self-vent (SRC-009 p. 15).
- Do not put the water pump at the lowest point in the circuit; particles settle there. Use a water filter if coolant is dirty (SRC-009 p. 15).
- Keep adequate clearance from hot vehicle parts (SRC-009 p. 15).
- Hydronic II: suggested location is the engine compartment; other locations allowed if they meet the same install requirements (SRC-009 p. 15).
- **This van (as-built / SRC-005):** heater underbody, driver side, behind the fuel-tank fill / outside wall behind the driver seat — common Sprinter living-loop mount. Nameplate must remain readable or duplicated.

Permissible tilt angles: SRC-009 p. 16. Heater horizontal→vertical range per diagram; do not use the "lying on its side" forbidden zone.

Sure Marine cabin heater `[SRC–019 craft]`

- Mount **level** on a flat surface. Angled mounts trap air away from the bleeder (SRC-019 p. 25).
- Propylene glycol **in bottom, out top** unless the unit's plate says otherwise. Plan an outlet-side bleeder.
- Keep the Sure Marine cabin heater below the 5 L expansion header fluid level.
- Cabin air only — draw living-space (or outside) air across the Sure Marine cabin heater; do not pull from a fume-laden bay.
- **This van (owner / Tyrah): under the fridge** (fits). SN **16401**.

Isotemp Slim Square 4.2 gal heat battery `[DEC–001, SRC–022]`

- Coil connections should sit below the 5 L expansion header level so the coil self-vents. If the tank is above that level, add a bleeder at the high point (SRC-022 §2).
- Horizontal, vertical, or other Slim Square positions in the manual are allowed; **drain / safety valve must be the lowest point**.
- **This van (owner):** mount **driver side under the new sofa/bed (low)**; expansion header in **sofa-bed backrest (high)**. Prefer ease of access and ease of construction over a complex hidden void.

- Cap the **factory mixer forever** — AM100-1LF is the only tap mixer (DEC-008).
- Keep hydronic hoses to the coil short. Leave ~30 cm (1 ft) in front for element/thermostat service.
- Tank full of fluid is extra mass (~16 L static glycol plus the unit; empty weight ~35 lb per SRC-022 §8.3). Mount to a shelf or bulkhead rated for that.
- Envelope about 22-1/4 × 7-3/16 × 15-3/4 in (SRC-022 / National Luna Slim Square 16 L table). `[MEASURE]` against the actual bay.

5 L expansion header `[DEC–005, SRC–019]`

- Highest circulating-loop point. Acts as expansion volume and air "burp" tank (SRC-019 p. 26–27).
- Teed on the return in this build (DEC-010) — not in series, not off the Isotemp coil.
- Unrestricted return path from that tee back to the Hydronic D5S diesel heater's own pump.
- **This van (owner):** mount in the **sofa-bed backrest (high)**; Isotemp stays **under the bench (low)**. Confirm the expansion header top still wins after framing; raise the backrest cradle or fall back to a higher wall mount only if it loses the height check.

Factory Sprinter sketch `[NOT THIS BUILD]`

SRC-014 shows a factory engine-bay heater with MB pump and controls. That is Candidate B / OEM — not this independent living-space loop (DEC-009).

Out of scope

SRC-020 floor reconstruction / insulation. Not used for heater or DHW layout.

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- ☐ **MEASURE:** Pump height vs coolant level; Isotemp Slim Square 4.2 gal heat battery mount and ~30 cm service clearance; confirm heater tilt within SRC-009 diagram

Coolant Loop

Scope

Coolant chemistry, flow, pressure, and plumbing. Fresh water is heated in the Duda B3-12DW-20 plate heat exchanger (DEC-003). The Isotemp Slim Square 4.2 gal heat battery (DEC-007): circulating glycol in the coil, static glycol in the chamber.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-009	p. 13, 15, 20–23, 31–32	II D5S: 250 l/h; 2.5 bar; 50% max; staging; bleed; rejected engine-circuit options
SRC-021 / SRC-022	Product + I&O §4–5, §8.3	coil in the Isotemp Slim Square 4.2 gal heat battery in glycol; chamber static glycol (DEC-007)
SRC-027 / SRC-029	Retail + SDS 506	Prestone LowTox AF555 concentrate (PG 90–100%)
SRC-028 / SRC-030 / SRC-031	Product + SDS 317XX + RTU contrast	Star brite Bio-Safe year-round concentrate (PG 80–100%); not SKU 30300
SRC-018	Water system p. 25–32	Bleeders, fill/drain, pump orientation, fan-heater flow – marine reference only
SRC-019	Water system p. 5–7, 22–30; thermostat p. 7	Same family of loop practice as SRC-018, from the Sure Marine TSL-17 guide (this van's SC1600B / REAL-style cabin heat lineage). Workmanship only – not Eberspächer specs

Content

Chemistry and limits (this heater – Hydronic D5S diesel heater, SRC-009 p. 13)

PARAMETER	VALUE
Medium	Water + coolant, **max 50% coolant**
Max overpressure	2.5 bar
Water volume in heater	~0.18 l
Min flow through heater	250 l/h
12 V pump delivery	680 l/h @ 0.1 bar, <12 W

Overheat / low-flow is a lockout condition: coolant must drop below 70 °C before restart after overheat (SRC-009 p. 32). Normal staging (SRC-009 p. 31): Power → High ~65 °C; High → Low ~80 °C; Low → Off/pause ~85 °C; restart from pause ~75 °C. **During pause the circulating pump keeps running** so the Isotemp coil can still exchange heat with the static chamber (The Numbers; Isotemp value study).

Hydronic II C and D5WS tables are **not** this heater (DEC-000) – do not use their min-flow or wattage columns.

On-hand fluid `[Q–019]`

BRAND	BASE	CLASS	CONCENTRATE?	MIX RULE
Prestone LowTox AF555	Propylene glycol	Automotive engine coolant	**Yes – concentrate.** SDS 506: PG **90–100%** (SRC-029). Retail UPC lists Concentrate (SRC-027).	Dilute with water. Heater cap **≤50%** coolant (SRC-009). Do not pour undiluted.
Star brite Bio-Safe year-round (31700 / 317XX)	Propylene glycol	Phosphate-free engine coolant	**Yes – full strength.** SDS 317XX: PG **80–100%** (SRC-030). Not Star brite's 50/50 RTU 30300 (SRC-031).	Dilute with water to **≤50%** coolant (SRC-009). Do not pour undiluted.

How we know they are concentrates: a ready-to-use 50/50 coolant SDS lists roughly 40–60% glycol plus water. Both of these SDS sheets list propylene glycol in the **80–100%** band with no water as a major ingredient. That is full-strength propylene glycol.

Same PG base is necessary, not sufficient. Inhibitor packages differ. **No manufacturer mix approval found.** Do not combine them in the loop. Prefer one product for the fill.

Hydronic operating pressure (≤ 2.5 bar / ~36 psi) is **not** the Isotemp Slim Square 4.2 gal heat battery potable safety-valve setting (5.5 bar / 80 psi). Do not size the glycol loop to the potable PRV.

Circuit options — engine splice `[NOT SELECTED — DEC-009]`

Independent living loop does not use these. Retained only so nobody reopens Candidate B from a manual skim.

SRC-009 p. 20–23:

- Inline with non-return valve: heat to engine first; cabin blower comes in around 30 °C.
- Thermostat + T-piece: small circuit to cabin HX until ~70 °C; full large-circuit switchover ~75 °C.
- 5-port combination valve + T-piece, or 6-port combination valve when flow and return run parallel. Parking-heater small circuit to ~67 °C, then some heat to the engine.

SRC-010 p. 12: 5-port Order No. 330 00 177 + T-piece 20 1673 80 11 00; 6-port 330 00 176.

SRC-008 (vendor): likely 25 2014 80 62 00. Do not assume that equals 330 00 176 (Q-007).

Independent living loop (DEC-009) does not use these engine valves; it still must meet min flow, self-vent height, and 2.5 bar.

Isotemp Slim Square 4.2 gal heat battery `[DEC-001, DEC-007]`

Glycol through the corrugated 316 SS coil (circulating loop). The ~4.2 gal / 16 L chamber holds the **same mix, sitting static** — not freshwater, not loop flow. That static mass is the Isotemp Slim Square 4.2 gal heat battery. Factory design was coil + potable water; this build keeps that split and puts glycol on both sides because the tank has no DHW job left. Bosch is gone.

The coil always sees circulating glycol from the Hydronic D5S diesel heater's own pump (DEC-013 removed any separate auxiliary glycol pump). The static chamber can also be heated by the tank's 750 W / 115 VAC element from the Paneltronics AC panel, but that element does not pump the loop or replace the D5S pump. Hot water through the Duda plate still depends on circulating glycol.

The 5 L expansion header (DEC-005) is the **circulating** glycol header. It does not replace the static 4 gal mass.

Hydronic / coil

- 1/2 in BSPP engine-water connections. Use heat-rated hose; brass or stainless fittings, not ordinary plastic (SRC-022 §4.1–4.3).
- Coil below the glycol header so it self-vents; bleeder if the tank is mounted high (SRC-022 §2).
- On first fill, add coolant for **coil + hose** volume on the circulating side, and fill the **chamber** separately (static).
- Keep coil hoses short to limit restriction (min Eberspächer flow 250 l/h, SRC-009 p. 13).
- Coil stays in circulating flow so the battery is charged and discharged (Q-017). Do not tee the chamber into that loop.

Tank / element

- Drain/safety valve at the lowest point (SRC-022). Do not energize the 750 W element on an empty chamber (SRC-022 §1). Element and thermostat were built for water; they now sit in static glycol (HAZ-006).
- factory mixer on the Isotemp Slim Square 4.2 gal heat battery is **inert** in this build: it only mixes potable hot and cold (SRC-022, 38–65 °C). The chamber is static glycol with no tap flow, so that valve has nothing to do. Cap unused ports. It is **not** the DHW mixer (DEC-008).
- Winter drain of this chamber also removes the Isotemp Slim Square 4.2 gal heat battery.

DHW — Duda B3-12DW-20 plate heat exchanger + owner mixing valve `[DEC-003, DEC-008]`

Glycol on one side of the double-wall plate exchanger (B3-12DW-20); fresh water on the other. Mixer is already on the hot-water outlet (SRC-024). Unmixed Duda B3-12DW-20 plate heat exchanger outlet can be as hot as the glycol loop — scald hazard (HAZ-005). Tap hoses must come off the mixed outlet, not a glycol barb. Do not use the Isotemp Slim Square 4.2 gal heat battery factory valve as the tap mixer.

Not selected: Bosch glycol buffer (DEC-001); SureCal tank (SRC-016); coil in the Isotemp Slim Square 4.2 gal heat battery as DHW exchanger.

Transferable loop practice from marine guides `[REFERENCE, not spec]`

SRC-018 (Webasto DBW) and **SRC-019 (Sure Marine TSL-17 with SureWire, June 2017)** document the same installer craft this build is reinventing for a van. They are **not** heater-family specs for the Hydronic D5S diesel heater. Where they conflict with SRC-009 (mix %, fuel, exhaust, pump identity), SRC-009 wins.

Layout (SRC-019 p. 5–6, 26–27)

- Expansion/header must be the **highest** point. It holds head on the pump and acts as a "burp" tank for air. Any component above it becomes an accidental air trap and can starve that branch.
- Unrestricted path from the expansion header toward the circulating pump. Do not put restrictive loads (cabin heater, tank coil) between the header tee and the pump.
- Fan heaters / space heat exchangers below the header; typically **bottom-in / top-out**, mounted **level**, with a bleeder on the outlet. Angled mounts trap air away from the bleeder.
- Domestic heat exchanger on a path that is always open when you want hot water. Marine "summer valves" that bypass cabin loops are **rejected here** (DEC-010); summer hot water is a short EasyStart Timer call instead.
- Diesel boiler → domestic water requires an **anti-scald / thermostatic mixing valve** (this build: AM100-1LF thermostatic mixing valve on the Duda B3-12DW-20 plate heat exchanger).

Hose and fittings (SRC-019 p. 22–23)

- Reinforced heater hose (not garden hose, not ordinary radiator hose for long runs).
- Brass / copper / stainless barbs; stainless clamps; double-banded clamps preferred.
- Hose fully over the barb; adhesive on beaded fittings; metal elbow for tight 90° turns; ~1/8 in clearance through bulkhead holes; support clamps along the run.
- In-line bleeder at hose high points that can trap air. Fill/drain at the **lowest** point.

Fill / bleed / first heat (SRC-019 p. 27–30)

- Estimate circulating volume before buying concentrate. Example capacities in that guide (order-of-magnitude only): ~3 gal per 100 ft of 3/4 in heater hose; ~0.2 qt per REAL-style fan heater; a 5 L-class header holds on the order of a gallon when full.
- Prefer a **water-only flush and leak check** before committing propylene glycol. Cap off the header while filling. Do not run the pump dry; do not run the pump during the first flush if debris may still be in the hose.
- Bleed until a **steady** stream (sputter = air). Run the pump briefly, then bleed again.
- Check header level only when the system is **cool**. Hot open-cap checks can scald.
- Optional diagnostic after first heat (marine practice): supply-to-return ΔT around 10°F is healthy circulation; much larger ΔT suggests flow is too low. Confirm against SRC-009 min flow (250 l/h) if anything feels lukewarm while the burner runs.
- Mix chemistry: that guide discusses ~40% antifreeze and a 50% ceiling. **This heater** is capped at **≤50% propylene glycol** by SRC-009; use Prestone LowTox AF555 only (DEC-017).

Thermostat placement (SRC-019 p. 7)

- No mercury thermostats (vibration). Trial-mount the SC1600B on a temporary lead if needed to find a spot free of drafts and direct sun — same advice as SRC-009 room-sensor placement.

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- [ ] **CLOSED:** Q-019 / DEC-017 — Prestone LowTox AF555 only, ≤50%
- [ ] **MEASURE:** Static height of Eberspächer and coil in the Isotemp Slim Square 4.2 gal heat battery vs the 5 L expansion header; circulating fill volume (coil + hoses + heater)

Fuel System

Scope

Eberspächer metering-pump and fuel-line rules. Do not use Webasto/TSL-17 fuel pages (SRC-018 / SRC-019) as specs.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-009	Fuel supply p. 26–30; fuel quality p. 29	Hydronic II diesel rules, pump orientation, heads (**this heater**)
SRC-005	Priming notes	Owner hand-bulb siphon — blog; manufacturer still requires vented fuel

Content

Fuel quality

- Hydronic D5S diesel heater (this build): commercial diesel EN 590; **blending with max 20% FAME** EN 14214 is permitted (SRC-009 p. 13 tech data).
- Pure biodiesel / FAME as the sole fuel is **not** approved (SRC-009 p. 29 “Operation with biodiesel”).
- Do not use Hydronic II C or D5WS fuel pages for this heater (DEC-000).

Safety (SRC-009 fuel chapter)

- Switch off engine and heater before refuelling or fuel work.
- No naked flame, no smoking, avoid skin contact and inhalation.
- Cut hose/pipe with a sharp knife; no crushed or burred ends.
- Butt-join pipe inside hose to avoid bubbles.
- Protect fuel parts from heat; never route or fasten fuel pipes to the exhaust. Provide heat clearance at crossings; heat shields if needed.
- Dripping/evaporating fuel must not collect on hot parts or electrical gear.
- Clamp lines (~50 cm guideline) against vibration and mechanical damage.
- Fuel pipe from metering pump to heater should have a continuous rise.
- In buses/coaches: fuel pipes and tanks must not run through the passenger compartment or driver’s cab (statutory example in SRC-009 — treat occupied van living space with the same caution).

Metering pump (Hydronic II, SRC-009)

- Allowed installation 15°–90°; preferred 15°–35°; **0°–15° not allowed**.
- Pressure head tank → pump: max 3000 mm.
- Suction head, pressureless tank: max 1000 mm diesel (500 mm petrol).
- Suction head with 0.03 bar tank-cap valve: max 400 mm diesel (150 mm petrol).

D5WS suction/pressure numbers in SRC-010 are **not** this heater (DEC-000).

Transferable fuel-line workmanship from SRC-019 `[REFERENCE, not spec]`

The TSL-17 guide is a different heater system, so its tubing sizes, lift limits, and fuel-pump geometry do not apply here. Its workmanship advice is still useful: keep fuel components individually supported and accessible for inspection or service; protect the line from rubbing and heat; use fuel-rated sealant where a threaded fitting requires it; and do not let a hose or tube carry the weight of a filter, valve, or pump. For this Hydronic D5S diesel heater, SRC-009 remains authoritative for fuel type, pump angle, heads, routing, and all heater-specific limits.

Commissioning

Coolant circuit and the whole fuel supply must be carefully vented before first fire (SRC-009 p. 31). SRC-005 describes a hand-siphon prime and fuse-pull reset after failed starts — blog only; manufacturer lockout rules are in SRC-009 p. 32 (diesel: restart after 70 s miss; lock after two further failed attempts in the safety time). Do not switch off/on more than twice into a fault.

Priming the fuel line (first fill)

Manufacturer requires the fuel system vented before first fire (SRC-009 p. 31).

HOLD — confirm the tool on the actual van before opening the fuel line: measure the existing line outside diameter and the accessible joint where the temporary primer will connect. Select hose-barb adapters and the union to match those measured sizes and fuel compatibility. Record the sizes and photograph the assembled tool. Do not guess a 6 mm, 8 mm, 3/16 in, 1/4 in, NPT, or BSP size from a generic listing; SRC-009 and the owner notes do not establish the van's installed adapter dimensions.

Owner priming tool (SRC-005, documented): hand-bulb siphon (~$15 parts — O'Reilly hand siphon + Ace hose barbs + union). Procedure: open the fuel line at a convenient joint, insert the barb, squeeze the bulb to pull diesel past the metering pump toward the heater inlet, then reconnect the line before first start. Goal: avoid burning the diesel start-attempt budget on a dry line.

Lockout (manufacturer wins over blog): SRC-009 p. 32 — diesel: restart after ~70 s miss; lock after two further failed attempts in the safety time; do not switch off/on more than twice. Clear locks via EasyStart Timer diagnostic path (or other approved tools listed in SRC-009). SRC-005's "pull the fuse to reset" and "three chances" are community notes — use only if the timer path is unavailable, and still obey the off/on limit.

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- ☐ **MEASURE:** Pump angle, suction/pressure heads, fuel take-off method
- ☐ **MEASURE:** Photograph / label the priming-tool adapters for the van toolbox

Exhaust and Combustion Air

Scope

Routing, diameters, lengths, and CO/fire constraints from Eberspächer hydronic manuals. Marine thru-hull/water-box practice in SRC-018 / SRC-019 does not apply to a van.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-009	p. 24–25	Hydronic II exhaust and combustion-air kit (**this heater**)
SRC-005	Field notes	Optional 90° elbow for height / condensate – confirm as-built
SRC-007	Disclaimer	CO lethality reminder (air heater blog; still valid as hazard)

Content

Exhaust – Hydronic II kit (SRC-009 p. 24–25)

- Flexible exhaust inner Ø 24 mm, 900 mm, plus 300 mm tail with end sleeve, plus silencer.
- Whole exhaust run: **min ~200 mm, max ~2 m** (SRC-009 p. 25 figure).
- Silencer on L or Z bracket, always fixed to the vehicle (bracket torque 6+0.5 Nm; pipe clips 7+0.5 Nm).
- Tail should be much shorter than the heater-to-silencer run.
- Must end in open air; must not protrude beyond the vehicle side.
- Slope slightly down; ~Ø 5 mm drain at the lowest point for condensate.
- Mouth must not point into the airstream, clog with dirt/snow, or feed combustion-air intake.
- Clearance from fuel lines (plastic or metal), electrical cables, brake hoses, and other heat-sensitive parts.
- Recommended support spacing ~50 cm against vibration.
- Do not work on exhaust while the heater is running; wait until cool.

Field note (SRC-005): a stainless 90° exhaust elbow at the heater can reduce hanging height under a low Sprinter and give condensate a low drain point. Still obey slope, drain, clearance, and max-length rules in SRC-009. Confirm what is actually fitted on this van before redesigning.

Combustion air – Hydronic II (SRC-009 p. 25)

- Draw from an area that stays ≤ 25 °C and is not spray/dust laden; not from the passenger compartment.
- Kit hose: double-wall sound-insulating, inner Ø 20 mm, **max 760 mm**, hose clip 3+0.5 Nm.
- May be shortened; cut clean so chips cannot jam the combustion fan.
- Opening always free; not into wind blast; not clogged; slope down with ~Ø 5 mm condensate drain if needed.
- Keep exhaust and intake from recirculating (silencer clearances min 150 mm / 200 mm – SRC-009 p. 25).

CO / intrusion

HAZ-002: exhaust must not enter the interior through ventilation, warm-air intakes, or openings (SRC-009 danger callout).

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- ☐ **MEASURE:** Installed exhaust length, support spacing, drain hole, intake location temperature; note whether a 90° elbow is fitted

Electrical and Controls

Scope

Pin tables and fuse sizes for this van's Hydronic D5S + EasyStart Timer + master / fans / Paneltronics. Part one Controls is the plain-English story: **12 V diesel heat** (master lockout) and **120 V Isotemp chamber** (Paneltronics **WATER HEATER**) are separate worlds. Manufacturer diagrams still win at the connector.

Wire size note: Eberspächer manuals list **mm²**. This book uses **AWG** for shop work. Same conductors.

MANUAL (MM²)	USE (AWG)
4.0	**12**
2.5	**14**
1.0	**18**
0.5	**20**
0.35	**22**
0.22	**24**

Sources

SOURCE	USED FOR
SRC-009 pp. 13, 32–33	Voltage window, fuses, battery cable, heater connector
SRC-003 pp. 8–9, 17	EasyStart 10-pin; Hydronic circuit
SRC-002 pp. 3–5, 8–10	Altitude kit adapter
SRC-033	SC1600 R–W limits (family; SC1600B not proven identical)
SRC-038 / DEC-020	Paneltronics WATER HEATER
SRC-019	Thermostat = signal; fans need a relay
SRC-040	Owner house-electrical intent (this van)
SRC-041	FarOutRide reference architecture (class of system)

Heater envelope (SRC-009 p. 13)

Voltage window	10.5–16 V
Start / run (without pump)	~120 W start; ~40 / 37 / 12 W Power / High / Low
Water pump	<12 W · ~680 L/h @ 0.1 bar
Battery → heater	**12 AWG** if round trip ≤ 20 ft / 6 m
Main fuse	**20 A**
EasyStart fuse	**5 A** (insert last)
Vehicle blower fuse	**25 A** – unused on this van

Emergency: control off, pull fuse, or disconnect battery. ≤ two off/on cycles into a fault.

Heater connector (SRC-009 p. 33)

Seen from the cable-inlet side.

PIN	COLOUR	AWG	FUNCTION	THIS VAN
1	red	14	Battery +	Land
2	brown	14	Battery –	Land
3	black/red	20	Vehicle fan	**Tape off**
4	green	18	Metering pump +	Land
5	blue/white	20	Diagnosis	Land
6	blue	20	Aux-heat criterion	Tape off unless used
7	yellow	20	Wake + (S+)	From EasyStart pin 6
8	violet	20	Water pump +	Land
9	brown	18	Water pump –	Land
10	brown/green	18	Metering pump –	Land

Seal unused chambers. Kit blower-relay leads unused.

EasyStart Timer 10-pin (SRC-003 p. 9)

PIN	SIGNAL	COLOUR	AWG
1	Battery +	red	22
2	Lighting	grey/black	24
3	Battery –	brown	22
4	Diagnosis	blue/white	24
5	DAT	violet	24
6	Wake out (S+)	yellow	22
7	ADR	brown/yellow	24
8	ADR	white/red	24
9	Temp – (optional sensor)	brown/white	24
10	Temp + (optional sensor)	grey	24

Heater-side leads into this connector are **20 AWG** on red / brown / blue-white for Hydronic II (manual 0.5 mm²). Pins 9–10 are for Eberspächer's temperature **sensor** – not the SC1600B dry contact.

Master switch (DEC-012)

Off/On switches + to EasyStart, thermostat signal path, and fans. Heater **20 A** feed stays separate. Representative face: Sure Marine **W005-378K**.

SC1600B / fans (Q-022 open)

SC1600B: battery-powered; R–W dry contact ≤1 A. Fans need a relay. Fan dial Low/Med/High without hard Off (DEC-018). Until Q-022 closes: EasyStart Timer calls only. Fan-path relay reference: TE Connectivity **V23134-A3052-X540** (or any ISO mini 12 V coil / ≥10 A contacts) – DigiKey.

Altitude kit 22 1000 33 22 00

Adapter on the control harness (SRC-002). Land at HOLD 4 while the EasyStart harness is open (Q-015 / DEC-006). Required before high camps; not on the post–first-fire wait list with Q-022 / Q-009.

House electrical context (SRC-040 / SRC-041 / photo extract)

Two hydronic worlds. (1) 12 V from the **house battery** → heater + EasyStart + fans. (2) 120 V from shore or **PROwatt SW** via one **Blue Sea 9009** → Paneltronics → **WATER HEATER** → Isotemp 750 W. A **second 9009** feeds only the Mean Well charger (shore/J1772)—not heat. Both 9009s are hand rotaries, not automatic. This appendix does not size the house bank, solar, B2B, or shore charger.

OBSERVED ON VAN / BUILD PHOTOS	ROLE FOR HYDRONIC
Xantrex LinkPRO (wood panel)	Battery monitor – watch voltage before long inverter + **WATER HEATER** runs
Xantrex PROwatt SW 2000 (wood panel)	Inverter – select via Paneltronics **9009** for off-grid AC
Paneltronics 72313 / 9972313B	AC distribution: MAIN · BATTERY CHARGER · **WATER HEATER** · OUTLETS
Blue Sea ST Blade (~12 circuit) + ML-RBS	House DC distribution / disconnect – hydronic master / EasyStart / fans land here **or** on a **second breakout** if slots are full
SmartPlug (120 V) + DUOSIDA J1772 32 A	Shore / Level-2 inlets feeding the two 9009 paths
Victron SmartSolar MPPT 100|50 (photos)	Solar charge – confirm still installed
Sterling B2B 60 A	Alternator → house
Mean Well RPB-1600-12	Shore/J1772 → house (early photos: IOTA DLS-55 – confirm removed)

Owner intent: SRC-040. Same *class* of architecture: SRC-041. Photo dump: photo-extract-electrical. Confirm path on the van (Q-025). Do not copy FarOutRide parts lists into this BOM.

Isotemp element (DEC-004 / DEC-020)

115 VAC / 750 W ≈ 6.5 A on Paneltronics **WATER HEATER**. Upstream GFCI/ELCI required (Q-009). Fill chamber before HOLD 9. Path: shore or PROwatt SW (via **9009**) → GFCI/ELCI if present → Paneltronics MAIN → **WATER HEATER** → element. Label the Paneltronics **9009** face **SHORE** / **INVERTER**.

Verification

- ☐ Heater and EasyStart pins match the tables above
- ☐ Blower leads taped off
- ☐ Master Off kills three cabin branches
- ☐ 5 A fuse inserted last
- ☐ Blade block has free slots **or** second breakout installed for hydronic branches
- ☐ Q-022 documented before claiming SC1600 auto

Open

- ☐ Q-009, Q-015, Q-022
- ☐ Fan-branch and branch-B fuse amp ratings
- ☐ Free fuse slots vs second breakout choice
- ☐ Approved SC1600 → heater wake landing (do not invent)

Installation Procedure

Scope

Hydronic D5S diesel heater install sequence (DEC-000 / SRC-009). Isotemp Slim Square 4.2 gal heat-battery rules from SRC-022 (DEC-001 / DEC-007). Floor insulation is out of scope. The serial install path for the vehicle is `../install-in-this-order.md`.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-022	§2–6	Isotemp Slim Square 4.2 gal heat battery locate, mount, plumb, fill; element full-chamber and AC safety rules
SRC-021	Product notes	USA BSP-to-NPT / PEX swivel fittings
SRC-009	p. 31	Eberspächer first-fire bleed
SRC-019	p. 22–25, 27–30	Hose/clamp craft; cabin heater bleed; water-first leak check before glycol fill
DEC-010	Decision log	Loop flow path: Hydronic D5S diesel heater's own pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger → coil in the Isotemp Slim Square 4.2 gal heat battery → return
DEC-013	Decision log	No auxiliary glycol-pump path; Isotemp element remains a Paneltronics AC load

Content

Isotemp Slim Square 4.2 gal heat battery (do in this order)

1. Mount so the drain/safety valve is the lowest point; leave service access in front (SRC-022).
2. Plumb the hydronic **coil** into the circulating glycol loop, below the 5 L expansion header or with a high-point bleeder. Do not tee the 4.2 gal chamber into that loop.
3. Fill the **chamber** with the same glycol mix as the loop. Leave it static. Do not pipe those ports to taps. Leave the PRV drain unblocked (SRC-022 §5). Factory mixer on this tank is unused.
4. Fill/bleed circulating glycol including coil volume. Prefer water-first leak check, then ≤50% propylene glycol; bleed Sure Marine cabin heater outlet until a steady stream (SRC-019; see `../install-in-this-order.md`).
5. Route the 115 VAC element to the Paneltronics AC panel. Do not energize until the chamber is full and the AC protection/grounding checks are complete. Chamber still gets filled with static glycol for thermal mass.
6. The AM100-1LF thermostatic mixing valve is already on the Duda B3-12DW-20 plate heat exchanger freshwater outlet (DEC-008 / SRC-024). Run tap hoses from the mixed outlet only. Set and label it before anyone showers. Do not use the Isotemp Slim Square 4.2 gal heat battery factory mixer as the tap mixer.

Eberspächer mounting, fuel, exhaust, EasyStart Timer wiring, Altitude kit 22 1000 33 22 00, and first fire: follow SRC-009 and `../install-in-this-order.md` (DEC-010 loop order).

Verification

- ☐ Requirements traced to authoritative sources
- ☐ Dimensions and routing checked against the actual vehicle
- ☐ Conflicts and assumptions recorded
- ☐ Relevant inspection hold points completed

Open items

- [ ] **OPEN:** Owner fuel-priming tool procedure (07-fuel-system.md)
- [ ] **OPEN:** Q-009 — Paneltronics breaker, GFCI/grounding, cable route, and control/enable details (HOLD 9). Q-016 architecture closed (AC-only panel).

Commissioning Procedure

Scope

Isotemp Slim Square 4.2 gal heat battery commissioning (DEC-007). Eberspächer first-fire per SRC-009 p. 31–32. Plain-English twin: `book/install-in-this-order.md` steps 8–9.

Inputs and sources

SOURCE ID	PAGE/SECTION	INFORMATION USED
SRC-022	§5 Start up/test	Fill chamber, test PRV, bleed coil, then AC (fluid is glycol, not potable)
SRC-009	p. 31–32	Bleed coolant and fuel; leak check; staging; lockout rules
SRC-019	Water system p. 27–30	Water-first leak check; bleed until steady stream; cool-level check; ΔT as flow hint (reference)
SRC-005	Priming	Hand-bulb fuel prime before first fire (blog; manufacturer still requires vented fuel)

Content

Before first fire (hold points)

- [ ] Fuel routing/clamps inspected (HAZ-001 / Q-010) — factory aux pickup OK; line still needs a look
- [ ] Fuel line primed with hand-bulb tool; not relying on dry start attempts
- [ ] Exhaust ends open-air, away from intake and cabin openings; slope + drain (HAZ-002)
- [ ] **HOLD:** water-first circulating-loop leak test passed before glycol is committed (zero visible leaks at rest and during a brief pump run; drain water clear); then fill ≤50% propylene glycol, one brand; bleed until steady stream at expansion header and Sure Marine cabin heater outlet; pump briefly; bleed again
- [ ] Heater, pumps, and coil in the Isotemp Slim Square 4.2 gal heat battery below fluid level in 5 L expansion header
- [ ] Mix ≤50% coolant; one brand only
- [ ] AM100-1LF thermostatic mixing valve on the Duda B3-12DW-20 plate heat exchanger set (~120°F target); tap hoses from mixed outlet only before anyone showers
- [ ] EasyStart Timer can command the heater; Altitude kit 22 1000 33 22 00 preferred at HOLD 4 while the harness is open (required before high camps; DEC-006)
- [ ] Fuel-prime adapter sizes and access joint confirmed on the actual van; no dry-start experiment used
- [ ] Q-022 automatic thermostat mode either verified and documented, or explicitly deferred; manual EasyStart Timer operation available
- [ ] Sure Marine cabin heater mounted level, bottom-in / top-out, with outlet bleeder reachable (SRC-019)

Isotemp Slim Square 4.2 gal heat battery `[DEC-001, DEC-007]`

- [ ] Chamber filled with static glycol mix (not freshwater)
- [ ] PRV lever test; drain hose open
- [ ] Circulating glycol bled through the coil; coil hoses not air-locked

- [ ] Chamber left static — not teed into loop flow, not piped to taps
- [ ] 115 VAC element routed to Paneltronics; AC protection/grounding verified; element not energized until chamber is full; no aux pump path
- [ ] No mixing of chamber glycol with Duda B3-12DW-20 plate heat exchanger freshwater at fittings

Eberspächer first fire `[DEC–000 / SRC–009 p. 31–32]`

- [ ] Coolant and fuel systems vented
- [ ] Trial run from EasyStart Timer: watch underside and new joints
- [ ] Water and fuel connections leak-checked while hot
- [ ] If fault: do not switch off/on more than twice; use EasyStart Timer diagnostic path (not a D5WS code list)
- [ ] After overheat: cool below 70 °C before restart
- [ ] Expect Power → High ~65 °C → Low ~80 °C → pause ~85 °C; pump keeps running in pause; restart from pause ~75 °C

After first fire / before calling it done

- [ ] First run watched end-to-end; no wetness or unexplained smell
- [ ] 5 L expansion header level rechecked after cool-down (cool only — never open hot)
- [ ] Optional: supply-to-return ΔT near ~10°F while running; much larger suggests low flow (SRC-019 hint; confirm vs SRC-009 250 l/h)
- [ ] AM100-1LF thermostatic mixing valve setpoint confirmed before any shower
- [ ] Master switch Off verified as lockout
- [ ] Altitude kit 22 1000 33 22 00 landed at HOLD 4 (or explicitly documented exception before a low-elevation-only first fire); required before high camps (Q-015 / DEC-006)
- [ ] Isotemp Slim Square 4.2 gal heat battery element operation tested only after Paneltronics checks and full-chamber confirmation

Verification

- [ ] Requirements traced to authoritative sources
- [ ] Dimensions and routing checked against the actual vehicle
- [ ] Conflicts and assumptions recorded
- [ ] Relevant inspection hold points completed

Open items

- [ ] **OPEN:** Q-009 Paneltronics AC protection, cable route, and control/enable details (HOLD 9). Q-016 is closed on architecture (AC-only panel); remaining work is Q-009.
- [ ] **MEASURE:** Confirm Altitude kit 22 1000 33 22 00 is mounted and wired at HOLD 4; required before high camps (Q-015)

Isotemp value study

Why this note exists

Most van guidance sizes **air** heaters (Airtronic D2 ≈ 2 kW, D4 ≈ 4 kW). This build uses a **Hydronic D5S** whose lowest documented stage is still **2.1 kW**, plus a marine calorifier (Isotemp Slim Square) as **static glycol mass**—not as a potable tank. Owner and installer forums have long said hydronic loops without enough mass **short-cycle** badly; the 2018 design notes for this van called for roughly **8.5 gal** of thermal mass (SRC-004). This build kept the idea and changed the hardware to a **4.2 gal** Isotemp with the chamber on glycol (DEC-007).

This note estimates whether that smaller chamber still earns its complexity: van heat-loss bands, stored energy, predicted starts per hour with and without the chamber, a coil-coupling factor, and what AC preheat can (and cannot) do for dish water.

Where this sits in the book. Day-to-day language: User's Guide (especially AC chamber and dishes). Staging and pause/pump: The Numbers. Architecture and finish lines: System Design · Acceptance. Manufacturer truth: SRC-009 and SRC-022. Isotemp coil guidance for engine-calorifier service suggests roughly **0.5 gal/min** through the coil for efficient transfer (SRC-022 family manuals); this loop's heater pump is sized for the D5S minimum **250 L/h**, so treat coil UA as a measured unknown—hence the planning coupling factor below.

Question

Does the Isotemp Slim Square 4.2 gal heat battery improve this build enough to justify keeping it, or would selling it simplify the system without giving up meaningful performance?

Short answer

The Isotemp provides a real but limited benefit. The static chamber adds thermal mass, which can absorb surplus heat when the Hydronic D5S diesel heater is producing more heat than the cabin and domestic-hot-water loads require. The stored heat can then return through the Isotemp coil while the heater is paused or while a short demand continues.

The Isotemp does **not** make the Duda B3-12DW-20 plate heat exchanger produce hot water while the diesel heater pump is stopped. The Duda remains a flow-through exchanger. A hot tap still requires circulating glycol from the Hydronic D5S diesel heater pump. The 750 W Isotemp element can charge the static chamber, but the D5S pump must circulate glycol through the coil before that stored heat can reach the Duda.

The current evidence supports keeping the Isotemp if the priorities are quieter, steadier low-load heating and the ability to preheat the loop from available AC power. The evidence does not support keeping the Isotemp solely as an always-ready hot-water tank. The practical keep-or-sell decision should be based on measured cycling and hot-water recovery after the loop is commissioned.

What the Isotemp adds

The Isotemp contributes three possible benefits:

1. **Thermal buffering.** The static chamber absorbs heat through the coil and returns heat to the circulating loop later. Buffering matters most when the cabin heat demand is below the D5S low output.
2. **Short holdover.** Stored heat can reduce the time between a heater pause and the next heat call, provided the pump continues to circulate and the chamber remains warmer than the loop.
3. **AC preheating.** The 750 W element can add heat while AC power is available. The element heats the static chamber; the D5S pump and the Duda plate still control delivery to the cabin and taps.

The Isotemp does not add a second pump, increase the D5S heat output, improve the Duda plate's maximum transfer rate, or make the system safe to operate with an empty chamber.

Field benchmark for heater sizing

Insulated camper vans commonly run well on heaters near **2 kW**, while builders sometimes choose **4–5 kW** for reserve or colder use. That pattern appears across installer and owner write-ups (for example Sparks Overland on 2 kW vs 4 kW, Vanlife Customs, and Sprinter-oriented sizing notes such as VVKB). Those sources are not controlled laboratory tests. Layouts, climates, windows, and leakage vary. The shared signal is still useful: a well-insulated van often has a **normal operating heat load near or below 2 kW** for part of the season.

The Hydronic D5S low stage is **2.1 kW** (SRC-009). An Airtronic D2 can modulate well below that; the D5S cannot. If this van needs about 0.8–1.5 kW on an ordinary cold night, Low already has about **0.6–1.3 kW of surplus**. Without thermal mass, that surplus mostly raises loop temperature until the D5S pauses. Installer forums make the same qualitative point for hydronic vans: without enough mass, the heater short-cycles hard (for example Sportsmobile Forum — Plumbing Espar Hydronic D5). That is the physical reason a buffer can matter here even when a 2 kW air heater would simply run steady.

Estimated heat loss from the van

The exact heat loss cannot be known until the actual van dimensions, insulation coverage, window area, outside temperature, wind, and air leakage are measured. The following estimate is a planning range for a 170-inch Sprinter with a conventional Thinsulate conversion. The estimate assumes a high-roof living space, approximately 350–500 ft² of conditioned volume, approximately 350–500 ft² of opaque insulated envelope, and approximately 45–70 ft² of windows and door glazing.

The estimate uses the kind of values commonly reported for automotive Thinsulate installations: roughly R-4 to R-6 for the effective opaque wall and ceiling assembly after metal ribs, seams, gaps, and compressed insulation are included, and roughly R-1.5 to R-2.5 for the windows and glazed doors. The effective assembly values are deliberately lower than the best-case material rating. A standard Thinsulate job does not insulate the van as continuously as a new house wall.

HEAT-LOSS COMPONENT	PLANNING RANGE AT A 1°F TEMPERATURE DIFFERENCE
Opaque walls, ceiling, doors, and floor	80–125 Btu/h·°F
Windows and glazed doors	18–47 Btu/h·°F
Metal ribs, framing, floor edges, and other thermal bridges	15–35 Btu/h·°F
Infiltration at roughly 0.3–1.0 air changes per hour	2–9 Btu/h·°F
Estimated total	**115–216 Btu/h·°F**

For comparison, the Hydronic D5S produces approximately 7,165 Btu/h at its 2.1 kW low stage, 17,061 Btu/h at its 5.0 kW high stage, and 17,744 Btu/h at its 5.2 kW power stage.

INDOOR TEMPERATURE 70°F	OUTDOOR TEMPERATURE	TEMPERATURE DIFFERENCE	ESTIMATED VAN HEAT LOSS
70°F	40°F	30°F	3,450–6,480 Btu/h (1.0–1.9 kW)
70°F	20°F	50°F	5,750–10,800 Btu/h (1.7–3.2 kW)
70°F	0°F	70°F	8,050–15,120 Btu/h (2.4–4.4 kW)
70°F	−10°F	80°F	9,200–17,280 Btu/h (2.7–5.1 kW)

These figures exclude solar gain, occupants, cooking, engine heat, and temporary door opening. Wind can increase infiltration substantially. A cold metal floor, uninsulated rear doors, or uncovered windows can move the actual result toward or beyond the high end. The estimate is therefore useful for deciding whether the heater has surplus output, not for sizing a final installation or predicting a precise warm-up time.

What the estimate says about the Isotemp

At 40°F outside, the estimated heat loss is often below the D5S low-stage output. The Isotemp may then absorb some surplus heat and reduce short cycling. At 20°F outside, the low stage is close to the estimated demand: the Isotemp may help at the low-loss end, but the Isotemp will have little opportunity to charge at the high-loss end. At 0°F and below, the van may need most or all of the D5S high-stage output, so the Isotemp is more likely to discharge than charge.

The heat-loss estimate therefore strengthens the case for the Isotemp as a shoulder-season and moderate-cold buffer. The estimate does not support claiming a major benefit during the coldest conditions, when the heater's output is already needed by the van. The estimate also explains why an Isotemp test performed only in mild weather could overstate the benefit: mild weather creates the largest surplus available for storage.

Energy capacity: the upper bound

The main chamber holds approximately 16 L of static glycol mix. For a first-order estimate, treat 16 L as 16 kg. The tables below use a specific heat near **4 kJ/kg·K** (water-like) so the arithmetic stays transparent. A ≤50% propylene glycol mix is closer to about **3.5–3.6 kJ/kg·K** near room temperature and rises toward roughly **3.6–3.8 kJ/kg·K** in the 60–80°C band (Engineering Toolbox and related glycol property tables). That lowers stored energy by roughly **10–15%** versus the water-like tables. Treat every kWh figure as slightly optimistic until measured.

TEMPERATURE SWING IN THE CHAMBER	STORED ENERGY (WATER-LIKE (C_P))	EQUIVALENT AT THE D5S 2.1 KW LOW OUTPUT
10°C / 18°F	0.18 kWh	5 minutes
20°C / 36°F	0.36 kWh	10 minutes
30°C / 54°F	0.53 kWh	15 minutes
40°C / 72°F	0.71 kWh	20 minutes

These are storage limits, not guaranteed delivered benefits. Heat-transfer resistance in the coil, cabin heater, Duda plate, hose flow, control logic, and temperature limits will reduce the usable amount. The AC working thermostat also caps element charging near **75°C** (SRC-022). The table is a scale check, not a performance guarantee.

A 20°C chamber swing can absorb about 0.36 kWh. If the cabin and tap loads together consume only 1.0 kW while the D5S is delivering its 2.1 kW low output, the surplus is about 1.1 kW and would fill that theoretical buffer in roughly 20 minutes. If the loads consume 2.0 kW, the surplus is only about 0.1 kW and the buffer provides little additional value. If the loads exceed 2.1 kW, the buffer discharges instead of charging.

Heater cycling

The Isotemp can reduce cycling only when all of the following conditions are true:

- The D5S is operating at its low output or is about to enter a pause.
- The cabin and domestic-hot-water loads are smaller than the heater output.
- The coil can transfer the surplus into the static chamber.
- The stored heat can later return through the coil before the next burner start.

The Isotemp cannot eliminate cycling when the system is correctly calling for more heat than the D5S can provide. The Isotemp also cannot guarantee fewer starts because the D5S control algorithm, coolant sensor location, pump behavior during pause, cabin thermostat behavior, and actual heat load determine the cycle.

A useful expectation is **longer runs and longer pauses**, not a specific percentage reduction. A buffer can reduce short burner restarts if the stored heat bridges the gap between the D5S pause temperature and the next restart temperature. The buffer can also make a low-load run less abrupt. The buffer cannot be credited with a measured reduction until the installed system is logged.

Engineering estimate of cycling with and without the Isotemp

The following calculation takes a deliberate first stab at the question. The calculation is not a substitute for a temperature log, but it gives a useful scale for the expected effect.

Assumptions

- The D5S is operating at its documented 2.1 kW low output after the initial Power and High stages.
- The D5S pauses near 85°C and restarts near 75°C, giving an approximately 10°C control band (The Numbers; SRC-009). During control pause the **burner is off** and the **circulating pump keeps running**, so the coil can still exchange heat with the chamber while the flame is out.
- Loop order puts the Isotemp **last** (after cabin heater and Duda). Surplus heat hits the cabin first. The chamber mostly stores leftovers. That is intentional for daily comfort; it also caps how strongly the buffer can fight short cycling.
- The circulating loop without the Isotemp contains approximately 4–6 L in the heater, cabin heater, plate, hoses, and fittings. Use 5 L as the midpoint.
- The Isotemp adds approximately 16 L of static glycol. If the chamber follows the circulating loop closely, the combined effective mass is approximately 21 L.
- Glycol mixture specific heat is approximated as 4 kJ/kg-K, and 1 L is treated as 1 kg. Actual glycol properties, tank stratification, and coil transfer will change the result.
- During a low-stage cycle, the van heat load is treated as constant. The estimate uses 1.0, 1.5, and 2.0 kW of load. Those loads map roughly onto the heat-loss table above (cabin held at 70°F): about **40°F outside** for 1.0 kW on a better-insulated build, about **40–20°F** for 1.5 kW, and about **20°F** (or a leakier build near 40°F) for 2.0 kW. Colder than that pushes load above 2.1

kW, where this low-stage charge/pause model no longer applies.

For a 10°C control band, the thermal energy in the temperature swing is:

$$E = mc_p\Delta T$$

That gives approximately 0.056 kWh for a 5 L loop without the Isotemp and approximately 0.233 kWh for a 21 L loop with the Isotemp fully participating. During burner-on time, the net charging rate is $2.1 - Q_{load}$. During pause, the discharge rate is Q_{load}.

VAN HEAT LOAD	ROUGH OUTDOOR AIR (CABIN 70°F)	ESTIMATED CYCLE WITHOUT ISOTEMP	ESTIMATED CYCLE WITH ISOTEMP FULLY COUPLED	STARTS/H WITHOUT ISOTEMP	STARTS/H WITH ISOTEMP
1.0 kW	~40°F on a better-insulated build (low end of the 40°F loss band)	3.3 min on + 3.3 min paused = 6.6 min	14.0 min on + 14.0 min paused = 28 min	9.1	2.1
1.5 kW	~40–20°F (mid 40°F band → low 20°F band)	5.6 min on + 2.2 min paused = 7.8 min	23.3 min on + 9.3 min paused = 32.6 min	7.7	1.8
2.0 kW	~20°F, or a leakier build near 40°F (near D5S low-stage output)	33.3 min on + 1.7 min paused = 35 min	140 min on + 7.0 min paused = 147 min	1.7	0.4

Outdoor temperatures are a **planning correlation** from the heat-loss ranges above, not a forecast for this van. Wind, windows, and floor losses can move the same outdoor temperature into a different row. At about **0°F and below**, the estimated loss is often already **above 2.1 kW**, so the D5S is no longer in a low-stage surplus cycle and this starts-per-hour table does not apply.

The 2.0 kW row is a boundary case. At a 2.0 kW load, the D5S has only 0.1 kW of surplus at low stage, so the D5S may remain at low output for a long period rather than follow a clean charge-and-pause cycle. At loads above 2.1 kW, the Isotemp cannot charge during low stage and cannot reduce cycling by storing surplus heat.

The idealized calculation predicts about three to five times longer cycles and roughly 75–80% fewer starts per hour when the Isotemp is fully coupled. Those percentages are an **upper-bound estimate**, not a claim about the installed van. The coil cannot make 16 L of static glycol behave like instantaneous circulating volume.

Coil coupling estimate (why "fully coupled" is too rosy)

Define a coupling factor f as the fraction of the 16 L chamber that effectively joins the circulating mass on the cycle timescale:

$$m_{\text{eff}} = 5 + f \cdot 16 \quad \text{(liters)}$$

$f = 1$ is the table above. Real f is lower because heat must cross the corrugated 316 SS coil, the chamber is static (stratified), and this build puts the Isotemp **last** (after the cabin heater and Duda), so coil inlet fluid is already cooler than heater outlet.

No published UA for the Slim Square coil was found. A planning estimate uses a thermal time constant $\tau = C_{\text{chamber}}/UA$:

- Chamber capacitance $\approx 16\,\text{kg} \times 4\,\text{kJ/kg}\cdot\text{K} = 64\,\text{kJ/K}$.
- Marine calorifier coils of this class are "extra long / corrugated," but still a single pass into a tank. A plausible UA band is about **20–80 W/K** (weak stratification / low ΔT → strong well-mixed coil).
- That gives $\tau \approx$ 13–53 minutes (mid ≈ **25–30 minutes** at ~40 W/K).

Cycle half-periods without the buffer are only about **2–6 minutes** at 1.0–1.5 kW load — much shorter than τ — so the chamber cannot stay isothermal with the loop. Power also limits coupling: at a few kelvin coil-to-chamber ΔT, $UA \cdot \Delta T$ is only a few hundred watts, while Low-stage surplus at 1.0 kW load is about **1.1 kW**. Only part of that surplus enters the chamber each minute; the rest still races the D5S sensor through the small circulating volume.

Planning stab

COUPLING f	MEANING	EFFECTIVE MASS	USE AS
1.0	Chamber tracks loop instantly	21 L	Ceiling only
0.35–0.50	Mid UA (~40 W/K), Isotemp last, some stratification	**~11–13 L**	**Expected band**
0.25	Weak coil / low ΔT / air	~9 L	Pessimistic
0.55	Strong coil, good mixing, large ΔT	~14 L	Optimistic installed

Point estimate for predictions: $f \approx 0.40$ (about 40% of the chamber participates). That is a judgment call, not a measurement. Recompute E and starts with $m_{\text{eff}} = 11.4\,\text{L}$ ($E \approx 0.127\,\text{kWh}$):

VAN HEAT LOAD	ROUGH OUTDOOR (CABIN 70°F)	STARTS/H WITHOUT	STARTS/H AT $f = 1$ (CEILING)	STARTS/H AT $f \approx 0.4$ (PLANNING)
1.0 kW	~40°F (better insulation)	9.1	2.1	**~4.1**
1.5 kW	~40–20°F	7.7	1.8	**~3.4**
2.0 kW	~20°F (or leaky near 40°F)	1.7	0.4	**~0.8**

So the honest Mild-weather expectation is closer to **cutting starts roughly in half** (about **9 → 4/h** at 1.0 kW), not the 9 → 2/h ceiling. Still a large change; still depends on surplus at Low stage.

The most defensible expectation before commissioning is therefore:

- **Mild weather (~1.0 kW / ~40°F band):** planning starts drop from about **9/h to about 4/h** ($f \approx 0.4$); ceiling about **2/h** if the coil were perfect.
- **Moderate cold (~1.5 kW):** planning about **8/h → 3–4/h**; ceiling about **2/h**.
- **Near 2.0 kW / ~20°F:** little surplus left; starts are already low; Isotemp helps less.
- **At or above 2.1 kW load (often ~0°F and colder on the loss table):** no Low-stage surplus to store; cycling benefit → ~0.

Measurement plan

Measure the same operating condition with the Isotemp coil connected and, only if practical and safe, with the Isotemp bypassed according to a documented temporary test configuration. Do not disconnect or re-plumb the installed system casually.

For each test, record:

- outdoor temperature and approximate cabin temperature;
- starting coolant temperature and expansion header level while cool;
- cabin fan setting and thermostat or EasyStart setting;
- heater run time, pause time, and number of starts per hour;
- supply and return temperatures near the D5S, and temperature at the Isotemp coil connections;
- whether a hot tap was opened and for how long;
- battery voltage and AC-element state.

Run at least three one-hour tests: low cabin demand, normal cabin demand, and a short hot-water draw. Compare starts per hour, burner-on minutes per hour, pause duration, and cabin-temperature variation. A convincing cycling benefit would be a repeatable reduction in starts per hour or a repeatable increase in pause duration without worse cabin comfort or slower hot-water recovery.

Hot water on hand

The current plumbing makes the Duda plate the domestic-hot-water exchanger. The Isotemp chamber contains glycol, not drinking water. The arrangement therefore behaves in three different ways:

OPERATING CONDITION	HOT-WATER RESULT
D5S running and pump circulating	The Duda can provide hot water continuously, subject to heater output, glycol temperature, freshwater flow, and the AM100-1LF mixer. The Isotemp is secondary thermal mass.
D5S paused but pump circulating	The Isotemp can contribute stored heat through the coil if the static chamber is warmer than the circulating loop. Hot-water duration depends on the actual temperature difference and heat-transfer rate.
AC element on, D5S off and pump stopped	The Isotemp chamber warms, but the Duda does not receive that heat. The element alone does not provide hot water at the taps.

The Isotemp can improve convenience by allowing a preheat period before a shower or by reducing the time the diesel burner must run during a short draw. The Isotemp does not create an immediate hot-water reserve in the same way that a potable hot-water tank would. A user must start the D5S pump to move stored chamber heat through the coil and Duda.

For scale, heating 10 L of fresh water from 10°C to 40°C requires approximately 0.35 kWh before exchanger and plumbing losses. A 16 L chamber at a 30°C temperature swing contains approximately 0.53 kWh in the idealized calculation above. That amount could support one short draw in principle, but the result depends on how much of the chamber can transfer through the coil while maintaining the mixed outlet near 120°F. The study cannot claim a shower duration without measured freshwater flow and heat-transfer data.

AC preheating

The 750 W element supplies approximately 0.75 kWh per hour before losses. Raising the full 16 L chamber by approximately 30°C would require about 0.53 kWh in the idealized calculation, or about 40–45 minutes at 750 W if all element power entered the chamber. Real heating time will be longer because of thermostat behavior, heat loss, glycol properties, and the chamber's actual usable temperature range. The working thermostat cuts the element near **75°C / 167°F**.

AC preheating is most useful when the van already has permitted AC power (shore or solar through the house electrical — SRC-040 / SRC-041) and the user wants a short dish rinse or a shorter next diesel run. AC preheating is less compelling when the PROwatt SW inverter must supply the element from the house battery alone. A 750 W load is substantial for an inverter and battery system; conversion losses increase the required battery energy. The exact battery penalty depends on inverter efficiency and battery voltage, which are not yet measured in this build (Q-025).

Dishes after an AC charge — what to expect

Charge the chamber first. That often takes about **40–60 minutes** to reach ~**75°C** at 750 W. The taps stay cold until you run EasyStart.

Then turn the master **On**, make a short EasyStart start, and let the **pump** run. Heat moves from the chamber through the coil and the Duda to the sink. The pump always starts; the burner is separate.

If the glycol loop is still warm, the flame often stays off. Hot water at the sink usually arrives in about **30–60 seconds**. You get about **2–5 minutes** of rinse water (a few liters), then the flame lights or the water goes lukewarm.

If the glycol loop is cold, the flame almost always comes on. Hot water at the sink usually arrives in about **2–5 minutes**. The chamber shortens the diesel run; it does not skip the flame.

This is a dish rinse, not a shower. Skipping the flame requires a loop that is already warm.

Convenience and complexity tradeoff

Convenience gained

- A quieter and potentially steadier low-load heating pattern.
- A short thermal bridge during brief hot-water draws or burner pauses.
- The option to preheat the static chamber when AC power is already available.
- More thermal inertia during short control interruptions.

Complexity retained

- A larger fluid volume to fill, bleed, inspect, and protect from freezing.
- A tank that must be structurally restrained on the **driver side under the new sofa/bed** and remain serviceable (favor access and simple construction).
- Additional coil fittings and hose length that can introduce air, restriction, or leaks.
- A 115 VAC element that requires a full chamber, correct grounding, upstream GFCI/ELCI protection, and a verified Paneltronics circuit.
- A system that still requires the D5S pump for hot water at the Duda plate.

The Isotemp is therefore a comfort and control component, not a required component for basic diesel cabin heat and flow-through hot water. Selling the Isotemp would simplify the loop and remove the element hazard, but would surrender the buffer and AC-preheat options. The Duda plate and AM100-1LF would remain necessary for hot water.

Recommendation

Keep the Isotemp through initial commissioning if mounting and AC safety work remain reasonable. Do not treat the Isotemp as justified by hot-water availability alone. After commissioning, keep the Isotemp only if measured operation demonstrates at least one of these outcomes:

- fewer D5S starts per hour or materially longer pause periods during normal low-load heating;
- less cabin-temperature swing without unacceptable fan or control changes;
- a useful reduction in diesel run time after a permitted AC preheat;
- a repeatable short hot-water benefit that matters in the intended daily routine.

Sell the Isotemp if the measured system shows no meaningful cycling improvement, if the element cannot be given a properly protected AC circuit, or if the added mounting, fill, freeze protection, and inspection burden is not worth the modest holdover benefit. The current evidence justifies testing the Isotemp, but it does not justify claiming a guaranteed fuel saving, a guaranteed cycle reduction, or hot water while the D5S pump is stopped.

Decision record

Update this report after commissioning with measured values. A short operating copy of the same checklist lives under User's Guide — Occasional checks.

AFTER	RECORD
First diesel fire (mild night)	D5S starts per hour; average burner-on and pause duration
Same night, chamber participating	Chamber and loop temperatures before/after a short hot-water draw
Dish rinse (warm loop / cold loop)	Flow rate, mixed outlet temperature, usable draw time
HOLD 9 AC charge	Preheat time to ~75°C; inverter draw; battery energy used
Ongoing	Leaks, air problems, or maintenance added by the Isotemp

TL;DR

- **Why a buffer helps:** D5S Low is **2.1 kW**. A well-insulated van often needs **less than 2 kW** in mild/moderate cold, so Low has surplus heat to store.
- **Predicted starts/hour** (10°C band). Ceiling = chamber fully coupled ($f = 1$). Planning = about **40%** of the chamber participates ($f \approx 0.4$) – coil UA, stratification, and Isotemp-last loop order:

LOAD	ROUGH OUTDOOR (CABIN 70°F)	WITHOUT	WITH ($f = 1$)	WITH ($f \approx 0.4$)
1.0 kW	~40°F (better insulation)	**9.1**	**2.1**	**~4.1**
1.5 kW	~40–20°F	**7.7**	**1.8**	**~3.4**
2.0 kW	~20°F (or leaky near 40°F)	**1.7**	**0.4**	**~0.8**

- **Storage scale:** 16 L chamber, ~10–40°C swing → about **0.18–0.71 kWh** (roughly **5–20 min** at 2.1 kW), not hours.
- **AC / solar and dishes:** Charge the chamber first. That often takes about **40–60 minutes** to reach ~**75°C**; the taps stay cold during the charge. Then turn the master **On**, make a short EasyStart start, and let the **pump** run.
  - If the glycol loop is still warm, the flame often stays off. Hot water at the sink usually arrives in about **30–60 seconds**. You get about **2–5 minutes** of rinse water, then the flame lights or the water goes lukewarm.
  - If the glycol loop is cold, the flame comes on. Hot water at the sink usually arrives in about **2–5 minutes**. The chamber helps the diesel finish faster; it does not skip the flame.
  - This is a dish rinse, not a shower.
- **Decision:** keep if logs show fewer starts / steadier cabin / useful AC preheat for short draws; sell if those numbers do not show up.

Decision Log

DEC-000 — Heater identity confirmed: Hydronic D5S diesel heater

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The heater is an Eberspächer Hydronic D5S diesel heater, 12 V, order no. 25 2526 05 00 00 (per owner's original order confirmation), purchased with the Installation Kit for Hydronic II D4S/D5S and the EasyStart Timer controller. All Hydronic II C and D5WS branches in the design docs are no longer applicable — use only SRC-009 (and other Hydronic D5S diesel heater sources) for specs, wiring, fuel, exhaust, and coolant rules.
- **Reason:** Owner confirmed from original purchase record.
- **Alternatives considered:** Hydronic II C D5S (SRC-001, ruled out); Hydronic D5WS (SRC-010/SRC-011, ruled out).
- **Supporting sources:** SRC-009; SRC-023 (order line 252526050000); nameplate photo `20191117_154203.jpg`
- **Affected documents:** `01-system-requirements.md`, `02-system-architecture.md`, `03-parts-inventory.md`, `04-bom-and-missing-parts.md`, `05-mechanical-layout.md`, `06-coolant-loop.md`, `07-fuel-system.md`, `08-exhaust-and-combustion-air.md`, `09-electrical-and-controls.md`, `open-questions.md`, `research/verified-requirements.md`

DEC-001 — Isotemp Slim Square 4.2 gal heat battery replaces Bosch buffer

- **Date:** 2026-09-19
- **Status:** Accepted

- **Decision:** The Isotemp Slim Square 4.2 gal heat battery (16 L), 115 VAC / 750 W (Indel P/N 601623Q000003 / 601623QX00003) replaces the Bosch electric glycol buffer as **thermal mass** (DEC-007). It is **not** the domestic hot-water heat exchanger. Bosch is gone.
- **Reason:** Owner substitution. DHW heat transfer is the Duda B3-12DW-20 plate heat exchanger (DEC-003). The factory mixing valve on the Isotemp Slim Square 4.2 gal heat battery is **not** the DHW mixer (DEC-008).
- **Alternatives considered:** Bosch electric glycol buffer (SRC-004, withdrawn); SureCal calorifier (SRC-016, not selected); coil in the Isotemp Slim Square 4.2 gal heat battery as the DHW exchanger (not selected).
- **Supporting sources:** SRC-021, SRC-022
- **Affected documents:** `01-system-requirements.md`, `02-system-architecture.md`, `03-parts-inventory.md`, `04-bom-and-missing-parts.md`, `06-coolant-loop.md`, `09-electrical-and-controls.md`, `open-questions.md`, `research/verified-requirements.md`, `hazards.md`

DEC-002 — Cabin control is EasyStart Timer only

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** Cabin control is the Eberspächer EasyStart Timer (SRC-003), not the Heatmiser DT-B/PRT-B option.
- **Reason:** Owner choice, matching the original order (SRC-023 configured EasyStart Timer).
- **Alternatives considered:** Heatmiser DT-B/PRT-B + blower (SRC-017, not selected); both (not selected).
- **Supporting sources:** SRC-003; SRC-023
- **Affected documents:** `09-electrical-and-controls.md`, `open-questions.md`

DEC-003 — Duda B3-12DW-20 plate heat exchanger is the freshwater heat exchanger

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The Duda B3-12DW-20 plate heat exchanger is the glycol-to-**fresh water** heat exchanger. Domestic hot water is heated there, not in the coil in the Isotemp Slim Square 4.2 gal heat battery. The Isotemp Slim Square 4.2 gal heat battery is the thermal-mass buffer (DEC-007). Bosch is gone.
- **Reason:** Owner-confirmed 2026-09-19.
- **Alternatives considered:** coil in the Isotemp Slim Square 4.2 gal heat battery as the DHW exchanger (not selected); dropping the Duda B3-12DW-20 plate heat exchanger (not selected).
- **Supporting sources:** Owner-confirmed 2026-09-19
- **Affected documents:** `01-system-requirements.md`, `02-system-architecture.md`, `03-parts-inventory.md`, `04-bom-and-missing-parts.md`, `06-coolant-loop.md`, `open-questions.md`, `research/verified-requirements.md`, `hazards.md`

DEC-004 — 115 VAC element on the Isotemp Slim Square 4.2 gal heat battery feeds from the Paneltronics AC panel

- **Date:** 2026-09-19
- **Status:** Accepted; final Paneltronics breaker/GFCI details OPEN
- **Decision:** The Isotemp Slim Square 4.2 gal heat battery's 750 W / 115 VAC element is an active AC load in this build. Its supply lands on the vehicle's Paneltronics AC panel, not on the 12 V distribution panel. Confirm the exact Paneltronics source, breaker, GFCI/ground-fault protection, grounding/bonding, switching, and cable routing before energizing it.
- **Reason:** Owner-confirmed 2026-09-19: 115 V goes to the Paneltronics panel.
- **Alternatives considered:** Landing the element on a 12 V panel (incorrect); leaving the element permanently unwired (superseded).
- **Supporting sources:** Owner-confirmed 2026-09-19
- **Affected documents:** `09-electrical-and-controls.md`, `open-questions.md`

DEC-005 — 5 L expansion header

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The independent glycol loop's expansion vessel is a **5 L expansion header**. The jug on hand is Water Flow Tank brand, heatso MPN 65288A, the same duty as Eberspächer / Webasto 29 2100 01 77 41. **Not yet mounted.** The top of the tank must be the **highest circulating glycol point**. **Owner (2026-09-20):** the expansion header mounts in the **sofa-bed backrest (high)**; the Isotemp Slim Square 4.2 gal heat battery stays **under the bench/bed (low)** — same furniture stack, different heights (prefer driver side with the Isotemp). Fill/bleed through a hinged or removable backrest panel. Confirm height after framing; raise the backrest cradle or fall back to a higher wall mount only if the backrest loses the height check. Planned elevation order of magnitude ~20–30 in above the van floor is a starting guess, not an as-built fact.
- **Reason:** Owner already purchased this part; owner confirmed high-mount rule 2026-09-20.
- **Alternatives considered:** None evaluated for the part itself; location options remain open pending height + fill/bleed access.
- **Supporting sources:** heatso.com product page (retrieved 2026-09-19); owner layout discovery 2026-09-20
- **Affected documents:** `05-mechanical-layout.md`, `06-coolant-loop.md`, `04-bom-and-missing-parts.md`, `open-questions.md`, `house-electrical-understanding.md`

DEC-006 — High-elevation use is a design requirement

- **Date:** 2026-09-19 (timing note 2026-09-20)
- **Status:** Accepted
- **Decision:** The van will be used at high elevation early and often. Sustained heating above ~1500 m is in scope. Hydronic D5S diesel heater therefore requires the air-pressure / Altitude kit 22 1000 33 22 00 (also written 22.1000.33.2200) per SRC-009 p. 32. Short mountain-pass use without the kit is not the operating profile. **Install and wire the kit at HOLD 4** while the EasyStart harness is open (Q-015). A low-elevation first fire can run without it, but the kit is **not** on the post–first-fire wait list with Q-022 / Q-009.
- **Reason:** Owner-confirmed 2026-09-19; harness-open install preferred 2026-09-20; first high camp timing **ASAP** (owner 2026-09-20).
- **Alternatives considered:** Sea-level / below-1500 m use only (not selected); defer kit until after first fire (rejected as preferred practice).
- **Supporting sources:** Owner; SRC-009 p. 32; SRC-002
- **Affected documents:** `01-system-requirements.md`, `03-parts-inventory.md`, `04-bom-and-missing-parts.md`, `09-electrical-and-controls.md`, `open-questions.md`, `research/verified-requirements.md`, `hazards.md`, `book/install-in-this-order.md`, `book/front-matter.md`

DEC-007 — Isotemp Slim Square 4.2 gal heat battery is the thermal mass, glycol on both sides

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The Isotemp Slim Square 4.2 gal heat battery's ~4.2 gal / 16 L main chamber holds **static glycol/coolant mix**, not potable water — the tank never touches potable water at all (DHW is entirely the Duda B3-12DW-20 plate heat exchanger, DEC-003). The coil carries the actively **circulating** glycol from the Hydronic D5S diesel heater loop. This is the Isotemp Slim Square 4.2 gal heat battery used mostly as designed (static reservoir + coil heat exchanger), just with glycol instead of water on the static side, since there's no DHW role left for this tank to protect. Heat crosses the coil into the static glycol while the heater fires, and back out of it into the circulating glycol when the Hydronic D5S diesel heater is in pause and the pump keeps running (SRC-009 p. 31–32). The 115 VAC / 750 W element is a Paneltronics AC-panel load (DEC-004 / DEC-013); it heats the static chamber only and does not pump the loop — circulating glycol still runs through the coil via the D5S pump.
- **Reason:** Owner-stated intent 2026-09-19: keep the Isotemp Slim Square 4.2 gal heat battery mostly as designed (static reservoir + coil), but glycol on both sides avoids freeze risk from static potable water sitting unused, and matches that the tank has no DHW role left to protect. Hydronic D5S diesel heater holds only ~0.18 L in the heater (SRC-009 p. 13). Staging temperatures from SRC-009 p. 31: Power → High ~65 °C; High → Low ~80 °C; Low → Off/pause ~85 °C; restart from pause ~75 °C. Overheat restart requires coolant < 70 °C (SRC-009 p. 32). Q-020 resolved.

- **Alternatives considered:** Bosch glycol-volume buffer (withdrawn, DEC-001); Isotemp Slim Square 4.2 gal heat battery as DHW calorifier with potable water in the tank (not selected); filling the main chamber with actively circulating glycol instead of a static reservoir (considered, rejected — no efficiency gain over the coil design, and requires re-plumbing the tank's ports for full loop flow/pressure).
- **Supporting sources:** Owner; SRC-009 p. 13, 31–32; SRC-022
- **Affected documents:** `01-system-requirements.md`, `02-system-architecture.md`, `06-coolant-loop.md`, `open-questions.md`, `research/verified-requirements.md`, `hazards.md`

DEC-008 — DHW mixing valve is a separate unit, not the Isotemp Slim Square 4.2 gal heat battery factory mixer

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The freshwater / DHW mixing valve is a **AM100-1LF thermostatic mixing valve**, 1/2" NPT lead-free thermostatic mixing valve, factory range 70–145°F, per owner's supplyhouse.com order confirmation. Also ordered in that shipment: two 3/4" FIP x FIP brass elbows (BRE075) and three 3/4" hose-barb x 3/4" male brass pipe adapters (G25121) for the transitions to hose. This is a **different valve** from the Isotemp Slim Square 4.2 gal heat battery factory mixer (SRC-022, 38–65°C) — do not treat that factory mixer as the tap-water mixer. **Owner confirmed 2026-09-20: Isotemp factory mixer unused forever** — leave capped; omit from User's Guide operating steps.
- **Reason:** Owner-confirmed 2026-09-19 via order confirmation email/photo. Van photo `photos/IMG_20200706_184257_691.jpg` (SRC-024) shows that valve already on the Duda B3-12DW-20 plate heat exchanger freshwater hot outlet.
- **Alternatives considered:** factory mixing valve on the Isotemp Slim Square 4.2 gal heat battery as the DHW mixer (not selected — and with DEC-007 that valve has no potable flow anyway, so it does no work on the Isotemp Slim Square 4.2 gal heat battery).
- **Supporting sources:** Owner supplyhouse.com order confirmation, 2026-09-19; SRC-024; SRC-025; SRC-026
- **Affected documents:** `02-system-architecture.md`, `03-parts-inventory.md`, `04-bom-and-missing-parts.md`, `06-coolant-loop.md`, `open-questions.md`, `hazards.md`, `10-installation-procedure.md`

DEC-009 — Independent living-space glycol loop only (no engine connection)

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The hydronic system is an **independent living-space glycol loop**. There is **no connection** to the vehicle engine cooling system — no hose splice, no combination valve, no engine takeoff.
- **Reason:** Owner-confirmed 2026-09-19: "no engine connection."
- **Alternatives considered:** Engine-coolant integration with inline / thermostat / 5- or 6-port combination valve (SRC-009 p. 20–23; SRC-010; SRC-008) — not selected.
- **Supporting sources:** Owner 2026-09-19
- **Affected documents:** `01-system-requirements.md`, `02-system-architecture.md`, `04-bom-and-missing-parts.md`, `06-coolant-loop.md`, `open-questions.md`, `research/verified-requirements.md`, `book/front-matter.md`

DEC-010 — Glycol loop flow path locked: pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger → Isotemp Slim Square 4.2 gal heat battery → return

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The glycol loop is pure series (no manifold, no summer valve): the Hydronic D5S diesel heater's own circulating pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate heat exchanger (glycol side) → coil in the Isotemp Slim Square 4.2 gal heat battery → return to the pump, with the 5 L expansion header teed in at the high point on the return.

- **Reason:** Owner priority: cabin heat ("hot air") matters more day-to-day than DHW, so the Sure Marine cabin heater gets first crack at heater output. DHW (the Duda B3-12DW-20 plate heat exchanger) is placed ahead of the coil in the Isotemp Slim Square 4.2 gal heat battery so a shower always draws the hottest available fluid rather than whatever's left after the Isotemp Slim Square 4.2 gal heat battery charges — that tank's job is just storage, so it's fine to get leftover heat. Pure series (not a parallel manifold, unlike the Sure Marine cabin heater/Webasto SRC-018 reference design) was chosen because installation simplicity matters a lot to the owner; a manifold+summer-valve buys marginal thermal evenness at the cost of more fittings/joints.
- **Alternatives considered:** Sure Marine cabin heater → Isotemp Slim Square 4.2 gal heat battery → Duda B3-12DW-20 plate heat exchanger (rejected — risks the Isotemp Slim Square 4.2 gal heat battery taking a bite of heat before DHW while the Isotemp Slim Square 4.2 gal heat battery is charging); parallel manifold + summer valve per SRC-018 Webasto practice (rejected for install complexity); electrical-only mode switching instead of a plumbing summer valve (adopted — see book "what each piece does").
- **Supporting sources:** Owner 2026-09-19; SRC-018 (referenced as the alternative manifold design, not followed)
- **Affected documents:** `06-coolant-loop.md`, `open-questions.md`, `book/install-in-this-order.md`

DEC-012 — Master switch: single Off/On hard lockout, underneath the thermostat

- **Date:** 2026-09-19
- **Status:** Accepted (revised 2026-09-19 — superseded by DEC-014 on fan auto-behavior)
- **Decision:** One master 12V switch, two positions (Off/On), gates power to the EasyStart Timer, the SC1600B thermostat circuit, and the Sure Marine cabin heater circuit together. Off is a physical lockout (parked — nothing can start, regardless of cabin temperature or thermostat state). On just makes power available: within that, the SC1600B thermostat automatically calls the D5S and the fan together based on cabin temperature (DEC-014) — there is no separate manual fan on/off; the fan's own dial is **Low/Medium/High only** (no Off position) and purely sets speed once the thermostat has called for it.
- **Reason:** Owner wants cabin temperature to auto-trigger heat (DEC-014), while keeping one unambiguous hard-off for when the van is parked and unattended. A three-position dial with no Off position mirrors a car AC blower: speed is a separate, orthogonal control from whether the system is calling for heat at all.
- **Alternatives considered:** Fully manual fan (no thermostat tie) — this was the original decision, superseded 2026-09-19 once the owner clarified the SC1600B should auto-trigger the fan; a separate manual fan enable/disable switch in addition to the thermostat — rejected as redundant once the owner confirmed the thermostat alone (plus the master switch) fully covers on/off.
- **Supporting sources:** Owner 2026-09-19
- **Affected documents:** `09-electrical-and-controls.md`, `open-questions.md`, `book/install-in-this-order.md`

DEC-013 — No separate auxiliary glycol pump; Isotemp element remains a Paneltronics AC load

- **Date:** 2026-09-19
- **Status:** Accepted as withdrawal (revised 2026-09-19)
- **Decision:** There is no separate auxiliary 12 V glycol pump and no separate solar/electric circulation path. The Isotemp Slim Square 4.2 gal heat battery remains thermal mass: circulating glycol runs through the coil and static glycol sits in the chamber. The 750 W / 115 VAC element is retained as an AC heat source and is wired to the Paneltronics AC panel under DEC-004. It does not create a separate flowing glycol loop or replace the D5S's circulating pump.
- **Reason:** The extra pump, relay, and separate control path were not worth adding. The existing Paneltronics AC architecture supplies the element without adding a second glycol circulation circuit; its useful output remains limited by the static chamber's stored heat.
- **Alternatives considered:** Add a separate auxiliary 12 V glycol pump (rejected); leave the existing 115 VAC element permanently unwired (superseded); use the element as a substitute for the D5S loop pump (not selected).
- **Supporting sources:** Owner / design simplification 2026-09-19; earlier Sure Marine CM10P7 listing retained only as history
- **Affected documents:** `06-coolant-loop.md`, `09-electrical-and-controls.md`, `book/front-matter.md`, `book/install-in-this-order.md`, `book/03-living-with-it.md`, `book/01-what-this-system-does.md`

DEC-014 — SC1600B is a cabin dry-contact call; master remains the hard off

- **Date:** 2026-09-19 (corrected 2026-09-20)
- **Status:** Accepted — **heater wake landing still OPEN (Q-022)**
- **Decision:** The SC1600B (Sure Marine 531-SO193701) is a space-heating control. It runs on AA batteries. Terminals R–W are a dry contact that closes below cabin setpoint. That contact will eventually (a) drive a **relay** so cabin fans only run when heat is called, and (b) wake the Hydronic — but **not** by landing on EasyStart Timer pins 9–10. Those pins are for Eberspächer's optional temperature **sensor** (SRC-003), not a dry-contact thermostat. Approved wake landing stays under Q-022. Fan dial remains Low/Med/High speed only (DEC-015 / DEC-018). Master Off/On (DEC-012) still removes 12 V from EasyStart, the thermostat signal path, and the fans together.
- **Reason:** Owner wants cabin temperature to auto-call heat and fans, with one unambiguous hard off when parked. Sure Marine craft (SRC-019) treats these thermostats as low-current signals that need a relay for fans. EasyStart's published Hydronic diagram does not show a dry-contact thermostat input on pins 9–10.
- **Alternatives considered:** Wiring SC1600 into EasyStart pins 9–10 as if it were the factory room sensor (rejected — wrong electrical type); claiming auto complete before Q-022 (rejected).
- **Supporting sources:** Owner 2026-09-19; SRC-003; SRC-019; SRC-033; Q-022
- **Affected documents:** `09-electrical-and-controls.md`, `book/01-what-this-system-does.md`, `book/03-living-with-it.md`, `open-questions.md`

DEC-015 — Sure Marine cabin heater: two Noctua NF-F12 fans, one inline 12V speed controller, PWM/tach unused

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** Both stock fan outlets on the Sure Marine cabin heater are replaced with **Noctua NF-F12 PWM** fans (owner has two on hand). They are wired in parallel to a single inline 12V DC fan-speed controller (rotary knob or multi-position switch) on the shared positive lead. The fans' PWM and tach wires are not used — only the two power leads. One shared control drives both fans together at multiple speed levels; they are not individually addressable. This is the same "fan dial" referenced in DEC-012/014: on/off is the thermostat's job, this dial is speed only (Low/Medium/High) — confirm the chosen controller's low end doesn't itself cut the fan fully off (Q-021).
- **Reason:** Owner wants multiple speed levels (like a car dash fan control) without per-fan complexity. A simple inline 12V speed controller is the standard, idiomatic way to vary speed for parallel-wired 12V fans without needing true PWM signal generation/wiring — fewer wires, fewer failure points, still gives continuously variable speed. Both fans draw ~0.14 A each at 12V (~0.28 A combined), trivial for any basic inline controller.
- **Alternatives considered:** True PWM signal from a PWM controller to both fans' 4th pin (rejected — more wiring, no real benefit at this scale); independent per-fan switches (rejected — owner does not need or want individual control).
- **Supporting sources:** Owner 2026-09-19; SRC-005 (blog previously replaced only one of two factory outlets with a single Noctua; this build replaces both)
- **Affected documents:** `09-electrical-and-controls.md`, `open-questions.md`

DEC-016 — Template

- **Date:** YYYY-MM-DD
- **Status:** Proposed
- **Decision:**
- **Reason:**
- **Alternatives considered:**
- **Supporting sources:**
- **Affected documents:**

DEC-017 — Circulating fill brand: Prestone LowTox AF555 only

- **Date:** 2026-09-19

- **Status:** Accepted
- **Decision:** Fill the circulating loop and the Isotemp chamber with **one brand only**: Prestone LowTox AF555 concentrate diluted to **≤50%** propylene glycol by volume (equal parts concentrate + water is the maximum). Cap the Star brite Bio-Safe jug and do not pour it into this system.
- **Reason:** Both on-hand products are concentrates (SRC-029 / SRC-030). Inhibitor packages differ; no manufacturer mix approval. Locking one brand removes a fill-day decision. Prestone LowTox is the documented automotive PG concentrate already matched to SRC-009's ≤50% rule.
- **Alternatives considered:** Star brite Bio-Safe only (acceptable if Prestone were unavailable — not chosen); mixing brands (rejected).
- **Supporting sources:** SRC-009, SRC-029, SRC-030, SRC-027; Q-019
- **Affected documents:** `book/install-in-this-order.md`, `book/05-the-numbers.md`, `open-questions.md`

DEC-018 — Fan dial: voltage Low/Med/High without hard Off; NA-FC1 not used

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** The Sure Marine cabin-heater speed control is a **12 V voltage-style** Low/Med/High dial (or 3-speed switch) on the shared positive lead per DEC-015. Prefer a switch **without** a hard Off detent (SRC-032). **Do not** use the Noctua NA-FC1 (or any PWM-only controller) on this build unless DEC-015 is formally revised — NA-FC1 can command 0% duty and some fans stop below ~20% (SRC-034). At install, meter that the lowest detent still spins both fans (HOLD 4).
- **Reason:** Closes the Q-021 product fork before parts shopping. Matches power-leads-only wiring already locked.
- **Alternatives considered:** NA-FC1 with "no stop" mode (rejected — wrong signal path for DEC-015); switch with Off (rejected as default — Off belongs to master/thermostat).
- **Supporting sources:** SRC-032, SRC-034; DEC-015; Q-021
- **Affected documents:** `book/install-in-this-order.md`, `book/03-living-with-it.md`, `open-questions.md`

DEC-019 — Expansion-header relief needs a safe catch path

- **Date:** 2026-09-19
- **Status:** Accepted
- **Decision:** When mounting the 5 L expansion header, arrange the ~1.2 bar cap so any relief discharge of hot propylene glycol goes into a **catch bottle**, drip pan, or hose to a safe exterior/garage floor path — never onto bedding, wiring, or living-space fabric. Exact bottle/hose hardware is chosen at the rear-bench mount (Q-024 hardware detail).
- **Reason:** Relief fluid is hot and pressurized. Leaving discharge "wherever" fails HAZ-003 / common sense.
- **Alternatives considered:** Free drip over the bench (rejected).
- **Supporting sources:** SRC-009 (heater ≤2.5 bar; expansion header ~1.2 bar); SRC-019 (header as burp/expansion point); Q-024
- **Affected documents:** `book/install-in-this-order.md`, `open-questions.md`, `hazards.md`

DEC-020 — Paneltronics 9972313B / face 72313: WATER HEATER branch; GFCI not on the panel

- **Date:** 2026-09-19
- **Status:** Accepted (face photo confirmed; GFCI/route still open for HOLD 9)
- **Decision:** The installed Paneltronics AC panel matches wiring sheet **9972313B** (face mark **72313**): **MAIN 30 A** + labeled branches **BATTERY CHARGER**, **WATER HEATER**, **OUTLETS**, reverse-polarity LED. Land the Isotemp 750 W element (~6.5 A) on **WATER HEATER** only. This panel does **not** include GFCI/ELCI — provide ground-fault protection upstream (shore/inverter ELCI or a GFCI device) before HOLD 9. The 12 V DC panel must **never** carry the 750 W load (DEC-004).
- **Reason:** On-van face photo 2026-09-19 closes panel identity and names the element branch.
- **Alternatives considered:** Assuming GFCI is built into the panel (rejected); feeding element from 12 V (rejected — DEC-004); inventing a free unlabeled branch (rejected — WATER HEATER is labeled).
- **Supporting sources:** Face photo; SRC-038, SRC-035; DEC-004; Q-009 / Q-016
- **Affected documents:** `book/install-in-this-order.md`, `book/02-where-things-are.md`, `open-questions.md`

DEC-021 — Sofa/bed: EZTube frame, Rostek corners, sapele deck

- **Date:** 2026-09-20
- **Status:** Accepted
- **Decision:** Rebuild the sofa/bed as a **hybrid**: an aluminum **seat-base frame** first, a short **aluminum** Isotemp cradle/restraint bolted to floor structure, and a sapele deck after the footprint is measured. The seat frame is EZTube 1 in square tube (0.050 in wall, 6063-T5) with Rostek 3-way corners. Both are accepted. The deck and the day panels are Chesapeake Light Craft premium sapele marine plywood. The frame bridges the wheel well and lands on the floor. The well sheet metal and the freshwater tank do not carry the seat. Backrest shape, cushions, and the sleeper mechanism stay out of scope until the frame footprint is measured. Provisional length about **75 in** and a recalled **21 in** height are placeholders, not cuts. The van is the **regular 170 body, not the 170 EXT**. The passenger well holds the freshwater tank; the driver bay holds the Isotemp. By day (owner intent): **outboard panels become seat backs**, **inboard panels become seat butts**. Do **not** build the whole furniture in T-slot extrusion; do **not** rely on all-wood alone for Isotemp crash restraint. Updated 2026-10-17.
- **Reason:** Weighted against easy-to-work (**35%**), strength (**25%**), light (**20%**), cost (**20%**) — hybrid scored **4.55 / 5** vs all-wood 4.15, all-extrusion 3.30, steel 2.75. Matches owner priority (access + construction ease). Owner revised day motion away from a carry-out full-length lift-out toward stay-attached panels that form seat backs and butts (2026-09-20). Field check: full aluminum bed frames save only ~7 lb over wood at ~3× frame cost; plywood deck dominates mass either way.
- **Alternatives considered:** All wood (rejected for tank restraint / revise); all 80/20 furniture (rejected — cost, rattle, slow first build); steel tube (rejected — weight / revise); Baltic birch everywhere (optional for visible faces only, not required as structure); one full-length lift-out + rear stash pocket (superseded by multi-panel seat-back / seat-butt geometry).
- **Supporting sources:** Owner priorities 2026-09-20; FarOutRide bed-platform cost/weight comparison; Working design understanding
- **Affected documents:** `book/06-the-sofa.md`; `book/install-in-this-order.md`; `book/01-what-this-system-does.md`; `book/research/house-electrical-understanding.md`

Open Questions

Questions for you, in order

Answer what is still open. Skip the rest. Manuals win on how; these answers only place parts and set preferences.

Already locked (do not re-ask) — see also Locked facts checklist:

- Altitude kit P/N **22 1000 33 22 00** + heater **H-Kit** — confirmed; kit is in the garage → land at HOLD 4 with the EasyStart harness.
- Paneltronics **WATER HEATER** — empty; reserved for Isotemp 750 W; factory Isotemp mixer unused forever.
- Master switch — in the garage; mount next to EasyStart at HOLD 4 (does not cut Isotemp AC).
- Locations: Sure Marine **under fridge**; Duda **under sink**; expansion header → **backrest (high)**; Isotemp → **under bench (low)**; day sofa **outboard → seat back** / **inboard → seat butt**.
- AC: two **Blue Sea 9009** rotaries (not automatic); SmartPlug + DUOSIDA J1772; Mean Well not via Paneltronics BATTERY CHARGER.
- 12 V: blade block may need a **second breakout** for hydronic master / EasyStart / fans.

May wait after a low-elevation diesel first fire (only these):

- Q-022 — SC1600B auto call (use EasyStart until closed)
- Q-009 / HOLD 9 — Isotemp AC path (required for complete system acceptance)

The altitude kit is **not** on that wait list.

Still useful:

1. **5 L expansion header** — **In sofa-bed backrest (high)**; Isotemp under bench/bed (low). Confirm the top is the highest glycol point after framing (Working design understanding).

2. **Day sofa** — Night flat. Day intent: **outboard → seat back**, **inboard → seat butt** + aisle. Panel count / hardware OPEN. See Working design understanding / DEC-021.
3. **SC1600 fans** — Reference relay: TE **V23134-A3052-X540** (or any ISO mini 12 V / ≥10 A) — DigiKey. Match on shelf or buy.
4. **Cabin fans** — Both Noctua NF-F12; speed = voltage Low/Med/High dial (not NA-FC1). Meter Low still spins at HOLD 4.
5. **Shore / inverter ground fault** — Inverter *might*; shore unknown; Paneltronics has no GFCI. Leave for HOLD 9.
6. **House electrical snapshot** — **200 Ah** / **650 W** solar / Mean Well RPB-1600-12 / **Sterling 60 A** B2B / PROwatt SW 2000 / two **9009** rotaries. Blade fuse block may need a **second breakout** for hydronic. Photo extracts: photo-extract-electrical. Still compare to SRC-040 (Q-025).
7. **Mountains** — First high camp **ASAP →** altitude kit at HOLD 4 is urgent.
8. **Hot water** — ~80% sinks; showers minority — mention showers, don't center on them.
9. **Envelope** — Walls/ceiling already **Rattle Trap** + Thinsulate-class insulation behind grey panels / L-track (photo-extract-insulation) — plan HOLD S hatches through that stack.

Live layout/sofa discovery: Working design understanding. Photo knowledge dumps: electrical · heater · insulation.

ID	QUESTION	WHY IT MATTERS	SOURCE/OWNER	BLOCKS WORK?	RESOLUTION
Q-001	What heater is actually on this vehicle?	Heat flow, min coolant flow, voltage limits, wiring, and fault codes differ.	SRC-023; nameplate photo	No	**Resolved 2026-09-19 (DEC-000):** Eberspächer Hydronic D5S diesel heater, order no. **25 2526 05 00 00** (invoice SKU 252526050000), with Installation Kit for D4S/D5S and EasyStart Timer. Nameplate photo reads Hydronic II / D5S / 25 2526 / 5.2 kW / 12 V diesel / max 2.5 bar. Use **SRC-009** only.
Q-002	Independent living-space hydronic loop or splice into engine coolant?	Architecture, BOM, and combination-valve use all change.	Owner	No	**Resolved 2026-09-19 (DEC-009):** Independent living-space glycol loop only. **No engine connection.**
Q-003	Domestic hot water path	DHW plumbing, mixing valve, PRV	Owner	No	**Resolved 2026-09-19 (DEC-003 / DEC-008 / SRC-024 / SRC-026):** Duda B3-12DW-20 plate + AM100-1LF. Isotemp is static glycol mass only (DEC-007).
Q-004	Cabin control: EasyStart Timer only, Heatmiser, or both?	Harness, relays, and thermostat location.	Owner / parts on hand	Partial	**Resolved 2026-09-19 (DEC-002):** EasyStart Timer only (as primary). SC1600 is optional auto call once Q-022 closes.
Q-005	Vehicle platform and year?	OEM sheets apply only to those factory layouts.	Owner	Partial	**Resolved 2026-09-19, confirmed 2026-10-17:** 2017 Mercedes Sprinter 4x4, **regular 170** body, not the 170 EXT. Same wheelbase as the EXT; shorter cargo floor. Inches: Cargo floor.
Q-007	Confirm 6-port combination valve part stamp.	SRC-008 is a dealer listing.	Physical part	No	**N/A (DEC-009):** no engine splice; combination valve not in this build.
Q-008	Is the Duda plate still in the loop, or does the Isotemp coil replace it for DHW?	Two DHW exchangers vs omitting the plate.	Owner	No	**Resolved 2026-09-19 (DEC-003 / DEC-007):** Duda is the freshwater exchanger; Isotemp chamber is static glycol mass.

ID	QUESTION	WHY IT MATTERS	SOURCE/OWNER	BLOCKS WORK?	RESOLUTION
Q-009	Final Paneltronics breaker, GFCI/grounding, cable route, and element control?	750 W ≈ 6.5 A at 115 V; never on empty chamber.	Owner / SRC-038 / SRC-040 / `photos/ac-electrical.jpg`	Yes (HOLD 9 only)	**2026-09-20:** Inverter *might* cut on ground fault (unverified). Shore side: **owner does not know.** Paneltronics has **no** GFCI (owner doubt matches DEC-020). HOLD 9: verify inverter hardwire, decide shore story, cable/first energize.
Q-010	Is the fuel tank tap-in installed correctly?	Fuel leak / fire risk before further work.	Owner / physical inspection	Yes	**2026-09-20:** Tap OK; hose likely run; **needs priming**. **Fuel pump** under van: owner not yet confident – **future work** before first fire (HOLD 1 walk + HOLD 8 prime).
Q-011	Glycol loop plumbing order?	Hose runs, bleed points.	Owner	No	**Resolved 2026-09-19 (DEC-010):** pump → heater → Sure Marine cabin heater → Duda → Isotemp coil → return; expansion header teed on return high point.
Q-012	Exact heater mounting height vs expansion header and nearby lines?	Self-vent height; fuel-line clearance.	Owner / measurement	Partial	**Verify while mounting.** The expansion header goes in the **sofa-bed backrest (high)**; Isotemp under bench (low). Confirm the expansion header top is the highest glycol point after framing (DEC-005).
Q-013	Is heater 12 V wiring (EasyStart, pump, altitude) finished?	What's left before first fire.	Owner	Partial	**2026-09-19:** EasyStart **mounted** on wood panel; not fully wired. Altitude kit in garage. Pump/heater 12 V landing still open (HOLD 4).
Q-014	Is exhaust/combustion-air routing actually compliant?	HAZ-002 CO.	Owner / physical inspection	No	**2026-09-20:** Owner thinks exhaust is **OK**. Still a quick HOLD 2 look vs SRC-009 before fire – not expected to be a rebuild.
Q-015	Is altitude kit on the van and wired?	DEC-006 high camps; do at HOLD 4.	Owner / SRC-002 / SRC-039	Yes (high camps); preferred at HOLD 4	**Kit P/N and H-Kit confirmed: 22 1000 33 22 00 / H-Kit.**

ID	QUESTION	WHY IT MATTERS	SOURCE/OWNER	BLOCKS WORK?	RESOLUTION
					Location: garage. First high camp **ASAP** (owner 2026-09-20). Land at HOLD 4 while EasyStart harness is open. Still open: mount, wire per SRC-002, functional check.
Q-016	Does a 12 V panel position control the Paneltronics element circuit?	12 V must never feed 750 W.	Owner / DEC-020	Partial	**Raised 2026-09-19 (DEC-020):** panel is **AC-only**. Element → **WATER HEATER** branch. 12 V is never the load path.
Q-017	Is the Isotemp coil always in the circulating loop?	Buffer only works if coil sees flow.	Owner	No	**Resolved 2026-09-19 (DEC-010):** always in series; no summer bypass.
Q-018	What mixing valve is on the Duda hot outlet?	Scald control.	Owner / SRC-036	No	**Resolved 2026-09-19 (DEC-008 / SRC-024 / SRC-036):** AM100-1LF. Set ~120°F: loosen → lift handwheel → turn → lock; confirm at faucet; label.
Q-019	Glycol brand, concentrate vs premix, mix %?	SRC-009 ≤50%; no brand mix.	Owner / SDS	No	**Resolved 2026-09-19 (DEC-017):** Prestone LowTox AF555 concentrate only, ≤50%. Cap Star brite Bio-Safe for this loop.
Q-020	Hydronic D5S Power/High/Low staging temperatures?	Cycling estimates.	SRC-009 p. 31	No	**Resolved 2026-09-19:** Power→High ~65°C; High→Low ~80°C; pause ~85°C; restart ~75°C; overheat restart <70°C.
Q-021	Does the fan dial's lowest setting still spin (no hard Off)?	Owner's guide wording; DEC-015.	DEC-018	No	**Resolved 2026-09-19 (DEC-018):** voltage Low/Med/High **without** hard Off; NA-FC1 rejected. HOLD 4 meters that Low still spins.
Q-022	How does SC1600B wake the heater and enable fans?	Automatic cabin heat.	SRC-019 / SRC-033 / SRC-003	No	**2026-09-20:** Intent locked (DEC-014) – dry contact → fan relay + heater wake. **Landing open.** Do **not** use EasyStart pins 9–10 (those are for the factory temp sensor). Fans need a relay.

ID	QUESTION	WHY IT MATTERS	SOURCE/OWNER	BLOCKS WORK?	RESOLUTION
					Until tested: EasyStart Timer only.
Q-023	What controls the withdrawn auxiliary glycol-pump circuit?	Must not reappear accidentally.	Owner	No	**N/A 2026-09-19 (DEC-013):** no auxiliary glycol pump.
Q-024	Does the expansion header ~1.2 bar relief need a drip pan / routed drain?	Hot glycol under pressure.	DEC-019	No	**Resolved 2026-09-19 (DEC-019):** yes – catch bottle, drip pan, or hose to a safe path. Free drip over living space forbidden.
Q-025	Does the installed house electrical still match SRC-040?	Isotemp AC load planning; inverter vs shore behavior.	Owner / SRC-040 / SRC-041 / `photos/ac-electrical.jpg` / `photos/Tyrah portfolio_electrical/`	Partial (HOLD 9)	**2026-09-20:** Two **Blue Sea 9009** rotaries (as-built): Paneltronics (shore ↔ PROwatt SW 2000; poorly labelled) and Mean Well **RPB-1600-12** (shore ↔ J1772). Inlets: **SmartPlug** + **DUOSIDA 32 A J1772**. House: **200 Ah** / **650 W** / **Sterling 60 A** / Victron **100|50** (photos). Blade block may need **second breakout** for hydronic. Confirm IOTA DLS-55 removed. HOLD 9 ground-fault open. Extract: photo-extract-electrical.

Working design understanding

Scratchpad for Ben's questions. Not the Builder's Guide. Promote locked answers into part one / HOLD S when ready.

Hydronic mental model (promoted to Controls). Two worlds that never share a fuse panel: **12 V** diesel heat + fans (master Off lockout) vs **120 V** Isotemp chamber on Paneltronics **WATER HEATER**. Two **Blue Sea 9009** rotaries: one is Isotemp heat (shore/inverter → Paneltronics); the other is battery charge only (shore/J1772 → Mean Well).

Input from Ben

- Doesn't the Xantrex we own have GFCI on the outlets already? Doesn't that help answer some outstanding questions?
- Comparison table of sofa materials / build options.
- Frame mechanisms; Tyrah wants a sofa bed whose middle opens by day (hinge/slide panels).
- Layout: stove/oven at passenger door; 30+ gal water tank over passenger rear wheel well.
- Walls are plywood panels (not pine slats); need access behind them.
- Do we need to rip out the bed?
- Wiring: Paneltronics → Isotemp is obvious—don't complicate it. Don't debate GFCI/inverter feed validity here.
- **Does not love acronyms.** Prefer full names. The glycol high-point tank is the **5 L expansion header**. Water Flow Tank is only the brand on the jug (MPN 65288A).

House AC (owner – `photos/ac-electrical.jpg`)

Authority for this van's AC topology: **Van AC Electrical Diagram** (`photos/ac-electrical.jpg`).

```
J1772 240 V (L1/L2) ──2-pole breaker──┐
                                      ├──→ Blue Sea 9009 rotary ──→ Mean Well RPB-1600-12
120 V RV shore (L1/N) ─2-pole breaker─┤         (charger; NOT via Paneltronics)
                                      │
                                      └──→ Blue Sea 9009 rotary ──→ 120 V AC panel (Paneltronics)
Xantrex inverter (L1/N) ──────────────┘         ├──→ Coolant / water heater 120 V  (= Isotemp path)
                                                └──→ Outlets 120 V
```

SWITCH	PART (AS-BUILT)	SELECTS	FEEDS
Rotary A	**Blue Sea 9009** (owner: both are 9009s)	J1772 240 V ↔ 120 V RV shore	**Mean Well RPB-1600-12** only
Rotary B	**Blue Sea 9009** (owner: both are 9009s)	120 V RV shore ↔ **Xantrex PROwatt SW 2000**	**Paneltronics** MAIN → WATER HEATER + OUTLETS

As-built note (owner 2026-09-20): diagram draft showed an *automatic* transfer into Paneltronics; both paths **ended up rotary Blue Sea** – owner thinks **both are 9009s**. Confirm faceplates at HOLD 9. Hand-throw: shore vs inverter for the AC panel; shore vs J1772 for the Mean Well.

Mean Well does **not** use the Paneltronics **BATTERY CHARGER** breaker – it is fed only through its Blue Sea 9009.

Isotemp = diagram's "Coolant / water heater 120v" off the AC panel (**WATER HEATER** branch). Not on the 12 V panel.

GFCI (owner 2026-09-20): "The inverter might be capable of that." Treat as **unverified**. PROwatt SW 2000 face has dual GFCI receptacles; whether the **hardwired** feed into the Paneltronics 9009 / Paneltronics is also ground-fault protected is a manual + as-built check – do not assume face GFCI covers Isotemp. When the panel is on **shore** (not inverter), that inverter GFCI story does not apply anyway. HOLD 9: confirm inverter hardwire behavior, then state shore-side protection separately (or "breaker + chamber full" if none).

House DC snapshot (owner): 200 Ah battery bank (**LiFeBlue** LiFePO4 class in build photos – often 2×100 Ah); **650 W** rated solar on roof (**Victron SmartSolar MPPT 100|50** in build photos). Chargers: **Mean Well RPB-1600-12** (shore / J1772 via Blue Sea 9009; replaces early **IOTA DLS-55** in diagrams – confirm IOTA gone) and **Sterling 60 A** B2B. **Blue Sea ML-RBS** remote battery switch + ~12-circuit **ST Blade** fuse block in power bay. Owner: blade block may be **full** → hydronic may need a **second breakout**. PROwatt SW 2000 inverts for the AC panel when its 9009 is selected.

Shore inlets (build photos): exterior **SmartPlug** (120 V; cord often NEMA 5-15P) and underbody **DUOSIDA DSS-EV-32S-V3** J1772 (**32 A / 240 V**). Photo extract: photo-extract-electrical.

Photo extracts (2026-09-20)

FOLDER	NOTES FILE
`photos/Tyrah portfolio_ electrical/`	photo-extract-electrical
`photos/Tyrah portfolio_ heater stuff/`	photo-extract-heater
`photos/Tyrah portfolio_ insulation/`	photo-extract-insulation

---

12 V heat wiring (plain)

Two separate feeds from the house battery side (often via the **Blue Sea** fuse block):

1. **Always-hot to the Hydronic D5S** — heavy fused pair (book: 20 A). Does **not** go through the master. Heater brain can finish after-run.
2. **Master switch → EasyStart Timer (and later fans / thermostat path)** — master is **upstream of the 7-day EasyStart**. Master Off = EasyStart dead = no wake signal = no new heat call. Master On only unlocks the timer; it does not start the heater by itself.

Isotemp 120 V is unrelated to this master.

```
Paneltronics breaker labeled WATER HEATER  →  Isotemp 750 W element
```

Chamber must be full before energize. Never land this on 12 V. Master Off does not cut this branch.

Everything else about house AC stays out of this scratchpad unless you ask.

Layout constraints (owner)

FACT	IMPLICATION
Stove/oven at **passenger door**	Galley heat / clearance / hose paths stay clear of that door swing
Freshwater tank **30+ gal over passenger rear wheel well**	Matches ~35 gal intent (SRC-026). Driver-side sofa bay freer for Isotemp
Isotemp	**Driver side under bench/bed (low)** (owner)
Jump seat behind driver → sink/fridge aft of that	**Tyrah / as-built 2026-09-20:** Sure Marine cabin heater **fits under the fridge.** Duda B3-12DW-20 plate is **under the sink** now (keep unless a hose path forces a move).
5 L expansion header	**In sofa-bed backrest (high)**; Isotemp under the bench lower — same stack, different heights. Prefer driver side with Isotemp. Fill/bleed through hinged/removable backrest. Top of expansion header = highest circulating glycol point—verify after framing.
Blue Sea 12 V distribution block	**On van** (owner). Likely feed point for fused hydronic branches—confirm which circuits and fuse sizes later.

Rip out the bed?

Owner decision: the current bed is **coming out entirely.** Rebuild to meet HOLD S: night flat sleep deck; day **outboard → seat back** / **inboard → seat butt**; Isotemp cradle with restraint; ~30 cm service space at the element end; hose/cable access without draining when panels move; wall/floor bolts into metal (via plywood hatches as needed).

Materials comparison (sofa / bed structure)

What matters here (weights)

Owner priority is **ease of access** and **ease of construction**. Isotemp empty ~35 lb; full chamber adds ~16 L (~35 lb fluid). Day panels must stay light enough to hinge or slide daily.

CRITERION	WEIGHT	WHY
Easy to work with / revise	**35%**	Owner priority; first build; day-sofa will iterate
Strength / restraint	**25%**	Sleep load + Isotemp crash restraint into floor metal
Light (esp. day panels)	**20%**	Daily hinge/slide; payload
Cost	**20%**	Whole-bed all-extrusion is expensive for little gain

Scores 1–5 (higher = better for this van). Weighted = sum(score × weight).

OPTION	EASY 35%	STRONG 25%	LIGHT 20%	COST 20%	WEIGHTED
A. All wood (2× framing + plywood deck)	5	4	2	5	**4.15**
B. Baltic birch furniture ply everywhere	4	4	3	3	**3.60**
C. All T-slot aluminum (80/20 / similar)	3	5	4	1	**3.30**
D. Hybrid: wood shell + Al Isotemp cradle + light day panels	5	5	4	4	**4.55**
E. Steel tube frame	2	5	1	3	**2.75**

Field check (FarOutRide bed-frame only): aluminum frame ~45 lb / ~$585 vs wood ~52 lb / ~$200 — ~7 lb saved for ~3× frame cost. Plywood deck dominates weight either way. All-extrusion furniture is the wrong spend for this van.

Conclusion (locked for HOLD S)

Build hybrid (option D):

1. **Seat bases** — EZTube 1 in square frame and Rostek 3-way corners are accepted (T fittings for crossmembers). The frame bridges the wheel well and lands on the floor. Deck and day panels are Chesapeake Light Craft premium sapele, cut after the footprint is measured. Backrest, cushions, and the sleeper come after that. Provisional length about 75 in. The recalled 21 in height is not yet defined. Passenger well: S&B 35 gal tank. Driver bay: Isotemp cradle.
2. **Isotemp cradle** — short T-slot aluminum rails or a small bolted aluminum frame, strapped to floor structure. Metal holds the tank; wood does not.
3. **Sleep / day deck** — light panels flat at night; by day **outboard → seat back**, **inboard → seat butt** (owner intent). Stay attached; no carry-out as primary. Hardware TBD; hoses never through moving panels.
4. **Walls** — keep existing plywood skins; add **access hatches** where bolts must hit metal.

Do not build the whole sofa in 80/20. **Do not** hang the Isotemp on hose barbs or on furniture weight alone. **Do not** glue the only service path under a fixed mattress platform.

If T-slot stock is already on the shelf, use it for the cradle and panel ledgers only—same hybrid, lower buy cost.

Frame mechanisms (removable middle → day sofa)

What Tyrah needs

- **Night:** continuous sleep surface — panels flat across the sofa/bed span

- **Day (owner intent): outboard** panels become **seat backs**; **inboard** panels become **seat butts**; walk-through aisle between facing sides. Panels stay attached. Exact count / hinges / slides OPEN until measure.
- Under/behind: Isotemp + hoses + cable stay put when panels move

Mechanisms (how the middle attaches)

MECHANISM	HOW IT WORKS	PROS	CONS	DAY-SOFA FIT
Hinge-up (preferred candidate)	Panel swings up against a side box or backrest	Always attached; one motion	Needs swing clearance; latch when up	Matches owner "hinge up by day"
Slide (preferred candidate)	Panel slides under a side seat or along ledgers	Stays low; no tall latch	Track must stay clean; more hardware	Matches owner "slide by day"
Lift-out on ledgers	Center deck sits on side rails; lifts straight up	Simplest hardware	Needs stash space; carry daily	Fallback only if hinge/slide fail measure
Front-pull drawer / platform	Center rolls or slides forward then lifts	Can clear a low obstacle	Track must stay clean; more hardware	Possible hybrid with slide
Pin / clevis / quick pins	Side frames stay; panel pins into sockets	Positive location	Pins must not rattle	Useful with either hinge or slide

Optimum construction for this van (DEC-021 locked)

Hybrid wood + aluminum, three bays:

1. **Two seat frames** – EZTube bridges each wheel well and lands on the floor. Deck and front face come after the frame is measured. Bolt frame feet to the floor, not to the hump and not to the freshwater tank.
2. **Aluminum Isotemp cradle** in one side bay (prefer **not** the passenger wheel-well bay that already holds the freshwater tank)– strapped/bolted tank, coil fittings facing service aisle, ~30 cm clear in front of element end.
3. **Sleep / day deck** – light panels on the center span: **flat at night**; day **outboard → seat back**, **inboard → seat butt** (owner). Prefer stay-attached. Hinges/tracks land on the seat frames; hoses never through moving panels.
4. **Hydronic and freshwater hoses** run in fixed bays or along the wall **behind removable plywood hatches**–never through the moving day panels.
5. **5 L expansion header** – **in the sofa-bed backrest (high)**; Isotemp stays **under the bench/bed (low)**. Same driver-side furniture stack preferred. Fill/bleed via hinged or removable backrest. Confirm expansion header top is highest circulating glycol point after framing.
6. **AC element cable** exits the Isotemp bay toward Paneltronics in a protected chase; leave slack for service.

Do not put the only Isotemp service path under a glued-down mattress platform.

Attach to van (floor and wall)

ATTACHMENT	USE
Floor	Through-bolt or lag into reinforced floor / known Sprinter hard points; use large fender washers; isolate aluminum from paint where needed
Wall	Prefer bolts into **metal** behind plywood via access hatches or through-bolted cleats–not screws into plywood alone for crash loads
Isotemp	Cradle to floor structure; tank straps to cradle; hoses last

Exact Sprinter floor/wall hard-point map: still **OPEN**–measure before drilling.

Plumbing to the structure (sketch)

FLUID	FROM / TO	STRUCTURE RULE
Freshwater	Tank (passenger wheel well) → pump → Duda → AM100-1LF → taps	Keep runs short; don't trap under moving day panels
Glycol loop	Heater → cabin heater → Duda → **Isotemp coil** → return; expansion header high on return	Coil hoses below expansion header; service loops in fixed bay
Isotemp chamber	Static fill/vent on tank	Same mix as loop; not drinking water
Isotemp AC	Cable to Paneltronics **WATER HEATER**	Chase in fixed structure; dead until HOLD 9

Decisions from Q&A

#	QUESTION	ANSWER
1	Current bed keep or rip?	**Coming out entirely.**
2	Day layout when middle is out	**Revised 2026-09-20 (owner):** Night = flat sleep deck. Day intent: **outboard panels become seat backs**, **inboard panels become seat butts** (facing side seats + aisle). Exact panel count / hinges / slides still open — geometry first (see #24).
3	Isotemp bay + nearby layout	**Isotemp: driver side under bed/bench (low). Sure Marine:** under the fridge (Tyrah — fits). **Duda:** under the sink (as-built now). Hose routing: defer.
4	5 L expansion header height	**Owner 2026-09-20:** the expansion header sits in the **sofa-bed backrest** (high); Isotemp stays **under the bench/bed lower** — same furniture stack, different heights. The top of the expansion header must still be the highest circulating glycol point. Fill/bleed via hinged/removable backrest panel. Prefer driver-side stack with Isotemp unless framing forces otherwise.
5	Driver-side sofa back / hide Isotemp	**Superseded by #4.** Backrest's job is the **expansion header** (access panel), not hiding the Isotemp. Isotemp service stays at the **low under-bench** face (~30 cm element clearance). Keep construction simple.
6	Blue Sea 12 V block	**On van** (owner). House 12 V distribution. **Owner 2026-09-20:** may be **out of free fuse positions** for hydronic master / EasyStart / fans — plan a **second blade-fuse breakout** if needed. Photo evidence: ~12-circuit Blue Sea ST Blade in power bay (photo-extract-electrical).
7	Sofa/bed materials	**Hybrid locked (DEC-021).** Wood shell + Al cradle + light ply day/night panels (seat-back / seat-butt geometry).
8	Paneltronics MAIN feed	**Owner 2026-09-20: Blue Sea 9009 rotary** selects **120 V RV shore** or **PROwatt SW 2000** into Paneltronics (diagram draft said automatic; as-built is rotary). Isotemp = "Coolant / water heater 120v" off that panel — **not** 12 V.
9	Mean Well vs Paneltronics BATTERY CHARGER	**Mean Well RPB-1600-12** fed only via its own **Blue Sea 9009** (J1772 240 V ↔ 120 V shore). **Not** through Paneltronics **BATTERY CHARGER**. Owner: both rotaries are **9009s** (confirm faceplates).
10	Inverter identity	**Xantrex PROwatt SW 2000** P/N **806-1220** (Amazon B002LGEMOQ). Diagram "SW2000" = wood-panel PROwatt SW.
11	Ground-fault cutout on Isotemp	Inverter *might*; shore **unknown**; Paneltronics has **no** GFCI. HOLD 9 verifies.

#	QUESTION	ANSWER
12	House battery + solar + charge	**200 Ah** battery; **650 W** solar; **Sterling 60 A** B2B; Mean Well RPB-1600-12.
13	Hot water use mix	~**80% sinks**; showers minority – book mentions showers lightly.
14	Fan relay for SC1600B	Reference: TE **V23134-A3052-X540** or any ISO mini 12 V / ≥10 A – DigiKey.
15	Cabin fans	Both **Noctua NF-F12**; voltage Low/Med/High dial (not NA-FC1); meter Low at HOLD 4.
16	First high camp	**ASAP** → altitude kit at HOLD 4 urgent.
17	Fuel tap + hose	Tap OK; hose likely run; **needs priming**.
18	Exhaust	Thinks **OK** – quick HOLD 2 look before fire.
19	Fuel pump under van	**Future work** before first fire.
20	Isotemp + Duda roles	Chamber = static glycol; Duda heats drinking water; **hot** needs pump; cold anytime.
21	Fan dial	**Book pick:** Sure Marine **W002-912** Hi/Low **no Off**, 12 V **0.52 A** (-36). Owner may already have W002-911/912 – check bag. Two Noctuas ≈ 0.28 A.
22	Master Off vs Isotemp AC	OK they are **separate** controls.
23	Isotemp factory mixer	**Unused forever.** Cap only; AM100-1LF is the tap mixer.
24	Day panels	**Owner 2026-09-20:** Prefer **outboard → seat back**, **inboard → seat butt** (details still open). Night = panels flat as sleep deck. Not carry-out. Hinge/slide hardware TBD to serve that geometry.
25	Transfer switches as-built	**Both rotary Blue Sea; both believed 9009.** One → Mean Well; one → Paneltronics. Diagram draft's "automatic" path is obsolete.
26	Paneltronics 9009 use / labeling	**Owner:** throw **by hand**; switch is **poorly labelled** – hard to remember which position is shore vs inverter. May recover from an old Google photo. Book: add a clear face label at HOLD 4/9 (e.g. SHORE / INVERTER).
27	Hydronic fuse positions	**Owner:** may be **running out of positions** on the existing Blue Sea blade block – **may need another breakout** for master / EasyStart / fan dial.

12 V heat wiring (plain)

```
House battery
   ├─(20 A fuse)──→ Hydronic D5S pins 1–2     [always hot – after-run / memory]
   │
   └─→ Blue Sea 12 V distribution (on van)
         └─→ Master switch (positive only)
               ├─→ EasyStart Timer (7-day)     [YES – master is upstream of the timer]
```

```
├──→ Fan dial → cabin fans
└──→ Later: thermostat / relay path (Q-022)
```

Master **Off** kills the EasyStart Timer, so the heater gets no wake signal and will not start a new heat call. Master does **not** sit on the heater's heavy 20 A feed. Paneltronics / Isotemp AC is unrelated.

Fan dial spec (DEC-015 / DEC-018)

Job: one shared control for **both Noctua NF-F12** (~0.14 A each, ~**0.28 A** together). Speed only. **On/off is master (+ later thermostat relay)** – dial must not be the only Off.

Will a Sure Marine switch on the shelf work?

WHAT YOU LIKELY HAVE	WORKS WITH TWO NOCTUAS?	NOTES
Hi/Low (computer-fan) switch – Sure Marine family like W002-911, **sized by fan amps (0.52 A or ~0.95 A @ 12 V)**	**Probably yes** if total draw ≤ switch rating (0.28 A is fine for the 0.52 A SKU)	Built for REAL computer-style fans. Prefer a version **without Off** (SRC-032). If yours has Off, still usable: ignore Off, use only Low/High, and keep master as the lockout – or replace.
3-speed defroster blower switch (often 5 terminals: B / L / M / H / C)	**No by itself**	Expects a motor with separate Low/Med/High wires. Two Noctuas are single 12 V power pair – that switch alone does not create three speeds.

Shelf check (2 minutes): count positions (Off?); count wires/terminals; any printing like W002-911 / amp rating. Meter Low (or Hi/Low low) so both fans still spin.

Buy if you do not have a matching Hi/Low

Book pick (Sure Marine, matches DEC-018): W002-912 Hi/Low Switch for Fan Heater – **no Off** position. Order the **12 V / up to 0.52 A** option (suffix **-36**). Two Noctua NF-F12 ≈ **0.28 A** combined – that SKU fits. (~$50–56)

SURE MARINE SKU	WHAT IT IS	USE HERE?
W002-912	Hi/Low, **no Off**	**Yes – preferred**
W002-911	Hi/Low **with Off**	OK only if you ignore Off and keep master as lockout
W002-914	VIMAR Hi/Low, no Off	Same job, different faceplate
Defroster 3-speed switch (comes with squirrel-cage units)	Selects motor speed taps	**No** for two Noctuas alone

Shelf check: look for **W002-911** / **W002-912** / **W002-914** on the bag, or count positions (2 vs 3 + Off).

- [x] Day intent: **outboard → seat back**, **inboard → seat butt**; aisle between facing sides (stay attached)
- [ ] Exact panel count + hinge/slide hardware (measure; geometry above wins)
- [x] expansion header in **backrest (high)**; Isotemp **under bench (low)** – same stack; verify expansion header top is highest glycol point after framing
- [ ] Exact floor/wall bolt points
- [ ] Which plywood panels become access hatches
- [x] Sure Marine **under fridge** (Tyrah: fits); Duda **under sink** (as-built)
- [ ] Hose routing (later)
- [ ] Blue Sea fuse positions for master / EasyStart / fans
- [x] Shore vs inverter into Paneltronics: **Blue Sea 9009 rotary** (as-built; diagram draft said automatic)

- [x] Mean Well via its own **Blue Sea 9009** (J1772 ↔ 120 V shore) — **not** Paneltronics BATTERY CHARGER
- [ ] Label both **9009** faces clearly (Paneltronics: SHORE / INVERTER; Mean Well: SHORE / J1772) — owner: Paneltronics one is poorly labelled today
- [ ] Optional: recover position meaning from old Google photo
- [ ] Photograph Blue Sea blade block (used vs free); add **second breakout** if no free slots for master / EasyStart / fans
- [ ] Confirm IOTA DLS-55 removed; Mean Well RPB-1600-12 is the shore/J1772 charger
- [ ] Confirm SmartPlug (120 V) + DUOSIDA J1772 (32 A / 240 V) still feed the two 9009 paths as designed
- [x] Inverter: **Xantrex PROwatt SW 2000** (806-1220 / B002LGEMOQ) — diagram SW2000 = PROwatt SW
- [x] Master Off ≠ Isotemp AC — owner OK they are separate controls
- [x] Isotemp factory mixer — **unused forever** (cap only; AM100-1LF for taps)
- [ ] **Future work:** under-van check of **fuel pump** (confidence) + fuel hose clamps + prime before first fire
- [ ] Q-009: ground-fault on Isotemp — inverter maybe; shore unknown; Paneltronics has **no** GFCI (owner agrees / DEC-020)

Scratch notes

–

Locked facts checklist (2026-09-20)

Purpose: Contradiction-hunt authority for part one + appendix. When anything in the HTML/PDF disagrees with this list, **this list wins** until an owner answer revises it. Photo dumps are evidence, not alternate designs.

Promote changes here first, then Design → Numbers → Use → Status → Build → appendix in one pass.

Loop and fluids

FACT	AUTHORITY
Heater = Hydronic **D5S** 25 2526 / H-Kit; not II C / not D5WS	DEC-000
No engine connection	DEC-009
Loop: pump → D5S → Sure Marine → Duda (glycol) → Isotemp **coil** → return; expansion header tees on return high point	DEC-010
No summer bypass; no second glycol pump	DEC-010 / DEC-013
Isotemp chamber = **static** glycol; not DHW; element does not circulate the loop	DEC-007
Drinking water only via Duda + **AM100-1LF**; Isotemp **factory mixer unused forever** (cap)	DEC-003 / DEC-008
Fill: Prestone LowTox AF555 ≤50% PG	DEC-017
Early Star brite / Bosch / D5WS-engine diagrams = **history only**	photo-extract-heater; DEC-000/001/009

Locations (owner / as-built)

PART	WHERE
Sure Marine cabin heater SN **16401**	**Under the fridge**
Duda + AM100-1LF	**Under the sink**
Isotemp Slim Square	Garage → **driver side under bench/bed (low)**
5 L expansion header	Garage → **sofa-bed backrest (high)**; confirm top wins height
Freshwater tank	Passenger rear wheel well; S&B **10-3008** ordered; 48 × 16 × 16 in (±0.5). Drawing: Cargo floor
Current bed	**Coming out entirely**
Day sofa geometry	**Outboard → seat back**; **inboard → seat butt**; aisle; stay attached
Sofa structure	Hybrid (DEC-021): EZTube 1 in square seat-base frame and Rostek 3-way corners are accepted. T fittings for crossmembers. Frame bridges the well and lands on the floor. Deck and day panels: Chesapeake Light Craft premium sapele marine plywood, cut after the footprint. Backrest, cushions, sleeper later. Provisional length about 75 in; recalled 21 in height is not yet defined. Al Isotemp cradle in the **driver** bay. Regular 170 body, not EXT
Cabin envelope	FatMat Rattle Trap + Thinsulate-class behind grey panels / L-track

House electrical

FACT	NOTES
Two **Blue Sea 9009** rotaries (as-built); diagram "automatic" path obsolete	One → Paneltronics; one → Mean Well
Paneltronics 9009	Shore ↔ PROwatt SW 2000; **poorly labelled** – mark SHORE / INVERTER
Mean Well 9009	Shore ↔ J1772; mark SHORE / J1772; **not** via Paneltronics BATTERY CHARGER
Inlets	Exterior **SmartPlug** (120 V); underbody **DUOSIDA DSS-EV-32S-V3** 32 A / 240 V J1772
Isotemp 750 W	Paneltronics **WATER HEATER** only – never 12 V
Master Off	Cuts EasyStart / fans / thermostat path; **does not** cut Isotemp AC
House bank	**200 Ah** (LiFeBlue class); **650 W** solar; Victron **MPPT 100|50** (photos); Sterling **60 A** B2B; Mean Well **RPB-1600-12**
Early IOTA DLS-55	Build photos; **confirm removed**
Blue Sea ST Blade (~12) + ML-RBS	On van; hydronic may need **second blade breakout**
Fan dial	Book pick Sure Marine **W002-912** Hi/Low no Off
GFCI on Isotemp hardwire	OPEN (Q-009) – face GFCI ≠ hardwire

Use / install emphasis

FACT	WHERE IT MUST SHOW
~80% sinks / showers minority	Design + User's Guide
First high camp ASAP → altitude kit at HOLD 4	Status + HOLD 4A
Fuel pump under-van check = future work before first fire	Status Next + HOLD 1/8
HOLD S hatches pierce insulation to metal	Install HOLD S

Explicitly OPEN (do not invent)

- Exact hinge/slide hardware and panel count for seat-back / seat-butt geometry
- Both 9009 faceplates confirmed as 9009 (owner belief)
- Free fuse slot count vs second breakout purchase
- Q-009 ground-fault story; Q-022 SC1600 wake
- Hose routing details

Contradiction hunt commands

After edits, rebuild, then fail the pass if HTML still contains as **current** design:

- `automatic transfer` (unless "diagram draft said")
- `intended near jump seat`
- `exact mount OPEN (rear bench`
- `full-length lift-out` as the live plan (OK only as superseded)
- expansion header location without **backrest** in status/inventory tables

Photo extract – Tyrah portfolio electrical

Source folder: `photos/Tyrah portfolio_ electrical/`
Extracted: 2026-09-20
Status: Working notes from image review. Promote locked facts into part one / appendix; treat build-era parts as history unless confirmed still on van.

Owner 2026-09-20: Blue Sea 12 V fuse positions for hydronic master / EasyStart / fan dial may be **full** – may need another breakout. Paneltronics-side **9009** is poorly labelled (hand-thrown shore vs inverter).

Inventory of files reviewed

FILE	KIND	USEFULNESS
`FB_IMG_1542865718571.jpg`	AC topology diagram (same family as `photos/ac-electrical.jpg`)	High – design intent
`download_20181211_215128.jpg`	Earlier AC topology draft	High – shows rotary into AC panel; Mean Well off shore only
`Screen Shot 2017-10-04…png`	Starlight / LiFeBlue generic LiFePO4 RV diagram	Medium – reference architecture, not this van's BOM
`Screen Shot 2018-07-03…png`	FarOutRide logical diagram	Medium – same *class* of system (SRC-041 family)
`Screen Shot 2018-09-25…png`	EVTV J1772 pinout / pilot	High – J1772 knowledge
`received_2152621031658335.jpeg`	Interlocked J1772 + NEMA wiring (AVC2)	High – how shore/J1772 interlock was designed
`FB_IMG_1542002955893.jpg`	Labeled power-bay photo (build)	High – early as-built labels
`20181209_193653.jpg`, `20181210_214723.jpg`, `20190514_201538.jpg`, `20190518_202731_603.jpg`	Power bay / art-panel assembly	High – hardware identities
`20180925_170037.jpg`	Roof cable gland	Medium – solar entry
`20190331_131010.jpg`	T-slot frame corner	Low – mechanical only
`20190331_175017.jpg`	Art panel outdoors (Tyrah)	Context – same resin face used as electrical faceplate
`20190409_171348.jpg`, `20190514` related inlet parts	Multi-pin connector + boat cable	Medium
`20191023`–`20191025` series	DUOSIDA J1772 inlet + underbody mount	High
`20191113_130855.jpg`, `20191116_102113.jpg`	SmartPlug 120 V shore	High
`20191113_164727.jpg`	Van interior during install	Medium – bay location
`IMG_20180619_210238_180.jpg`	LiFeBlue BMS app screenshot	Medium – battery chemistry / BMS
`IMG_20190403_223716_174.jpg`	Shore cord plug assembly	Low–medium

AC topology (diagrams)

Design intent (late diagram ≈ `ac-electrical.jpg`)

Sources

- **J1772 Level 2** split-phase **240 V** (L1 / L2 / Ground) through a **2-pole breaker**
- **RV 30 A or standard wall 120 V** (L1 / N / Ground) through a **2-pole breaker**
- **Xantrex SW2000** inverter (PROwatt SW 2000 on van) – L1 / N / Ground

Two transfer paths

1. **Blue Sea 9009** rotary: J1772 240 V ↔ 120 V shore → **Mean Well RPB-1600-12** only (not via Paneltronics **BATTERY CHARGER**)
2. Shore ↔ inverter → **120 V AC panel** (Paneltronics) → sinks: **Coolant / water heater 120v** (= Isotemp path) + **Outlets 120v**

Diagram draft vs as-built (owner 2026-09-20): late drawing still labels path (2) as *automatic* transfer. Owner: **both ended up rotary Blue Sea; both believed 9009s.** Label faces poorly; mark **SHORE / INVERTER** and **SHORE / J1772**.

Earlier draft (`download_20181211_215128.jpg`)

- Single **rotary** already selecting shore vs **SW2000** into the **120 V AC panel**
- **Mean Well** fed from shore/J1772 path **after breaker**, not through the AC-panel rotary
- AC panel sinks listed: **120v charger**, **coolant / water heater**, **outlets** — confirms Isotemp-class load was planned on AC panel from early on

J1772 interlock (`received_…jpeg` + EVTV screenshot)

- **AVC2** controller (Modular EV Power) with PROX / PILOT
- Two contactors: **240 V / 30 A** (J1772) and **120 V / 20 A** (NEMA); first energized disables the other (N.C. interlock)
- Combined output: **120 or 240 VAC to charger**
- Pins: AC L1/L2, ground, pilot, proximity
- Pilot state voltages (EVTV sheet): ~12 V not connected / ~9 V connected not ready / ~6 V ready; vehicle-side diode + 2.7 kΩ / 1.3 kΩ network

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Shore / charge inlets (as photographed)

INLET	EVIDENCE	NOTES
SmartPlug 120 V shore	Interior orange body ETL 4001038; exterior chrome cover plugged with orange SmartPlug; cord ends in **NEMA 5-15P** right-angle	Driver-side body panel above black molding; white sealant at interior plate
DUOSIDA DSS-EV-32S-V3	Cap label **32A 240V AC 3S**	Mounted in grey enclosure under van (near wheel well); flexible conduit into chassis
Custom breakout	Grey box + metal plate with **insulated studs**; multi-conductor white jacket (red/black/white/green/yellow seen in build photos)	Supporting J1772 / high-current AC landing

UL 1426 "Boat Cable" appears in connector build photos (marine-grade stranded).

House DC power bay (build-era photos)

Mounting: plywood (later **Tyrah resin art face**) in **aluminum T-slot** frame — passenger-side / rear bay area in interior shots.

Components identified in photos (build era)

ROLE	PART SEEN	LATER BOOK STATUS
Inverter	**Xantrex PROwatt SW 2000** (labeled SW2000 / PROwatt)	Still on wood panel — VERIFIED class
House battery	**LiFeBlue LiFePO4** — annotated photo: **200 Ah** as **2 × 100 Ah** parallel	Owner still says **200 Ah**
Solar	**Victron SmartSolar MPPT 100|50**	Confirm if still this model
Alternator charge	**Sterling** B2B (white/red)	Owner: **Sterling 60 A**
Shore AC→DC charger	**IOTA DLS-55 Series M** (108–132 VAC in, ~13.4 V / **55 A** out, ~900 W)	**Superseded in AC diagram by Mean Well RPB-1600-12** — confirm IOTA removed or retained
Battery disconnect	**Blue Sea ML-RBS** remote battery switch (yellow knob; ~500 A class / 12 V)	Confirm still present
DC branch fuses	**Blue Sea ST Blade ~12-circuit** fuse block (clear cover)	On van — hydronic may need **extra breakout** (owner)
High-amp positive	**~4-position** AMI/MIDI-style fuse holder; **200 A** waterproof ignition-protected breaker; **~40 A** breaker feeding blade block	Measure as-built
Monitor	Brass **shunt** on negative path (LinkPRO era)	LinkPRO on wood panel today
Busbars	Positive and negative stud bars (≈4–6 studs each in various photos)	Negative studs may congest before positive blade slots

Annotated photo text: **"3 × chargers"** (Sterling + Victron + IOTA) and **"2 × draws"** toward fuse/inverter area; ML-RBS annotated roughly as preventing overcharge path issues (wording imprecise — treat as owner note, not engineering claim).

BMS app (`IMG_20180619…`)

- 4-cell LiFePO4 pack UI: SOC example **67%**, pack ~**13.2 V**, cells ~**3.29 V** balanced
- Protections shown: HV/LV, OCC/OCD, LTD/LTC/HTD/HTC — **Standby / Perfect** in that screenshot

Roof entry

- Dual black cables through sealed roof gland → interior (solar pair class)

Implications for hydronic 12 V (HOLD 4)

1. Hydronic **always-hot** heater pins 1–2 and **master → EasyStart / fans** need fuse positions on house DC.
2. Owner: **may be out of positions** on the existing Blue Sea blade block → plan **second ST Blade breakout** (or equivalent) fed from a spare high-amp / bus stud with its own feeder breaker/fuse.
3. Do **not** land Isotemp 750 W on any 12 V breakout — AC only via Paneltronics **WATER HEATER**.
4. Before cutting hydronic feeders: photograph **current fuse labels** and open slots; record which ML-RBS / 200 A / 40 A path feeds the blade block.

Reference diagrams (not this van's BOM)

- **Starlight LiFeBlue** sheet: generic bus-bar + multi-branch fuse philosophy; absorb/float notes; storage SoC procedure — useful class reference only.

- **FarOutRide** logical diagram: solar + B2B + shore charger → house battery → fuse box + inverter → 12 V / 120 V loads – matches SRC-041 *class*.

Open checks from this set

- ☐ Confirm **IOTA DLS-55** removed and **Mean Well RPB-1600-12** is the only shore/J1772 charger
- ☐ Confirm both transfer switches are **9009**; label faces
- ☐ Photograph **current** Blue Sea blade block: used vs free slots; add second breakout if needed for hydronic
- ☐ Confirm Victron **100|50** and Sterling **60 A** still installed
- ☐ Map SmartPlug vs DUOSIDA paths to the two 9009s on the van today
- ☐ Recover 9009 position meaning from Google Photos if possible

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Owner Q&A tied to this folder

#	TOPIC	ANSWER
25–26	Transfers / labels	Both rotary Blue Sea (~9009); Paneltronics one poorly labelled
27	Fuse positions for master / EasyStart / fans	**May be full** – likely need **another breakout**

Photo extract – Tyrah portfolio heater stuff

Source folder: `photos/Tyrah portfolio_ heater stuff/`
Extracted: 2026-09-20
Status: Build-era evidence (2018). Compare to current DEC/BOM; do not treat superseded heater-family diagrams as live design.

Files

FILE	WHAT IT SHOWS
`20180222_133050.jpg`	**REAL** cabin fan heater (white box): one large + two smaller duct outlets; copper coil visible; "MADE IN THE USA"; marking near label area **10301**
`20180223_143534.jpg`	Kit layout: diesel heater body + harness, REAL air matrix, coolant circ pump, fuel dosing pump, **Star brite** PG gallon
`IMG-20180220-WA0002.jpg`	Schematic: **"D-5 WS HEATING SYSTEM PLUMBING LOOP"** (RIXENS ENT. INC., 2-28-02)
`IMG-20180402-WA0009.jpg`	Labeled parts spread: thermal storage, expansion tank, diesel heater, air HX, water plate, circ pump, fill/drain, altitude kit, antifreeze

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What this confirms (and what it does not)

Cabin air heater

- Early purchase / unboxing of a **REAL** hydronic air handler (Sure Marine–class cabin heater family).
- Outlet pattern (large + two small) matches ducted multi-zone intent.
- Serial/marking **10301** in that photo ≠ current van nameplate **16401** (book / DEC). Treat **16401** as as-built authority; **10301** is historical stock or a different unit.

Diesel heater kit contents (2018 layout)

Visible together:

- Heater body (silver/black) with multi-pin harness
- Fuel **dosing / metering pump**
- Coolant **circulation pump** (black, 90° ports — kit pump; later design may use heater's own pump only — see DEC-013)
- **Altitude kit** bag
- Brass **fill/drain** / isolation ball valves
- PTFE tape / sealant

Thermal mass + DHW path (labeled photo)

Owner labels on `IMG-20180402…`:

LABEL IN PHOTO	MAPS TO TODAY'S CONTROLLED NAME
Thermal storage tank w/ electric element (Bosch-branded face in photo)	**Isotemp Slim Square 4.2 gal heat battery** lineage (electric element on AC)
Expansion tank (translucent, dual bottom ports)	**5 L expansion header** class — exact SKU may differ from early jug
Diesel heater	**Hydronic D5S** family on van today (not D5WS — see below)
Air heat exchanger	**Sure Marine / REAL cabin heater**
Water heat exchanger (plate)	**Duda** plate class
Circ pump	Circulating pump (architecture later locked to **heater's own pump** only)
High altitude kit	**Altitude kit 22 1000 33 22 00** class
Antifreeze	Propylene glycol — early jug **Star brite BioSafe**; book fill standard is **Prestone LowTox AF555 ≤50%** (DEC-017)

Coolant chemistry (early)

- **Star brite** "ECO-FRIENDLY BIOSAFE" **non-toxic PG** gallon in kit photos.
- Current authority for fill: **DEC-017** / install guide (Prestone LowTox AF555, ≤50% PG). Do not specify Star brite as the fill fluid unless owner re-confirms leftovers.

Superseded schematic — D-5 WS loop

`IMG-20180220-WA0002.jpg` is a **D5WS-era** plumbing cartoon:

- Expansion / "AC back up" tank → pump → furnace → water-to-water HX → **two** fan units in series → **engine heat exchanger** → return
- Explicit **engine loop** and **domestic water** taps on separate exchangers
- Dimensions called out for tank, furnace, plates, fans

Book rule: Hydronic II C / **D5WS** are **ruled out** (DEC-000). This van is **Hydronic D5S**, **no engine connection** (DEC-009), **one** Sure Marine cabin heater, **Duda** + **Isotemp coil**, **no second glycol pump** (DEC-013).

Use this image only as **design history** (why early notes mentioned engine / dual fans). Do not copy its loop order into part one or HOLD plumbing.

Implications for the live build

1. Parts intent (diesel + air matrix + plate + storage + expansion + altitude + PG) was present by **spring 2018**.
2. Electric element on the storage tank was planned early → supports Paneltronics **WATER HEATER** / Isotemp AC story.
3. Dual-fan air units on the D5WS drawing ≠ current single Sure Marine under fridge.
4. Kit circ pump in photos may be leftover or unused if DEC-013 holds — verify shelf vs van before BOM claims.

Open from this folder

- [ ] Confirm whether the REAL unit in 2018 photos is the same physical unit as SN **16401** or a prior SKU
- [ ] Confirm kit circ pump kept or discarded
- [ ] Confirm leftover Star brite vs Prestone for first fill

Photo extract — Tyrah portfolio insulation

Source folder: `photos/Tyrah portfolio_ insulation/`
Extracted: 2026-09-20
Status: Build-era evidence (Oct–Nov 2017). Describes envelope behind today's plywood / grey panels — relevant to heat retention, sound, and access hatches (HOLD S).

Also present: `20171101_164145.mp4` (not transcribed here).

Files (stills)

FILE	VIEW
`20171017_191844.jpg`	Passenger-side wall work; FatMat on upper recesses; fibrous insulation on floor
`IMG_20171017_194043_791.jpg`	Same phase; black cavity pads + white batts + Rattle Trap
`20171017_212823.jpg`	Driver-side rear wheel well fully Rattle Trap–clad; black + white insulation in cavities
`20171018_200834.jpg`	Ceiling / upper walls — Rattle Trap patches; white fibrous starting; conduit along ceiling
`20171018_200841.jpg`	Driver-side wall + rear door; upper cavities filled black; Rattle Trap grid on door
`20171024_175617.jpg`	Ceiling install: Thinsulate-class sheet (white batting + black scrim); red/black wires clipped along roof edge
`20171025_154805(0).jpg`	Mid build: grey lower wall panels on; Thinsulate roll; window cover; Rattle Trap on uppers
`20171025_164549.jpg`	Fitting Thinsulate strips into B-pillar / cab-adjacent cavity
`20171102_170524.jpg`	Later: grey wall panels + dual **L-track / airline track** rows; ceiling bays filled black insulation; rear doors panelled

Materials stack (as photographed)

LAYER	PRODUCT / APPEARANCE	WHERE APPLIED
Vibration / sound	**FatMat Rattle Trap** (foil face, XXX / blue X print)	Flat metal panels: walls, ceiling bays, **wheel wells**, rear door uppers
Thermal / acoustic bulk	Black fabric-faced pads + white fibrous batting (consistent with **3M Thinsulate** SM-class)	Wall cavities between ribs; ceiling bays between roof ribs; pillar strips
Quilted reflective	Silver quilted sheet (seen on one wheel-well stage)	Wheel well (may be interim or additional)
Finish	Light **grey** interior panels; black fasteners	Walls and lower rear doors (by Nov 2017)
Cargo hardware	Horizontal **L-track** (two rows per side)	Side walls after paneling

Sequence observed

1. Bare white cargo shell
2. Rattle Trap on resonant flats + wheel wells
3. Stuff Thinsulate-class material into cavities / ceiling
4. Grey lower panels and tracks
5. Ceiling insulation between exposed white ribs (ribs often left visible)

Wiring note: early **red/black** pair already run along roof edge in clips during ceiling insulation — house DC / lighting class, not hydronic-specific.

Implications for hydronic / furniture (2026)

1. **Heat retention:** Cabin already has a serious insulation + deadening envelope — diesel / Sure Marine loads less heat into bare metal than a raw cargo shell.
2. **Access:** Book already requires **plywood access hatches** to metal for crash bolts (HOLD S). Insulation and panels are behind that story — hatches must reach metal without destroying the whole cavity pack.
3. **L-track:** Side-wall tracks may help strap furniture / Isotemp cradle / cable chase — measure before ignoring.
4. **Wheel wells:** Heavy Rattle Trap + insulation on rear wells — hose / cable penetrations need planned clear paths (freshwater tank already over **passenger** rear well).
5. **Noise:** Rattle Trap helps with road noise and may slightly quiet cabin fan / pump vibration transmission — not a substitute for soft mounts.

What this folder does *not* show

- Final wood furniture / sofa rebuild
- Hydronic hose routing
- Current electrical faceplate art panel (that lives in the electrical portfolio)

Open from this folder

- [ ] Confirm whether ceiling ribs stayed exposed or got a later headliner
- [ ] Confirm Thinsulate product code if a receipt exists
- [ ] Map which L-track segments remain usable after sofa rebuild

Verified Requirements

Heater identity is locked (DEC-000): Hydronic D5S diesel heater, **25 2526 05 00 00**. Numeric heater specs from SRC-009 unless noted.

ID	REQUIREMENT	SUBSYSTEM	SOURCE ID	STATUS	VERIFICATION
REQ-001	Heater is Hydronic D5S diesel heater, 25 2526 05 00 00, 5.2/5.0/2.1 kW, max 2.5 bar. Use SRC-009 only — not II C or D5WS	Heater	SRC-023; nameplate; SRC-009 p. 13; DEC-000	VERIFIED	Re-read nameplate if swapped
REQ-002	Coolant mix **≤50%** Prestone LowTox AF555 only (DEC-017); Star brite Bio-Safe capped for this loop	Coolant	SRC-009 p. 13; SRC-029; DEC-017	VERIFIED	Dilute concentrate ≤50% before fill
REQ-003	Allowable operating pressure ≤ 2.5 bar; 5 L expansion header relief ~1.2 bar	Coolant	SRC-009 p. 13; DEC-005 / heatso 65288A	VERIFIED (spec)	Cap stamp / leak test
REQ-004	Minimum heater water flow 250 l/h	Coolant	SRC-009 p. 13	VERIFIED (spec)	Flow check after plumbing
REQ-005	Heater and pumps below coolant level in header; do not put a pump at the absolute lowest point	Coolant / layout	SRC-009 p. 15	VERIFIED (rule)	Layout inspection
REQ-006	Fuel: diesel EN 590; max **20% FAME** blend; not pure biodiesel	Fuel	SRC-009 p. 13, 29	VERIFIED (spec)	Fuel type on vehicle
REQ-007	Metering pump 15°–90° (prefer 15°–35°); 0°–15° forbidden; diesel suction head ≤1000 mm (pressureless)	Fuel	SRC-009 fuel chapter	OPEN	Measure mount / heads (Q-010)
REQ-008	Exhaust open-air, not into interior/intake; slope + ~Ø 5 mm drain; no closed-garage run	Exhaust	SRC-009 p. 24–25; HAZ-002	OPEN (as-built)	Routing + CO check
REQ-009	Exhaust Ø 24 mm; run min ~200 mm / max ~2 m	Exhaust	SRC-009 p. 24–25	OPEN	Measure installed length
REQ-010	Combustion air ≤25 °C area, not cabin; hose Ø 20 mm, max ~760 mm; clean cut	Combustion air	SRC-009 p. 25	OPEN	Routing inspection
REQ-011	Sustained heat above ~1500 m needs kit **22 1000 33 22 00** + EasyStart Timer; H-Kit on heater. Prefer install at HOLD 4 while harness open	Controls	SRC-009 p. 32; SRC-002; DEC-006; Q-015	OPEN (kit in garage)	Land at HOLD 4; required before high camps

ID	REQUIREMENT	SUBSYSTEM	SOURCE ID	STATUS	VERIFICATION
REQ-012	12 V window 10.5–16 V	Electrical	SRC-009 p. 13	VERIFIED (spec)	Measure at heater
REQ-013	**Independent living-space loop only — no engine connection**	Architecture	DEC-009	VERIFIED (decision)	No engine hoses cut
REQ-014	Isotemp Slim Square 4.2 gal = circulating glycol in coil, **static** glycol in chamber; not DHW; Bosch gone	Isotemp / thermal mass	SRC-021; SRC-022; DEC-001; DEC-007	VERIFIED (role)	P/N photo; chamber not teed into loop
REQ-015	Isotemp Slim Square 4.2 gal heat battery element is 115 VAC / 750 W and feeds from Paneltronics, not the 12 V panel; fill chamber first and verify AC protection/grounding before energizing	Isotemp / electrical	SRC-022; DEC-004; DEC-007; DEC-020	PARTIAL — GFCI/route OPEN	Q-009 (HOLD 9)
REQ-016	DHW in Duda B3-12DW-20 plate heat exchanger; AM100-1LF thermostatic mixing valve on that hot outlet (~120°F set); not factory mixer on the Isotemp Slim Square 4.2 gal heat battery	DHW	DEC-003; DEC-008; SRC-024; SRC-026	VERIFIED (roles); taps OPEN	Land taps; label setpoint
REQ-017	coil in the Isotemp Slim Square 4.2 gal heat battery below the 5 L expansion header (or high-point bleeder); add coil volume on circulating fill	Coolant	SRC-022 §2, §5	VERIFIED (rule)	Height vs 5 L expansion header; fill procedure
REQ-018	Cabin control = EasyStart Timer only (DEC-002); in van, wiring unfinished	Controls	SRC-003; SRC-023; DEC-002	VERIFIED (selection); wiring OPEN	Finish landing (Q-013)
REQ-019	No auxiliary glycol-pump path; Isotemp element remains a separate Paneltronics AC load	Electrical	DEC-004; DEC-013	VERIFIED (architecture); AC details OPEN	Q-009
REQ-020	Glycol flow path: Hydronic D5S diesel heater's own pump → Hydronic D5S diesel heater → Sure Marine cabin heater → Duda B3-12DW-20 plate	Architecture	DEC-010	VERIFIED (decision)	Hose routing inspection

ID	REQUIREMENT	SUBSYSTEM	SOURCE ID	STATUS	VERIFICATION
	heat exchanger → coil in the Isotemp Slim Square 4.2 gal heat battery → return; expansion header tank tees at return high point; no summer bypass				
REQ-021	Master Off/On is intended to gate EasyStart Timer + SC1600B + Sure Marine cabin heater power; SC1600B is intended to call heat/Sure Marine cabin heater; dial is speed only	Controls	DEC-012; DEC-014	PARTIAL (design intent; Q-022 interface and switch verification open)	Wire per DEC-012/014; Q-021; Q-022
REQ-022	No separate solar/shore auxiliary glycol pump; hot water through the Duda loop remains diesel-loop dependent; Isotemp element is a separate Paneltronics AC load	Coolant / electrical	DEC-004; DEC-013	VERIFIED (architecture); AC details OPEN	Leave CM10P7 out; resolve Q-009
REQ-023	Staging ~65 / 80 / 85 / 75 °C (Power→High→Low→pause→restart); **pump continues in pause**; overheat restart <70 °C; ≤2 off/on into fault	Controls	SRC-009 p. 31–32; Q-020 resolved	VERIFIED (spec)	Observe first fire / diagnostics
REQ-024	5 L expansion header at high point – **sofa-bed backrest (high)** planned; not yet mounted; confirm top wins after framing	Coolant / layout	DEC-005	VERIFIED (part); height check OPEN	Mount top = highest loop point
REQ-025	Fuel takeoff uses factory aux connection; still inspect line before fire	Fuel	Q-010; HAZ-008	PARTIAL	Photo routing/clamps

Still blocking first fire (summary)

Fuel-line inspection (REQ-007 / REQ-025), exhaust confirm (REQ-008), fill/bleed (REQ-002–005, REQ-017), EasyStart Timer landing (REQ-018). Altitude kit 22 1000 33 22 00 (REQ-011) blocks high camps, not a low-elevation first fire—but land it at HOLD 4 while the harness is open. Paneltronics element circuit (REQ-015 / REQ-019 / REQ-022) is required for **complete** system acceptance (HOLD 9), not for diesel-only cabin heat / diesel hot water at the Duda.

Source Register

Manufacturer documentation outranks OEM vehicle sheets, marine installer guides, blogs, vendor listings, and photo albums. The installed heater is Hydronic D5S diesel heater, order no. 25 2526 05 00 00 (DEC-000 / SRC-023). Do not mix in Hydronic II C (SRC-001) or D5WS (SRC-010) specifications.

Local files in `sources/hydronic-install/` ingested 2026-09-19.

Local file map

FILE	SRC
15046148183.pdf	SRC-009
Eberspacher_High_Altitude_Kit_221000332200_Installation_Instructions.pdf	SRC-002
Eberspacher_Heater_Hydronic_D5WS_Technical_Manual.pdf	SRC-010
Eberspacher_Heater_D5WS_Workshop_manual.pdf	SRC-011
Eberspacher_Heater_Hydronic_D3WS_&_D5WS_Parts_List.pdf	SRC-012
Eberspacher_Heater_Hydronic_D3WS,_D4WS_&_D5WS_Fault_Codes.pdf	SRC-013
Eberspacher_Heater_Mercedes_Sprinter_VW_Crafter_D5WS_B5WS_information.pdf	SRC-014
Eberspacher_Heater_Mercedes_Vito_D5WS_D5WZ_Hydronic_25_2126_Information.pdf	SRC-015
15046148181.pdf	SRC-016
15046148182.pdf	SRC-017
Eberspacher_EasyStart_Timer_Installation_Instructions_221000341500.pdf	SRC-003
hydronic-installation-guide.pdf	SRC-018
TSL17SW_07-08-17-Install-Email.pdf	SRC-019
SureMarine_Marine_Heating_Catalog.pdf	SRC-032
floor-steps	SRC-020
../coolant/Prestone-LowTox-AF555-SDS.pdf	SRC-029
../coolant/Star-brite-BioSafe-317XX-SDS-NA-EN.pdf	SRC-030
../controls/ICM_SC1600L_Installation_Operation_Application_Guide.pdf	SRC-033
../controls/ICM_SC1600L_VL_Thermostat_Manual.pdf	SRC-033
../controls/Noctua_NA-FC1_Manual_EN.pdf	SRC-034
../electrical/Paneltronics_Catalog.pdf	SRC-035
../electrical/Paneltronics_9972313B_Wiring.pdf	SRC-038
../plumbing/Honeywell_AM-1_Series_Installation_Instructions_62-3075EFS.pdf	SRC-036
../plumbing/Honeywell_SparcoMix_AM-1_Series_Submittal_68-3098.pdf	SRC-037

Local coolant SDS files in `sources/coolant/` ingested 2026-09-19.

Register

ID	SOURCE	COMPONENT/MODEL	REVISION/DATE	AUTHORITY	LOCATION/URL	NOTES
SRC-001	Eberspächer Hydronic II C technical description, installation, operating and maintenance instructions	Hydronic II C D5S 12 V (25 2506 05 00 00); 24 V (25 2507 05 00 00)	Doc 25.2506.90.9988.0 A; 03.2022	Manufacturer	https://www.butler-technik.com/downloads/Hydronic_II_Technical,_installation_&_operating_manual.pdf	Compact Hydronic II C, not the same unit as SRC-009 Hydronic D5S diesel heater. Heat flow 4800/2100/1200 W. Min water flow >250 l/h. Retrieved 2026-09-19.
SRC-002	Eberspächer installation instructions – Altitude kit 22 1000 33 22 00	Air pressure sensor kit; Rixens SKU E11-099; for EasyStart Timer / Remote / Remote+ / Select	Order no. 22.1000.33.2200; 13-page installation manual	Manufacturer	`sources/hydronic-install/Eberspacher_High_Altitude_Kit_221000332200_Installation_Instructions.pdf` · https://www.butler-technik.com/downloads/High_Altitude_Kit_221000332200_-_Installation_Instruction_Manual.pdf	Compatibility, installation location, electrical connection, functional check, diagnostics, and Hydronic/Airtronic wiring diagrams. Retrieved 2026-09-19.
SRC-003	Eberspächer installation instructions – EasyStart Timer	EasyStart Timer 12/24 V, order no. 22 1000 34 15 00	Order no. 22 1000 34 15 00; document date not stated on fetched copy	Manufacturer	`sources/hydronic-install/Eberspacher_EasyStart_Timer_Installation_Instructions_221000341500.pdf` · https://www.butler-technik.com/downloads/Eberspacher_Easy_Start_7_day_Timer_Installation_instructions_221000341500.pdf	20-page manual. Timer, wiring, optional temp sensor 22 1000 34 22 00. Circuit diagram covers Hydronic / Hydronic II / Hydronic II C. Downloaded 2026-09-19.
SRC-004	mylifewithoutdoor / Ben Racine, "DIY high-altitude hydronic van heating"	Espar Hydronic D5 + EasyStart Timer + HAK; independent living loop design intent	Mar 13, 2018	Owner blog	https://medium.com/@mylifewodoor/diy-high-altitude-hydronic-van-heating-3c68ef42dc47	**Intent still valid:** diesel + optional electric in; fan + PHE out; double-wall drinking-water separation; thermal mass; no engine (deferred then, locked DEC-009). **Superseded plumbing:** blog order Duda B3-12DW-20 plate heat exchanger→Sure

ID	SOURCE	COMPONENT/MODEL	REVISION/DATE	AUTHORITY	LOCATION/URL	NOTES
						Marine cabin heater→Bosch; summer valve; Bosch 1440 W / ~8.5 gal claim. Replaced by DEC-010 / DEC-001 / 750 W element on the Isotemp Slim Square 4.2 gal heat battery. Cost figures historical only.
SRC-005	mylifewithoutdoor / Ben Racine, "Espar D5 Installation Notes"	Hydronic D5S diesel heater underbody install field notes	Nov 15, 2019	Owner blog	https://medium.com/@mylifewodoor/espar-d5-installation-notes-6f837f2a3f92	Mount behind fuel fill / driver wall; 10-pin chambers 8–9 for water pump; HAK in series on group A; proprietary crimper; blower relay often unused (aligns DEC-012); hand-bulb fuel prime; optional 90° exhaust elbow. Always confirm pins against SRC-009/002/003. Not a substitute for manufacturer manuals. Photos: SRC-006.
SRC-006	Google Photos album	Unconfirmed; likely SRC-004/SRC-005 install photos	Unknown	Photo album	https://photos.app.goo.gl/AWn5eqMhnWwR3CuS6	Contents not fetched. Treat as visual reference only until captioned. Retrieved 2026-09-19.
SRC-007	RadVan Adventures, "Espar Heater Installation – DIY Guide"	Espar Airtronic D2 (air heater), 2014 Sprinter passenger-seat-base install	2017-02-12	Blog	https://radvanadventures.com/2017/02/12/espar-heater-installation-diy-guide/	Different heater family. Do not copy Airtronic dimensions, ducting, or BTU claims. Retrieved 2026-09-19.
SRC-008	Lubrication Specialist product page – Espar 6-port combination valve	Eberspächer 6-port combination valve; likely P/N 25 2014 80 62 00 (confirm against physical part)	Unknown	Vendor listing	https://www.lubricationspecialist.com/espar/6-port-combination-valve/	Dealer page, not manufacturer documentation. Confirm stamp on the actual valve. Retrieved 2026-09-19.
SRC-009	Eberspächer Hydronic II	Hydronic II D4S 12 V (25 2554 05 00	Doc 25 2557 90 00 01; 05.2012	Manufacturer	`sources/hydronic-`	Hydronic II (not II C). D5S heat flow

ID	SOURCE	COMPONENT/MODEL	REVISION/DATE	AUTHORITY	LOCATION/URL	NOTES
	technical description, installation, operating and maintenance instructions	00); D5S 12 V (25 2526 05 00 00); inlet-pressure-resistant D5S (25 2557 05 00 00)			install/15046148183.pdf	5200/5000/2100 W. Min water flow 250 l/h. Exhaust Ø 24 mm, combustion air Ø 20 mm. Altitude: unlimited to 1500 m; kit 22 1000 33 22 00 for stays 1500–3000 m. Ingested 2026-09-19.
SRC-010	Eberspächer Hydronic B5WS / D5WS technical description and mounting instructions	Hydronic D5WS 12 V (25 2100 / 25 2101 05 00 00); 24 V (25 2146 05 00 00)	Butler Technik distributor copy; document number not on title page	Manufacturer	sources/hydronic-install/Eberspacher_Heater_Hydronic_D5WS_Technical_Manual.pdf	Predecessor family to Hydronic II. D5WS 5000/2300 W; min water flow 300 l/h (stricter than SRC-009). Passenger-compartment install not permitted. 6-port combo valve Order No. 330 00 176. Ingested 2026-09-19.
SRC-011	Eberspächer Hydronic B5WS / D5WS troubleshooting and repair instructions	Hydronic D5WS 12 V (25 2031 / 25 2032 05 00 00); 24 V (25 2009 05 00 00)	Doc 20 1777 95 11 54; 05.1998	Manufacturer	sources/hydronic-install/Eberspacher_Heater_D5WS_Workshop_manual.pdf	Workshop/fault diagnosis for D5WS only. Do not use fault codes on Hydronic II / II C without confirming they match. Ingested 2026-09-19.
SRC-012	Eberspächer Hydronic parts list (Ersatzteilliste)	Hydronic D3WZ, D5WZ, B5WS, D5WS (several 12/24 V order nos.)	Doc 25 2009 95 11 85; 02.2004	Manufacturer	sources/hydronic-install/Eberspacher_Heater_Hydronic_D3WS_&_D5WS_Parts_List.pdf	Spare-parts exploded views for the D5WS-era heaters. Not a van BOM. Ingested 2026-09-19.
SRC-013	Eberspächer Hydronic D3WS / D4WS / D5WS fault-code sheets	Hydronic D3WS, D4WS, D5WS	Unknown; scanned Butler Technik copy	Manufacturer	sources/hydronic-install/Eberspacher_Heater_Hydronic_D3WS,_D4WS_&_D5WS_Fault_Codes.pdf	Image-only PDF; text extraction failed. Use with SRC-011 after heater family is confirmed. Ingested 2026-09-19.
SRC-014	Eberspächer factory-install information sheet — MB Sprinter NCV3 / VW Crafter	Hydronic B5WS / D5WS / D5WZ from MY 2006; D5WS 25 2453 05 00 00 (from 05.2008)	Doc 20 1885 95 20 71; 02.2010	Manufacturer (OEM vehicle)	sources/hydronic-install/Eberspacher_Heater_Mercedes_Sprinter_VW_Crafter_	Engine-compartment OEM layout using MB water pump and MB controls. Not an

ID	SOURCE	COMPONENT/MODEL	REVISION/DATE	AUTHORITY	LOCATION/URL	NOTES
					D5WS_B5WS_information.pdf	independent living-space loop. Use only if this vehicle has that factory heater. Ingested 2026-09-19.
SRC-015	Eberspächer factory-install information sheet — MB Vito / V-Class tandem heater	Hydronic D5WS 25 2126 05 00 00 and D5WZ 25 2124 05 00 00	Doc 25 2005 95 13 77; 03.2001	Manufacturer (OEM vehicle)	sources/hydronic-install/Eberspacher_Heater_Mercedes_Vito_D5WS_D5WZ_Hydronic_25_2126_Information.pdf	Vito tandem system. Do not use unless the vehicle is that platform. Ingested 2026-09-19.
SRC-016	SureCal calorifier connection instructions	SureCal calorifier with thermostatic mixing valve, PRV, 12 V/mains immersion	Undated 1-page sheet	Vendor	sources/hydronic-install/15046148181.pdf	Not the selected DHW tank. Retained as a related calorifier sheet only. Selected unit is SRC-021. Ingested 2026-09-19.
SRC-017	Heatmiser DT-B / PRT-B thermostat wiring for blower boxes	Heatmiser DT-B and PRT-B with Webasto/Eberspächer water heaters	Undated; vendor wiring sheet (01924 274774)	Vendor	sources/hydronic-install/15046148182.pdf	Six wiring options: 1- or 2-speed blowers </> 3 A, plus Webasto and Eberspacher water-heater diagrams. Confirm thermostat model on the vehicle. Ingested 2026-09-19.
SRC-018	Sure Marine Service Webasto DBW-2010 installation guide with SureWire	Webasto DBW-2010 marine boiler (not Eberspächer)	2008-06-09	Marine installer	sources/hydronic-install/hydronic-installation-guide.pdf	Different heater. Transferable loop practice only: bleeders at high points, expansion tank, pump orientation, fan-heater bottom-in/top-out, fill/drain at lowest point. Do not copy Webasto exhaust, fuel, or electrical values. Ingested 2026-09-19.
SRC-019	Sure Marine Service TSL-17 installation guide with SureWire	Sure Marine TSL-17 marine boiler (not Eberspächer); SC1600B / REAL fan-heater lineage	2017-06-08	Marine installer	sources/hydronic-install/TSL17SW_07-08-17-Install-Email.pdf	**Transferable craft (not heater specs):** expansion header as the highest "burp" point with unrestricted path toward the pump

ID	SOURCE	COMPONENT/MODEL	REVISION/DATE	AUTHORITY	LOCATION/URL	NOTES
						(p. 5–6, 26–27); reinforced heater hose, SS double-band clamps, brass/SS barbs, elbows instead of kinks, bleeders at hose high points, fill/drain at lowest point (p. 22–23); fan heaters level, bottom-in/top-out, outlet bleeder, bleed until steady stream (p. 24–25); diesel DHW requires anti-scald mixing valve (p. 6, 25); water-first leak check then glycol, never run pump dry, check level only when cool (p. 27–30); ΔT ~10°F healthy / ≫17°F low-flow hint (p. 29); thermostat away from drafts/sun, no mercury (p. 7). **Do not copy:** TSL/Webasto fuel, exhaust/water-box, summer-valve architecture (rejected DEC-010), engine HX, or their ~40% antifreeze target — this heater is SRC-009 ≤50% propylene glycol. Ingested 2026-09-19; mined in depth 2026-09-19.
SRC-020	Owner notes — van floor reconstruction	Vehicle floor / insulation / longitudinal lumber	Undated text file	Owner notes	`sources/hydronic-install/floor-steps`	Out of hydronic scope. Do not use for heater, plumbing, or DHW design. Ingested 2026-09-19.
SRC-021	Vanlife Outfitters product page — Isotemp Slim Square 4.2 gal heat battery water heater	Isotemp Slim Square 4.2 gal heat battery 16 (4.2 gal / 16 L), 115 VAC / 750 W, with mixing valve; Indel Marine P/N	Retrieved 2026-09-19	Vendor (selected unit)	https://www.vanlifeoutfitters.com/products/isotemp-slim-square-4-2-gallon-water-heater	Replaces the Bosch glycol buffer (DEC-001). AISI 316 tank and coil; 750 W element; factory mixing valve. USA units typically

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		601623Q000003 / 601623QX00003				include BSP-to-NPT adapters; VLO recommends 1/2 in female-to-PEX swivel fittings. Branding may say Isotemp or Isotherm.
SRC-022	Indel Marine / Isotemp Basic, Slim & Slim Square installation and operating instructions	Slim Square 16, type 601623Q000003 (115 V / 750 W, mixer)	Manufacturer I&O; Slim Square §8.3	Manufacturer	https://assets.exalto-emirates.com/documents/upload/248/24746/Water%20heater%20%20basic-slim-slim%20square-OM%20&%20IM.pdf	Install rules for SRC-021. Coil and tank AISI 316. Safety valve 5.5 bar / 80 psi (potable). Slim Square potable pump max 2.9 bar / 42 psi. Mixer 38–65 °C. Drain/safety valve at lowest point. Do not energize element until tank is full. Keep ~30 cm service space in front. Hydronic coil connections BSPP 1/2 in.
SRC-023	Owner purchase record — Eberspächer Hydronic D5S diesel heater	Hydronic D5S diesel heater, order no. 25 2526 05 00 00 (written 252526050000); Installation Kit for Hydronic II D4S/D5S; EasyStart Timer	Qty 1; £902.98 GBP on the heater SKU line	Owner / vendor invoice	Owner-supplied order line 2026-09-19	Confirms DEC-000. SKU is the heater; kit and EasyStart Timer listed as configured options. Matches SRC-009 D5S 12 V 25 2526 05 00 00. Nameplate photo `20191117_154203.jpg` also reads Hydronic II / D5S / 25 2526 / 5.2 kW.
SRC-024	Owner photo — Duda B3-12DW-20 plate heat exchanger as installed in the van	Duda B3-12DW-20 plate heat exchanger double-wall PHE; thermostatic mixer on one upper port; one lower port through cabinet; one lower port on 3/4 in heater hose	Dated 2020-07-06	Owner photo	`photos/IMG_20200706_184257_691.jpg`	Status photo. Label: B3-12DW-20, 0.44 m², 316L, UL listed, dudadiesel.com. Mixer is on the Duda B3-12DW-20 plate heat exchanger. Mixed-outlet barb has no hose to fixtures. One remaining upper barb is open.
SRC-025	Owner photo — Duda B3-12DW-20	Same B3-12DW-20 with mixer and	Dated 2020-06-16	Owner photo	`photos/IMG_20200616_173552_9`	Pre-install. Confirms mixer-

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	plate heat exchanger on the bench before install	four barbs, no hoses			`88.jpg`	on-PHE layout before the cabinet mount.
SRC-026	Owner layout photo – intended freshwater plumbing	35 gal tank → strainer → pump → check valves; Duda B3-12DW-20 plate heat exchanger "inlet bottom, outlet top"; city-water tee; hot sink / hot shower; TMV cold supply; cold sink / cold shower	Owner annotated photo, added 2026-09-19	Owner photo	`photos/freshwater-system.png`	Intended DHW topology, not a photo of the van. Caption: thermostatic mixing valve not shown (it is already on the Duda B3-12DW-20 plate heat exchanger, SRC-024).
SRC-027	Prestone LowTox AF555 – retailer product specs	Propylene glycol engine coolant AF555; UPC 00797496860996; listed as **Concentrate**; freeze −54 °F / boil 225 °F (undiluted claims on retail); mix guidance on related listings 50–67%	Retrieved 2026-09-19	Vendor retail	https://www.doitbest.com/product/573477/prestone-lowtox-gallon-concentrate-54-f-to-225-f-automotive-propylene glycol-573477/	Do it Best fields: Formulation = Concentrate; Concentrate or Ready To Use = Concentrate. Supporting, not SDS. Heater cap still **50%** (SRC-009).
SRC-028	Star brite Bio Safe Year Round Non-Toxic PG Propylene glycol & Coolant – product page	Phosphate-free propylene glycol **engine** coolant; P/N **31700NC**; UPC 71247317015	Retrieved 2026-09-19; owner confirmed jug says "Bio-Safe year-round coolant"	Manufacturer product page	https://www.starbrite.com/products/star-brite-bio-safe-pg-propylene glycol-coolant	Matches on-hand jug name. Not the winterize line. Contrast with Star brite 50/50 RTU SKU **30300** (SRC-031), a different product.
SRC-029	Prestone SDS 506 – LowTox Propylene glycol/Coolant AF555	Product numbers AF555, 70011, 70052, AF559; formula YA-2792PG; **Propylene Glycol 90–100%**; sodium nitrite <0.5%	SDS date of preparation 10/23/13; revision 04/19/23	Manufacturer SDS	`sources/coolant/Prestone-LowTox-AF555-SDS.pdf` (also mirrored by Shamrock Chicago)	**Proves concentrate:** 90–100% PG is full-strength propylene glycol, not a 50/50 premix (premixed coolants list ~40–60% glycol + water). CPID extract of SDS dated 2016-08-05 for UPC 7 97496 86099 6 also shows PG 90–100%: https://whatsinproducts.com/types/type_detail/1/2154

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						3/standard/16-007-040
SRC-030	Star brite SDS – Biosafe Year Round Propylene glycol & Coolant 317XX	Product name STAR BRITE BIOSAFE YEAR ROUND ANTIFREEZE & COOLANT; code **317XX**; **Propylene glycol 80–100%**; sodium nitrite 0.1–<1%	Issue date 2023-06-08 (NA EN)	Manufacturer SDS	`sources/coolant/Star-brite-BioSafe-317XX-SDS-NA-EN.pdf` · https://owm-asset-library.s3.us-west-2.amazonaws.com/STARBRITE/SDS/317XX_SDS_NA_EN.pdf	**Proves concentrate / full strength:** 80–100% PG, not prediluted. Matches Bio-Safe year-round product family (31700NC).
SRC-031	Star brite 50/50 Pre-Diluted Ready-To-Use Propylene glycol Coolant	Separate RTU SKU **30300**; "No mixing required"; "Do not add water"; ethylene glycol (Prop 65)	Retrieved 2026-09-19	Manufacturer product page	https://www.starbrite.com/products/star-brite-50-50-green-pre-diluted-propylene glycol-coolant	Shows what Star brite calls ready-to-use. **Not** the Bio-Safe 31700 jug. Do not confuse the two.
SRC-032	Sure Marine Service – Marine Heating Catalog (REAL fan heaters)	REAL hydronic air handlers / defrosters; 3/4 in NPT + bleeder; 3-speed switch options	Catalog PDF	Manufacturer catalog	`sources/hydronic-install/SureMarine_Marine_Heating_Catalog.pdf`	Cabin-heater model sheet / mounting options. Switches available with or without Off (relevant to Q-021). Not a substitute for the on-hand unit's exact model stamp. Downloaded 2026-09-19.
SRC-033	ICM Controls SC1600L/VL installation, operation & application guides	SimpleComfort SC1600 series; R/W/S1/S2; millivolt-30 VAC/VDC; 1 A/terminal; 2 A total	2-page guides	Manufacturer	`sources/controls/ICM_SC1600L_Installation_Operation_Application_Guide.pdf` · `sources/controls/ICM_SC1600L_VL_Thermostat_Manual.pdf` · https://www.sidharvey.com/documents/SC1600L_Install.pdf	Both manuals show the same SC1600-series interface: R/W heat-control contact and S1/S2 optional remote sensor. This strongly supports a shared family design, but does **not** prove SC1600B ≡ SC1600L or map the thermostat to the EasyStart/D5S. Downloaded 2026-09-19.
SRC-034	Noctua NA-FC1 PWM fan controller manual	NA-FC1; 0–100% PWM duty; optional "no stop" ~300 rpm floor	EN manual	Manufacturer	`sources/controls/Noctua_NA-FC1_Manual_EN.pdf` · https://cdn.noctua.at/preview/media	Candidate only. DEC-015 wires Noctua fans on **power leads** with an inline 12 V dial – NA-FC1 is PWM-path and may stop

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					/noctua_na_fc1_manual_en.pdf	fans at low dial (Q-021). Do not assume this is the on-hand controller until the part is identified. Downloaded 2026-09-19.
SRC-035	Paneltronics product catalog	Marine AC/DC distribution panels; breakers; bus bars	Catalog PDF	Manufacturer	`sources/electrical/Paneltronics_Catalog.pdf` · https://paneltronics.com/docs/Paneltronics_Catalog.pdf	Generic catalog – **not** a wiring diagram for this van's panel. Q-009/Q-016 still need the panel's part number / photos / model-specific drawing. Downloaded 2026-09-19.
SRC-036	Resideo / Honeywell AM-1 Series installation instructions (62-3075EFS)	AM-1 / AM100-1LF family thermostatic mixing valves; 70–145°F standard	Doc 62-3075EFS	Manufacturer	`sources/plumbing/Honeywell_AM-1_Series_Installation_Instructions_62-3075EFS.pdf` · https://customer.resideo.com/resources/Techlit/TechLitDocuments/62-0000s/62-3075EFS.pdf	Temperature set/lock commissioning for the Duda mixer. Downloaded 2026-09-19.
SRC-037	Honeywell SparcoMix AM-1 Series submittal sheet (68-3098)	AM100-1LF dimensions / Cv / ASSE 1017	Doc 68-3098	Manufacturer	`sources/plumbing/Honeywell_SparcoMix_AM-1_Series_Submittal_68-3098.pdf` · https://customer.resideo.com/resources/Techlit/TechLitDocuments/68-0000s/68-3098.pdf	Spec sheet companion to SRC-036. Downloaded 2026-09-19.
SRC-038	Paneltronics AC three-branch circuit breaker panel with main – wiring sheet 9972313B	Paneltronics **9972313B**: 30 A main + three 15 A / 120 V AC branches; reverse-polarity LED; pre-wired buses	Wiring sheet 9972313B; retrieved 2026-09-19	Manufacturer	`sources/electrical/Paneltronics_9972313B_Wiring.pdf` · https://www.paneltronics.com/docs/wiring/9972313B.pdf	Candidate for this van (DEC-020). **No GFCI on panel.** Element ~6.5 A → one 15 A branch after face photo. Retailer listings match 30 A main / 3×15 A.
SRC-039	Heatso order #6083 support email	Owner's purchased Hydronic II D5S 12	2018-02-25	Seller support / owner email	Email supplied by owner; related product page	Heatso support explicitly states: "These units are

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		V + kit + 7-day timer is compatible with High Altitude Kit			https://www.heatso.com/eberspacher-espar-hydronic-ii-d5s-12v-heater/	compatible with HAK." Supports heater/HAK compatibility only; does not prove physical H-Kit marking, kit presence, or completed wiring.
SRC-040	mylifewithoutdoor / Ben Racine, "Sprinter Van Electrical"	Owner design intent for this van's house electrical (solar / shore / alternator / inverter / distribution)	Owner blog (Medium)	Owner blog	https://medium.com/@mylifewodoor/sprinter-van-electrical-27b51f6bbe8f	**Intent / context for hydronic AC only.** Owner pointed to this post 2026-09-20 as the incomplete but governing write-up of the van electrical. Automated fetch blocked (Cloudflare). Do not invent Ah, solar watts, charger amps, or wire sizes from memory—confirm on van. On-van face evidence (Paneltronics **72313**, Xantrex **PROwatt SW**, **LinkPRO** on the wood panel) must still match this post. Not a substitute for SRC-038 / HOLD 9.
SRC-041	FarOutRide, "Electrical System Guide for DIY Van Conversion"	Reference architecture for off-grid van electrical: house battery, solar, alternator B2B, shore charger, inverter, 12 V fuse block, 120 V loads	Updated 2026-08-22	Installer / vanlife guide	https://faroutride.com/electrical-system/	**Reference pattern, not this van's BOM.** Owner 2026-09-20: house electrical is "something like" this guide. Useful vocabulary: charge sources → house bank → 12 V loads; inverter and/or shore feed 120 V AC. FarOutRide "Standard" keeps shore charger and inverter as separate devices (no transfer switch). Use only to explain how Isotemp AC sits in

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						a typical van; pinouts and amp ratings for *this* Sprinter still come from on-van labels + Q-009.
SRC-042	Upfit Supply — Mercedes-Benz Sprinter (170 WB) interior cargo measurements	Regular 170 body, not the extended body. Wheel well 36 in W × 12 in H × 9 in D. 53.1 in between wells. Page lists 2017.	Page updated 2025-09-26; retrieved 2026-10-17	Cargo-upfit sheet	https://upfitsupply.com/blogs/measurement-guides/mercedes-benz-sprinter-170wb-interior-cargo-measurements	W is the along-wall length in their shelving diagram. Their 72 in roof figure is the low-roof shelving height on that page — do not use it for this van's interior height. Measure the hump before cutting.
SRC-043	2016 Mercedes-Benz Sprinter competitive comparison	Cargo floor length: 170 high roof 178.2 in; 170 EXT high roof 193.9 in	2016 brochure PDF	Manufacturer comparison	https://di-uploads-pod4.s3.amazonaws.com/fletcherjonesimports/uploads/2016/08/Competitive_Comparison_MB_Sprinter.pdf	Same 170 in wheelbase. The EXT floor is 15.7 in longer, behind the rear axle. Other sheets call a shorter "cargo bed" length; do not mix the two.
SRC-044	S&B Filters 35 gallon wheel-well tank, part 10-3008	Over-wheel-well tank, driver or passenger, 2007+ Sprinter 144 / 170 / 170 EXT. Not dually. 48 × 16 × 16 in, ±0.5 in. About 35 gal. Dry 31.2 lb. Straps included. KUS sender S5U-15 is 15 in, not included.	Ordered 2026-10-17	Manufacturer / vendor	https://sbfilters.com/products/35-gallon-wheel-well-water-tank-both-sides · https://www.campervan-hq.com/products/s-b-filters-35-gallon-wheel-well-water-tank-driver-passenger-side-sprinter-transit-10-3008	Passenger side on this van. Bounding box, not a measured saddle profile. Confirm port height above the lid when the tank arrives.